SERVICE STATIONS
SECTION UIP 11

SMALL SELF SERVICE BOOTHS

Average costs per square foot, typical 8' wall height for complete booth, excluding all exterior equipment and improvements. Electrical costs are for booth lighting only; add other circuits from unit costs to the right. Canopies should be added from the table below, heat from this Section on a cost-per-ton basis or from Section UIP 3. For masonry booths, use comparable steel booth costs.

LOW COST – This is an older, open style, uninsulated booth with minimum electrical and no plumbing. Cost range can be used to price miscellaneous storage structures. Small tire display cabinet structures will cost $36.25 to $39.25 per square foot.

Siding-Stucco-Glass				Steel-Glass or Masonry			
Area	Cost	Area	Cost	Area	Cost	Area	Cost
25	306.00	75	189.00	25	338.00	75	212.00
50	227.00	100	173.00	50	251.00	100	189.00

AVERAGE STEEL – Typical of present day cashier booths, with good electrical and no plumbing or heat. Add 25% for bullet-proof glass or see Section UIP 5.

50	440.00	100	290.00
75	346.00	125	258.00

GOOD STEEL – Good security structure with bullet proof glass and two or three commercial plumbing fixtures. For intercom system, add $755.00 plus $196.00 per speaker.

75	625.00	150	407.00
100	525.00	200	338.00

NOTE – For small kiosk storage buildings, use Average booth costs, less 10%. For small separate restroom buildings, use Good booth costs, less 5%.

BUILDING IMPROVEMENT UNIT COSTS

All costs are for completely installed items. They include costs of design, engineering and contractors' profit and overhead, as well as a prorated share of miscellaneous ancillary costs.

CANOPIES – Cost per square foot of covered area including light fixtures and support. Wiring costs are included in electrical costs, if all circuits are counted. Add 10% for gable or ranch style, 25% for round. Add for roof covering from Section UIP 4. Individually designed or highly ornamented canopies can cost 100% more.

	Low Cost	Avg.	Good	Excellent
Concrete tees	35.50	40.75	48.75	57.00
Steel	30.50	39.25	50.00	63.00
Wood frame and sheathing	26.75	33.75	41.00	52.50

FLOOR AND FOUNDATION – Cost per square foot of floor area.

Concrete slab	11.10	13.05	15.45	18.40
Wood floor structure	15.15	17.15	19.40	21.85

Add for floor covering from Section UIP 2.

ROOF – Cost per square foot of roof area.

Steel prefab. frame/decking	35.00	38.75	44.00	50.00
Wood frame and sheathing	19.20	20.45	21.75	23.35

Add for roof covering from Section UIP 4. For ceilings under gable roofs, see Section UIP 2.

PARTITIONS – Cost per square foot of partition, including doors.

	Low Cost	Avg.	Good	Excellent
Concrete block	24.20	26.50	29.50	33.00
Metal	27.50	30.50	34.25	37.75
Metal/glass, security	41.00	51.50	60.50	72.50
Wood frame, drywall (plaster, add 15% to 20%)	17.60	19.65	21.40	23.35

Add for cabinetry from Section UIP 2.

ELECTRICAL

Base cost per station	10300.00	11900.00	13800.00	16000.00
Add per circuit	740.00	820.00	975.00	1110.00

WALLS – Cost per square foot of exterior wall area.

Steel and glass, painted	36.50	40.75	45.00	50.50
Steel and glass, porcelainized	40.75	45.00	51.50	57.00
Steel panels, masonry veneer	45.75	52.50	59.50	69.00
Steel panels, block backup	44.75	50.50	55.50	62.00
Wood frame, stucco or siding	32.50	35.00	36.50	39.50
Wood frame, brick veneer	39.50	42.50	47.00	51.50
Brick masonry	43.25	48.25	55.00	59.50
Concrete block	35.00	37.75	42.25	47.00
Overhead steel or aluminum doors	26.00	29.50	34.25	38.75
Sectional, roll–up	26.75	31.75	36.50	42.50
Overhead plastic doors	23.35	26.50	29.50	33.00
Overhead wood doors	21.40	24.20	26.75	30.50
Folding steel gates	28.00	31.75	35.50	39.50
Add for electric door operator	1760.00	2070.00	2260.00	2575.00

Add for ceramic tile from Section UIP 5. See Section UIP 6 for store front entries.

PLUMBING

Cost per fixture	2650.00	3175.00	3875.00	4700.00

Count fountains without cooling as 1/2 fixture. Hot water heaters count as one fixture.

HEATING – Average cost per square foot of heated area. For specific system costs not found below, see Section SEG 4 or UIP 3.

Forced air furnace	4.41	5.69	7.37	9.52
Space heaters, suspended	1.78	2.48	3.46	4.80
Wall furnace	2.14	2.61	3.17	3.86
Package A.C. (short ductwork)	8.12	10.70	14.05	18.55
Heat pump system	8.72	12.10	16.80	23.30
Evaporative cooler	3.79	4.52	5.42	6.48
Individual thru–wall heat pump	3.79	5.19	7.13	9.75

Small individual heat pumps cost $2,290 to $3,100 per ton of rated capacity.

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SERVICE STATIONS
SECTION UIP 11

YARD IMPROVEMENTS

PAVING

	LOW COST	AVG.	GOOD
Cost per square foot			
Concrete islands	15.00	17.60	20.45
Island pump shelters, including lighting/supports	75.50	99.00	128.00
5" - 6" concrete, approaches and drives	6.34	8.43	10.50
4" concrete, walks, etc	5.28	6.84	8.43
Apron channel drain and grate, per linear foot	99.00	128.00	164.00
Asphalt	3.25	4.84	5.85
6" curb, per linear foot	14.75	18.45	23.30
Precast concrete bumpers, per linear foot	7.93	10.10	13.20
Wood bumpers, per linear foot	7.34	10.50	14.30
Metal guard rail, pipe or posts, per linear foot	35.00	46.25	62.50

YARD LIGHTING

Cost per pole, 12'	1380.00	1700.00	2170.00
Cost per pole, 24'	2100.00	2500.00	3125.00
Add per fixture, incandescent	645.00	785.00	1060.00
Fluorescent or quartz iodine	1210.00	1460.00	1700.00
Mercury vapor	1340.00	1760.00	2460.00
High pressure sodium or metal halide	1510.00	2170.00	3050.00

SIGNS

Cost per square foot of signs includes installation, lighting and wiring, but not cost of poles or structural supports.

	COST RANGE
Illuminated plastic, add 35% for 2 sides	124.00 – 258.00
Metal, painted two sides	81.50 – 122.00
Painted one side	65.50 – 96.00
Add for porcelainized metal, per face	14.60 – 19.20
Add for neon tubing, per face	35% – 45%
Plastic interior lighting	100.00 – 144.00
Spheres, per foot of diameter, including post	1080.00 – 1630.00

Installation amounts to 18% to 25% of total cost.

SIGN POSTS OR POLES

Cost per linear foot of poles set in concrete and painted. For tapered poles, use the diameter at the base. For cantilevered posts, add 50% to the cost. Decorative pole covers cost $2,120 to $4,750 each.

4"	75.50 – 99.00	10"	149.00 – 236.00
6"	100.00 – 147.00	12"	173.00 – 290.00
8"	127.00 – 196.00	14"	196.00 – 338.00

PIPING

Average cost: $1,760 to $2,330 per pump or dispenser per product, plus $1,140 to $1,520 per tank, plus $600 to $785 for each air and water well or stand. Add 50% for double wall installations.

EQUIPMENT

Miscellaneous office and garage repair and lube equipment, cash registers, safes, fume exhausters, etc., not listed below, can be found in Section UIP 15. See Section UIP 11 for Tanks.

OFFICE OR BOOTH EQUIPMENT

Electronic remote control totalizer, per hose	2170.00 – 3875.00
Computer cabinet	2210.00 – 2950.00
Tank monitor console	6400.00 – 11600.00
Food booth shelving, gondolas, etc., per booth	5800.00 – 23200.00
Merchandise freezer, each	7300.00 – 10500.00
Walk in cooler, per square foot	164.00 – 268.00

AIR COMPRESSORS

HP	COST RANGE	HP	COST RANGE	HP	COST RANGE
1/3	1960.00 – 2330.00	1½	4225.00 – 5150.00	7½	8800.00 – 10400.00
1/2	2500.00 – 3025.00	2	4725.00 – 5700.00	10	9950.00 – 11900.00
3/4	3050.00 – 3600.00	3	5600.00 – 6650.00	15	12200.00 – 14500.00
1	3550.00 – 4075.00	5	6850.00 – 8150.00	20	14000.00 – 16700.00

If the cost without installation is desired, deduct 30% on small size; 25% on medium, 20% on large sizes.

HOISTS

Frame, lift (in-ground)	COST RANGE
Auto, 8,000 lb. single post	12400.00 – 14700.00
8,000 lb. double post	16900.00 – 20500.00
Truck, 11,000 lb. double post	24300.00 – 27300.00
16,500 lb. double post	15800.00 – 19300.00
Truck, 19,500 lb. double post	22600.00 – 25900.00
24,000 lb. double post	28300.00 – 32100.00
Bus or heavy truck, 36,000 lb. double post	35000.00 – 39600.00
Drive-on (surface mount)	
Auto, 7,000 lb. four post	15200.00 – 18800.00
8,000 lb. single post	17600.00 – 21000.00
Truck, 12,000 lb. four post	13600.00 – 15800.00

Large commercial-type grease pits with air and electric outlets cost $17.15 to $24.70 per cubic foot. Installation cost of hoists is approximately 20% to 30% of the total cost.

PUMPS AND DISPENSERS

Mechanical dispenser including vapor recovery, exclusive of submerged pump	
Single	5800.00 – 7550.00
Twin	8700.00 – 11100.00
Electronic dispenser including vapor recovery, exclusive of submerged pumps	
Single	9900.00 – 13400.00
Twin	13400.00 – 18000.00
Three hose	18800.00 – 27400.00
Add for double (two) sided operation	7050.00 – 8150.00
Add to all multiple types for mixed products, per hose	510.00 – 820.00
Add for point of purchase, per acceptor	4625.00 – 5800.00
Add to all types for integral suction pump, per dispenser	740.00 – 1060.00
Submerged pumps, one pump may serve several dispensers	
1/3 horsepower	2140.00 – 2525.00
3/4 horsepower	2500.00 – 3175.00
1 1/2 horsepower	3125.00 – 3875.00
Industrial or Commercial pumps	4100.00 – 5300.00
Add for ticket printer and counter	785.00 – 1110.00
Consumer pumps, electric	1630.00 – 3250.00
Utility pumps, electric, farm and ranch type	1110.00 – 1670.00
Hand pumps, farm and ranch type	535.00 – 805.00

Costs include 10% installation cost on aboveground items, 20% for submerged pumps.

AIR AND WATER SERVICE

Cost per unit	LOW COST	AVG.	GOOD
Air and water wells, disappearing hose	880.00	1110.00	1390.00
Automatic tire inflater	2100.00	2420.00	2875.00
Single swing arm stand	690.00	820.00	1030.00
Water or air hydrant	645.00	740.00	820.00

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CAR WASHES

SECTION UIP 11



DRIVE THRU CAR WASHES

Small single car drive thru roll over robot type automated car washes cost $152,000 to $312,000 including equipment and building shell. Large commercial truck and municipal fleet washes cost $484,000 to $944,000 per bay. Add yard improvements from Page 2.



SELF SERVE CAR WASHES

Small coin operated washes, for self serve user operation typically cost $70,000 to $118,000 per stall including equipment and building. An open 8 bay facility may go as low as $54,250 per stall. Costs do not include yard improvements which may run 15% to 25% of stall costs.

DRIVE-THRU CAR WASHES

CLASS	TYPE	COST/ SQ. Ft.	EXTERIOR WALLS	INTERIOR FINISH	LIGHTING, PLUMBING AND MECHANICAL	HEAT
C	Excellent	$ 192.00	Best block or concrete, masonry trim, good tunnel doors, roof and trim	Good finish, drains and sump, small storage, office and/or waiting area	Good electrical, good commercial fixtures and outlets	Space heaters
	Good	163.00	Decorative block or brick, tunnel doors, good roof and trim	Unfinished, concrete floor, good drains and sump	Good lighting and outlets, adequate water	Space heaters
	Average	134.00	Open ends, block or low cost brick, average roof cover, little trim	Unfinished, concrete floor, drains, sump	Adequate electrical and water service and outlets	None
	Low cost	114.00	Side walls only, concrete block, shed or flat roof, very plain	Unfinished, concrete floor, adequate drains	Adequate electrical and water service and outlets	None
D	Good	154.50	Good stucco, siding or brick veneer, tunnel doors, good roof	Unfinished, concrete floor, good drains and sump	Good lighting and outlets, adequate water	Space heaters
	Average	126.00	Open ends, stucco or siding, average roof cover	Unfinished, concrete floor, drains, sump	Adequate electrical and water service and outlets	None
	Low cost	107.00	Side walls only, low cost siding or stucco, shed or flat roof	Unfinished, concrete floor, adequate drains	Adequate electrical and water service and outlets	None
S	Excellent	191.50	Best steel, masonry trim, good tunnel doors, roof and trim	Good finish, drains and sump, small storage, office and/or waiting area	Good electrical, good commercial fixtures and outlets	Space heaters
	Good	160.50	Good metal and steel frame, tunnel doors, good roof and trim	Unfinished, concrete floor, good drains and sump	Good lighting and outlets, adequate water	Space heaters
	Average	129.00	Open ends, enameled siding on light frame, little trim	Unfinished, concrete floor, drains, sump	Adequate electrical and water service and outlets	None
	Low cost	108.00	Side walls only, low cost siding on steel frame, shed or flat roof	Unfinished, concrete floor, adequate drains	Adequate electrical and water service and outlets	None
	Cheap	89.50	Light pre-engineered metal bldg.	Unfinished, concrete floor, drains	Minimum electrical and water service	None

SELF-SERVE CAR WASHES

CLASS	TYPE	COST/ SQ. Ft.	EXTERIOR WALLS	INTERIOR FINISH	LIGHTING, PLUMBING AND MECHANICAL	HEAT
C	Excellent	$ 168.50	Best block or brick, masonry or EIFS trim, good tiled bays and roof	Unfinished, concrete floor, good drains and sump, equipment room	Good electrical, good commercial fixtures	Space heaters
	Good	135.50	Decorative block or brick, bay doors, good roof	Unfinished, concrete floor, good drains and sump, equipment room	Good lighting and outlets, adequate water	Space heaters
	Average	103.00	End and bay walls only, block or low-cost brick, average roof cover, trim	Unfinished, concrete floor, adequate drains and sump, equipment room	Adequate electrical and water service and outlets	None
	Low cost	82.00	End and half bay walls only, concrete block, shed or flat roof	Unfinished, concrete floor, adequate drains, sump, equipment room	Adequate electrical and water service and outlets	None
D	Good	127.00	Good stucco, siding or brick veneer, bay doors, good roof	Unfinished, concrete floor, good drains and sump, equipment room	Good lighting and outlets, adequate water	Space heaters
	Average	95.50	End and bay walls only, stucco or siding, average roof and trim	Unfinished, concrete floor, adequate drains and sump, equipment room	Adequate electrical and water service and outlets	None
	Low cost	75.50	End and half bay walls only, low-cost siding or stucco	Unfinished, concrete floor, adequate drains, sump, equipment room	Adequate electrical and water service and outlets	None
S	Good	131.00	Good metal and steel frame, bay doors, good roof	Unfinished, concrete floor, good drains and sump, equipment room	Good lighting and outlets, adequate water	Space heaters
	Average	97.00	End and bay walls only, enameled siding on light frame	Unfinished, concrete floor, adequate drains and sump, equipment room	Adequate electrical and water service and outlets	None
	Low cost	75.00	End and half bay walls only, low-cost siding on steel frame	Unfinished, concrete floor, adequate drains, sump, equipment room	Adequate electrical and water service and outlets	None

NOTE: For refinement notes, see bottom of page 4

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CAR WASHES
SECTION UIP 11

AUTOMATIC CAR WASHES



Full service or tunnel car wash service buildings include finished office/sales area, locker and rest rooms and basic equipment room. Canopies and ornamental pylons are priced separately. Add heat from Page 1.

CLASS	TYPE	COST/ SQ. Ft.	EXTERIOR WALLS	INTERIOR FINISH	LIGHTING, PLUMBING AND MECHANICAL	HEAT
C	Excellent	$ 305.00	Best stone or brick, masonry trim, good fenestration, ornamentation	Good finish, best work-manship, many built-in features, waiting area	Good electrical, good commercial plumbing fixtures	Package A.C.
	Good	233.00	Good block or brick, good storefront and trim	Good office and retail space, tiled floors, rest- rooms, glazed view area	Average commercial fixtures, adequate interior circuits	Package A.C.
	Average	170.00	Average block or brick, little trim, small storefront	Small office, storage, restrooms, locker room, vinyl and carpet	Adequate commercial plumbing fixtures, standard electrical	Forced air
	Low cost	126.00	Concrete block, inexpensive sash and doors	Minimum finishes, vinyl com- position tile, few built-in items	Minimum interior electrical and plumbing fixtures	Space heaters
D	Excellent	290.50	Best stucco, EIFS or masonry veneer, good fenestration, ornament.	Good finish, best workman- ship, many built-in features, waiting area	Good electrical, good commercial plumbing fixtures	Package A.C.
	Good	222.00	Good stucco or brick veneer, good storefront and trim	Good office and retail space, tiled floors, rest- rooms, glazed view area	Average commercial fixtures, adequate interior circuits	Package A.C.
	Average	161.50	Average stucco or siding, little trim, small storefront	Small office, storage, restrooms, locker room, vinyl and carpet	Adequate commercial plumbing fixtures, standard electrical	Forced air
	Low cost	118.50	Stucco or siding, inexpensive sash and doors	Minimum finishes, vinyl com- position tile, few built-in items	Minimum interior electrical and plumbing fixtures	Space heaters
S	Excellent	300.00	Best steel, masonry trim, good fenestration and ornamentation	Good finish, best workman- ship, many built-in features, waiting area	Good electrical, good commercial plumbing fixtures	Package A.C.
	Good	227.50	Good steel, good storefront and trim	Good office and retail space, tiled floors, rest- rooms, glazed view area	Average commercial fixtures, adequate interior circuits	Package A.C.
	Average	164.00	Average painted steel, little trim, small storefront	Small office, storage, restrooms, locker room, vinyl and carpet	Adequate commercial plumbing fixtures, standard electrical	Forced air
	Low cost	120.50	Painted steel, inexpensive sash and doors	Minimum finishes, vinyl com- position tile, few built-in items	Minimum interior electrical and plumbing fixtures	Space heaters

CARWASH CANOPIES

CLASS	TYPE	COST/ SQ. Ft.	EXTERIOR WALLS
CDS	Excellent	$129.00	Good tunnel walls and doors, concrete, built up or steel roof, concrete floor, lighting, drains, sump, no heat
	Very Good	105.00	Good tunnel walls and roof structure, open ends, concrete floor, good electrical and drains, sump, no heat
	Good	85.50	Some tunnel knee walls or column ornamentation, good roof and supports, electrical, concrete floor, drains
	Average	57.50	No walls, entrance, service canopy, metal or wood frame, finished soffit, lighting, concrete floor
	Fair	38.00	No walls, average canopy, decorative columns, adequate lighting, concrete floor
	Low cost	25.25	No walls, shade, patio cover, metal or wood, minimum electrical, concrete paving
	Cheap	16.75	No walls, light steel, fiberglass or shade netting roof on low-cost pipe, asphalt, minimum electrical, auto detail area

The base wall height is 12 feet (3.66 meters), excluding gables; add or deduct 2% for each foot (.305 meter) of deviation. Adjust for size and shape from page 5 and heat from tables on Page 1. For equipment costs, see Page 5. For floor heat, add $9.17 to $23.25 per square foot of heated area. For automatic door operators, add $164.00 to $2460.00 each.

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COSTS
SECTION UIP 11

EQUIPMENT COSTS

Equipment costs cover all equipment for standard tunnel type car washes, but do not include building improvements, service station equipment, paving, signs, etc. Number of cars washed per hour is a function of the length of the wash line and the quantity and quality of the equipment. Low Cost classification is for the semi automatic wash, while the Good car wash is fully automated with personnel only for interior cleaning and before and after service commensurate with the capacity (length) of the line. For a detailed breakdown of the equipment costs, see table below. The 30' to 50' cost range includes self-wash tunnels.

LENGTH OF LINE	LOW COST	AVERAGE	GOOD
30' (incl. self–console control)	101000.00	142000.00	201000.00
50'	199000.00	251000.00	317000.00
75'	278000.00	338000.00	412000.00
100'	333000.00	399000.00	476000.00
125'	377000.00	445000.00	529000.00
150'	413000.00	484000.00	570000.00

UNIT COSTS	COST RANGE
Vacuum station, complete	18300.00 – 31700.00
Conveyor 30'	25800.00 – 40400.00
Conveyor 50'	35200.00 – 54750.00
Conveyor 75'	44000.00 – 68000.00
Conveyor 100'	53000.00 – 80500.00
Conveyor 125'	59750.00 – 91250.00
Conveyor 150'	66750.00 – 99750.00
Tire brush washer	14600.00 – 18800.00
Tire solution applicator, inc. pump	5600.00 – 7050.00
Prep. hand gun	8150.00 – 14000.00
Undercarriage flush	3525.00 – 4875.00
Applicator arch (pre–final rinse or wax), each	5200.00 – 8150.00
Rinse and wax deluxe arch combo	14700.00 – 19300.00
Polish and wax arch combo	22100.00 – 34300.00
Mitting curtains	32500.00 – 45100.00
Brushes side panel	15400.00 – 23900.00
Side and top combo	53000.00 – 59750.00
Hydraulic power PAC, each	7550.00 – 12700.00
Motor control	19300.00 – 35000.00
Computer console	10600.00 – 21000.00
Solution feed, pump	9250.00 – 14600.00
Water reclamation/filtration	56000.00 – 98500.00
Air dry blower	35200.00 – 66750.00
Washing machine, extractor	9250.00 – 17600.00
Mitting trough, hand wash, each	1160.00 – 2330.00

SELF SERVE WASH AND DRIVE THRU

Self wash assembly equipment base, including hot water	15200.00 – 41900.00
Add per bay (including basic soap, wax, rinses)	9100.00 – 17600.00
Degreaser–foam brush cleaner, extra waxes, base cost each	3950.00 – 5150.00
add per bay	1340.00 – 2950.00
Roll over robot, self drive thru, equipment base	74000.00 – 116000.00
Deluxe, including brushless (touch free) system	128000.00 – 188000.00
Add arch applicators from table above	
Pay entry, computerized communication system and signage	9400.00 – 19600.00
Heat freeze protection	2330.00 – 7050.00
Air dry blower	30200.00 – 53500.00
Water softener	4625.00 – 13900.00
Water reclamation/filtration	11600.00 – 63750.00
Vacuum, per exterior station (interior installations, less 25%)	2775.00 – 6300.00
Change machine/automated pay station	5400.00 – 11600.00
Towel vending machine	895.00 – 1320.00

FLOOR AREA/ SHAPE MULTIPLIERS

AREA PER UNIT Sq. Ft.	MULTIPLIER Carwashes	AREA PER UNIT Sq. Ft.	MULTIPLIER Carwashes
400	1.118	2,600	0.891
600	1.064	2,800	0.883
800	1.027	3,200	0.869
1,000	1.000	3,600	0.856
1,200	0.978	4,000	0.846
1,400	0.960	4,400	0.836
1,600	0.945	4,800	0.827
1,800	0.932	5,200	0.819
2,000	0.920	5,600	0.812
2,200	0.909	6,000	0.805
2,400	0.900	8,000	0.775

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PREFABRICATED METAL BUILDINGS

SECTION UIP 11

SQUARE FOOT COSTS

Costs are for standard galvanized steel buildings engineered for a 20 lb. live load with minimum fenestration, erected on concrete footings, without floors, lights or heat. The normal cost range is from 70% (rural) to 150% (commercial) of the listed costs depending on type of frame, skin and fenestration. For buildings modified for grain storage, add 15% to 25%. All costs are based on professional labor supervised by a contractor. For amateur workmanship or work done by farm or ranch help, costs should be decreased by 15% to 30%.

QUONSET BUILDINGS

(Base Height = 20' Center of Arch)

LENGTH (feet)	WIDTH 30'	40'	60'	70'
30'	33.75	-----	-----	-----
36'	32.25	-----	-----	-----
48'	30.00	27.50	-----	-----
60'	28.25	25.75	24.60	-----
72'	27.25	24.65	23.60	22.65
84'	26.25	23.85	22.55	21.90
96'	25.25	23.00	21.90	21.10
108'	24.50	22.40	21.15	20.55
120'	23.85	21.80	20.65	19.90
160'	22.35	20.35	19.05	18.45
200'	-----	19.05	18.10	17.60
240'	-----	18.25	17.35	16.90



PRE-ENGINEERED STEEL BUILDINGS

WIDTH (feet)	HEIGHT (to eaves)	LENGTH/ WIDTH RATIO 1.0	1.5	2.0	3.0	4.0	5.0
20'	10'	28.75	27.25	26.25	24.60	23.75	23.00
30'	12'	24.50	23.55	22.65	21.75	21.00	20.40
40'	14'	24.65	23.25	22.35	21.00	19.95	19.35
50'	14'	22.05	21.15	20.80	19.95	19.45	19.05
60'	14'	20.25	19.60	19.15	18.75	18.25	18.00
80'	16'	20.55	19.80	19.45	18.85	18.30	18.10
100'	16'	20.25	19.35	18.85	18.00	17.45	17.05
140'	16'	18.00	17.45	17.00	16.40	16.10	15.65
160'	18'	17.70	17.15	16.90	16.30	15.70	15.55
200'	18'	16.90	16.40	16.10	15.65	15.40	15.05

Height: Add or deduct 2% for each foot of deviation from base.

Light: 12 lb. live load: Deduct 10%. heavy load: 30 lb., add 10%; 40lb., add 25%.

Sandwich panels: Add 60%. aluminum skin: Add 15%.

Enameled steel: Add 5%. Slant wall buildings: Deduct 10%.

Wood pole framed buildings : Deduct 20%.

Buildings constructed of composite plastics (FRP): Add 75%.

BUILDING IMPROVEMENT UNIT COSTS

FOUNDATIONS – Concrete column footings: Apply to total number of columns. For perimeter footings and floors or other interior components see Section SEG 3, SEG 4 or SEG 7 Unit-in-Place Cost sections.

FOUNDATIONS	Low Cost	Avg.	Good	Excl.
Steel columns, light pre-engineered frame	58.50	78.00	104.00	139.00
Wood columns, light pole frame	44.25	63.50	91.00	128.00

Add or deduct 2% for each foot of deviation from 14' base height.

FRAME – Apply to total floor area.

	Low Cost	Avg.	Good	Excl.
Steel, pre-eng. open web tapered truss	4.38	4.78	5.29	5.67
Post and open web truss	4.54	4.87	5.44	6.17
Post and beam	4.62	5.12	5.67	6.34
Tapered plate, Post/beam end walls	5.29	6.17	7.30	8.37
Tapered plate	5.84	6.82	8.12	9.47
Plastic, fiber reinforced	17.35	18.90	20.80	22.80
Wood, pole frame, untreated	2.63	3.30	4.56	6.18
Treated wood	2.70	3.38	4.72	6.34

Add or deduct 4% for each foot of deviation from 14' base height. Add 4% for high profile (4:12 roof slope) buildings.

FLOORS: See Section SEG 3 or SEG 4.

ELECTRICAL – Apply to total floor area. For rural utility lighting, deduct 50% to 80%.

Lighting, incandescent	1.62	2.19	3.40	4.78
Lighting, fluorescent	1.97	2.86	3.80	5.36

WALL COVER AND SUPPORTS – Apply to total wall area.

Wall girts, steel	1.49	1.97	2.68	3.50
Wood	0.64	0.91	1.37	1.91
Plastic, fiber reinforced	2.11	2.99	4.22	5.44
Aluminum, light (.0175" to .024" thick)	3.49	3.89	4.54	4.87
heavy (.032" to .050" thick)	5.84	6.42	7.57	8.37
Steel, light (30 to 26 gauge)	2.68	3.25	3.65	4.62
Heavy (24 to 18 gauge)	4.94	5.84	6.82	8.12
Plastic (FRP 8 oz. to 16 oz.)	4.78	6.66	9.09	12.65
Add for fenestration, per sq. ft. of opening (wood frame, deduct 25% – 50%).				
Doors, sliding	16.55	21.65	27.75	37.50
Overhead	24.80	30.25	36.50	43.50
Pedestrian	48.50	63.00	82.50	106.00
Windows or louvers	44.25	56.00	72.00	91.00
Add for canopies, per sq. ft. of canopy area	16.55	21.65	27.75	37.50
Add for enameled paint	24.80	30.25	36.50	43.50
Add for sandwich panel	48.50	63.00	82.50	106.00
Add for insulation	44.25	56.00	72.00	91.00
Add for exterior sheathing	1.62	1.97	2.11	2.52
Add for interior sheathing or liner, finished	2.77	3.25	3.65	4.54

ROOF COVER AND SUPPORTS – Apply to roof area.

Roof purlins, steel	1.71	2.11	2.92	3.65
Wood	0.94	1.27	1.86	2.29
Plastic, fiber reinforced	2.77	3.49	4.71	6.08
Aluminum, light (.0175" to .024" thick)	3.32	3.56	4.22	4.62
Heavy (.032" to.050" thick)	5.53	6.26	7.15	8.12
Steel, light (30 to 26 gauge)	2.19	2.92	3.49	4.30
Heavy (24 to 18 gauge)	4.78	5.53	6.42	7.72
Plastic (FRP 8 oz. to 16 oz.)	4.54	6.08	8.28	11.20
Add for enameled paint	0.37	0.54	0.74	1.28
Add for sandwich panel	4.62	6.75	10.15	15.10
Add for insulation	1.28	1.49	1.85	2.13
Add for interior metal liner	2.59	2.86	3.32	3.80
Add for overhang soffit, per sq. ft. of soffit	6.08	7.30	8.45	10.15

Add 6% for high profile (4:12 roof slope) buildings. Add for ventilators, fiberglass light panels or skylights from Section UIP 4.

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GREENHOUSES
SECTION UIP 11

RESIDENTIAL GREENHOUSES

The following are average costs per square foot for stock residential greenhouses with standard glazing of double strength glass with one end wall door. Foundations and vents are included but no floor, heat, electrical, plumbing or watering devices. Costs are based on professional labor. For amateur workmanship, decrease costs by 15% to 25%. The low end of the cost range represents wood or cheap aluminum greenhouses with plain stem walls while the high end is a weatherproofed, concealed connection, tubular framed structure. The high end good colored frame may be full length or set on a high cost masonry stem wall. Custom designed installations can run 25% higher. Cheap pipe frame structures can run 25% lower. For polyethylene covers, deduct 15% to 25%.

AREA	EVEN SPAN COST RANGE	PLAIN GABLE END WALL DEDUCTION	LEAN-TO COST RANGE
50	91.00 – 140.00	377.00 – 510.00	88.00 – 149.00
100	77.00 – 121.00	800.00 – 1060.00	73.50 – 122.00
150	70.50 – 110.00	800.00 – 1060.00	66.00 – 110.00
200	66.00 – 104.00	1030.00 – 1320.00	61.50 – 103.00
250	62.50 – 99.00	1030.00 – 1320.00	57.50 – 96.00
300	60.50 – 94.50	1030.00 – 1320.00	55.50 – 91.00
400	57.00 – 88.00	1290.00 – 1790.00	51.50 – 83.00
600	52.00 – 81.50	1290.00 – 1790.00	45.75 – 75.50
800	49.50 – 75.50	1290.00 – 1790.00	42.25 – 70.00
1,000	46.25 – 72.00	1610.00 – 2290.00	----- – -----

For gable end doors, add $925 to $2,120 each. For commercial doors, add 25%.

For tempered or laminated safety glass or structural polycarbonate, add 25%.

For tinted or heat reflective glass, add 15%. For insulated glass, add 60%.

Heaters – $1,030 to $1,670; Humidifiers – $805 to $2,330;
Coolers – $1,630 to $2,575; Ventilating fans – $510 to $1160;
Partitions, glazed, per square foot of partition – $17.15 to $23.35.
Planting benches, per square foot of bench:
Plastic grid, $5.11 to $8.15; wood slat, $8.08 to $9.21
Steel mesh, $8.15 to $13.85

SOLAR ROOMS

The following are average costs per square foot for three sided lean to glass solar rooms with curved eaves attached to a permanent structure used for living space or commercial applications. Costs include one end wall door, foundations and vents or windows. Floor, heat, electrical and plumbing are not included. The low end of the cost range represents tempered glazing in a good metal tubular frame while the high end has insulated, coated and tinted safety glass. Custom designed installations can run up to 50% higher depending on the quality of finish work.



AREA SQ. FT.	COST RANGE	AREA SQ. FT.	COST RANGE
50	164.00 – 440.00	**300**	104.00 – 283.00
100	137.00 – 369.00	**400**	97.50 – 258.00
150	124.00 – 330.00	**600**	88.00 – 236.00
200	114.00 – 314.00	**800**	81.50 – 220.00
250	110.00 – 290.00	**1,000**	77.00 – 212.00

For gable end and door adjustments, see Residential Greenhouses.

Extra tall bays, add 15%. For laminated wood framing, add 10%.

Straight eaves, deduct 7%.

For corner hips and valleys, add $44.00 to $75.50 per square foot to corner area.

For decorative lights incorporated into frame members, add $28.75 to $42.50 per linear foot.

For built in shades, add $19.65 to $41.00 per square foot of covered area. For motorized operation, add $1,360 to $3,100 per operator.

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PREFABRICATED BUILDINGS

SECTION UIP 11

OFFICE STRUCTURES

The following are average costs per square foot for prefabricated office structures. The low quality prices represent “mobile” type offices like those found at construction sites. Minimum fenestration electrical and very basic finishes (paneling) are included, but plumbing is not. The average costs represent “mobile” type offices also, but of a more permanent nature, like temporary office headquarters or remote branch offices. Average exterior and interior finishes, average fenestration, suspended ceilings and one bathroom (two fixtures) are included. The high quality costs represent good modular offices or commercial structures. Good exterior and interior finishes, better than average fenestration and average restrooms (five fixtures) are included. Foundation costs, heating, ventilating or air conditioning and utility hookups are not included in any of the costs.

ADJUSTMENTS



Foundation, piers,
$17.80 to $68.00 each.
Perimeter wall,
$23.90 to $75.00 lin. ft.
Concrete slab,
$7.48 per sq. ft.
Plumbing,
$820.00 per fixture.
Steps, $164.00 per step.
Landing, $43.25 per sq. ft.

AREA	LOW COST	AVERAGE	HIGH COST
100	72.00	-----	-----
200	59.00	81.00	106.00
300	52.50	73.00	94.00
400	48.50	66.50	86.50
500	45.00	63.00	81.00
600	42.75	60.00	76.50
800	39.00	55.00	71.50
1,000	-----	52.00	66.50
1,500	-----	45.75	59.00
2,000	-----	42.00	54.50

For H.V.A.C., see Section UIP 3.

INTERIOR MODULAR OFFICES

The following are average installed costs per square foot for four walled prefabricated modular in plant offices. Included in costs are vinyl wall covering, suspended acoustical ceiling, fluorescent lighting, required doors, windows and electrical switches and outlets. Floor structure, covering, plumbing and heating, ventilating or air conditioning are not included. Add 5% for 50 lbs. per sq. ft. storage capacity on top of offices. For two story units with structural floor, add 15%. For three walled units, deduct 17%. For two walled units, deduct 30%. For fire sound rated panels, add 5% to 10%. For each foot of height over an 8’ base, add 8%.

FLOOR AREA (SQ. FT.)	COST	FLOOR AREA (SQ. FT.)	COST
50	119.00	**300**	59.00
100	91.00	**400**	53.00
150	77.50	**600**	45.00
200	69.50	**800**	40.25
250	63.50	**1,000**	36.75

For environmental machine enclosures, add 125% to 150% to all costs above.

AIR-SUPPORTED STRUCTURES

The following are average installed costs per square foot for standard air supported storage structures with minimum doors, including anchoring supports, primary fans and back up inflation units. Add for perimeter grade beam, flooring, lighting, heat and other interior features from Section SEG 4 or Unit-in-Place costs. For interior thermal liners, add 10% to 20%.

Fabric shell lives can range widely depending on the membrane material, the method of installation and appropriate maintenance.

The lives listed represent averages under standard applications. A replacement membrane generally may cost 35% to 50% of the initial cost of the structure.

TYPICAL STRUCTURE LIVES

TYPE	YEARS
Good (welded seams)	10 – 14
Average	5 – 8
Low cost (sewn seams)	3 – 5

QUALITY AND TYPE	AREA IN SQUARE FEET 3,000	5,000	10,000	15,000
Good vinyl polyester or Tedlar fabric	31.00	27.25	22.50	20.40
Average reinforced vinyl fabric	21.05	19.15	16.60	15.40
Low cost, woven polyethylene	17.45	15.35	12.85	11.60
TYPE	**20,000**	**30,000**	**50,000**	**70,000**
Good vinyl polyester or Tedlar fabric	18.85	17.05	14.85	13.55
Average reinforced vinyl fabric	14.45	13.45	12.30	11.50
Low cost, woven polyethylene	10.90	9.72	8.57	7.95

For recreational (tennis/golf) structures, see Section UIP 17. Roof structures, see Section UIP 4.

PREFABRICATED MEZZANINES

The following costs are average per square foot costs for prefabricated steel mezzanines for inside use, not to exceed a design load of 150 pounds per square foot. The costs include all structural supports, stairs and flooring for the mezzanine level. The costs do not include any footing or foundations. For mezzanines constructed of wood deduct 7%.

AREA (SQ. FT.)	COST	AREA (SQ. FT.)	COST
100	91.00	**500**	54.50
200	73.00	**600**	51.50
300	64.00	**800**	47.25
400	59.00	**1,000**	44.75

Stair landings cost $125.00 to $156.00 per square foot of platform area.

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PREFABRICATED BUILDINGS AND BULK OIL FACILITIES

SECTION UIP 11

ATM STRUCTURES

The following are average costs for free standing Automatic Teller Machine (ATM) structures. Costs include steel frame construction, membrane roof, welded glass and appropriate floor and ceiling finishes. Foundations, air conditioning, signs and ATM machines are not included.





Enclosed ATM Structures with lobby:	COST RANGE
1 ATM (approximately 140 sq. ft.)	69000.00 – 85750.00
2 ATM's (approximately 240 sq. ft.)	85750.00 – 96500.00
Drive–up/walk–up ATM structures:	
1 ATM (approximately 50 sq. ft.)	51500.00 – 57750.00
2 ATM's (approximately 100 sq. ft.)	57750.00 – 65500.00
Air conditioning/heating:	
Enclosed equipment areas only (all structures):	
1 ATM	4400.00 – 4875.00
2 ATM's	6350.00 – 7150.00
add for lobby area	2675.00 – 3175.00

For automated teller machines, see Section UIP 2.

GUARD HOUSES AND TOLL BOOTHS

The following are square foot costs for small, free standing, portable exterior buildings that are normally used for guard, gate and pump houses, toll booths, etc. Costs include standard aluminum frame, insulated walls and ceiling, flooring, windows and door with hardware. Foundations, heating and air conditioning are not included. Add 10% to 15% for a 24" overhang. See Unit-in-Place Section UIP 3 for air conditioning costs. For all steel construction, add 25% to 50%. For interior non weatherized use, deduct 20% to 40%.

AREA	COST RANGE	AREA	COST RANGE
15	374.00 – 453.00	**80**	211.00 – 258.00
30	296.00 – 366.00	**90**	204.00 – 243.00
40	273.00 – 335.00	**100**	188.00 – 226.00
50	258.00 – 311.00	**120**	173.00 – 211.00
60	243.00 – 290.00	**200**	131.00 – 156.00
70	226.00 – 281.00	**400**	91.00 – 111.00

For bullet resistant models, add per square foot:

Level 1 (small-caliber pistol rated)	258.00 – 499.00
Level 4 (high-power rifle rated)	545.00 – 880.00

BULK OIL PLANTS



Small bulk oil plants for truck or tank car operation cost $99,500 to $161,000 for a two rack arm assembly and $158,000 to $262,000 for a four arm assembly including rack, all equipment and an average amount of piping, but excluding tanks and spill containment.

UNIT COSTS	COST RANGE
Loading rack platform, per square foot installed.	
Single sided	273.00 – 350.00
Double sided	328.00 – 475.00
Add for roof	11.05 – 31.25
Add for stairway, each	1340.00 – 3175.00
For explosion resistive const., add	7000.00 – 8200.00

For each foot of height above 10 feet, increase costs by 3%.
For platforms with adjustable height operation, add 125%, adjustable gangways only, add 75%.

Loading rack equipment, per arm assembly (includes rack arm, nozzle and hose, closing throttle, gate valves, direct read meters and pumps)	30400.00 – 50750.00
Equipment at tanks, per tank assembly (includes miscellaneous valves and line strainers)	1060.00 – 2550.00

See Section UIP 11 for Tanks – Section UIP 12 for Pipe.

SPILL CONTAINMENT SYSTEMS

The following are average costs per square foot for installed above ground spill containment systems and include platform ramps, top grating, collector pans, support pans and leakproof drain connectors to a central collection system. Cost of piping to the central collection point or tanks is not included.

GRATING TYPE	TRUCK	RAIL
Fiberglass	219.00 – 422.00	265.00 – 453.00
Galvanized steel	281.00 – 483.00	343.00 – 585.00

PRECAST CONCRETE BUILDINGS

The following are average square foot costs for small prefabricated concrete modular buildings for weather and vandal resistant equipment storage. Costs include all concrete wall, roof and floor panels, steel wall vents and entry door, and minimum electrical. Foundations, air conditioning and equipment power panel and wiring are not included.

AREA	COST RANGE	AREA	COST RANGE
100	152.00 – 219.00	**500**	82.50 – 111.00
150	130.00 – 180.00	**750**	72.00 – 92.50
200	118.00 – 164.00	**1,000**	64.00 – 82.50
300	101.00 – 138.00	**1,500**	54.50 – 69.50

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EQUIPMENT COSTS
SECTION UIP 12

EXPLANATION

Providing extensive equipment costs is not within the scope of this manual, but the following tables list some common pieces of equipment with replacement cost ranges. Prices are on a cost each basis unless otherwise noted. The Low Cost and Good prices listed are not the lowest or highest costs to be found but are indicative of typical cost ranges. For built-in cabinets and fixtures, see Section UIP 2.

Janitorial items and cash registers, which are found in many business establishments, are listed separately, followed by major groupings of equipment. If certain pieces of equipment are not found under the type of business being appraised, they may be found under another type using similar equipment.

The quoted price ranges are, of necessity, very rough. For more accuracy, purchase prices, subcontractors' costs, installers' costs or manufacturers' prices should be obtained, where possible, by make and model.

SECURITY EQUIPMENT

	COST RANGE		
Walk through metal detector	4150.00	–	14000.00
handheld wand	306.00	–	620.00
Conveyorized x–ray scanners	57250.00	–	84750.00
tabletop	37600.00	–	62500.00
fluoroscopic letter/parcel scanners	7050.00	–	17400.00

JANITORIAL

Buffing machines	1840.00	–	3050.00
Cleaning carts	217.00	–	585.00
Floor scrubbers, electric	1530.00	–	9200.00
High pressure washers, electric	805.00	–	2300.00
Ladders, extension, 16' to 28', aluminum	330.00	–	605.00
fiberglass	495.00	–	915.00
step, 4' to 12', aluminum	165.00	–	209.00
fiberglass	172.00	–	240.00
step, 8' to 12', aluminum	284.00	–	479.00
fiberglass	299.00	–	660.00
dual access, 4' to 12', aluminum	202.00	–	374.00
fiberglass	306.00	–	389.00
dual access, 8' to 12', aluminum	510.00	–	665.00
fiberglass	525.00	–	805.00
Lobby pans	16.50	–	76.50
Misc., mops, push brooms, squeegees	21.45	–	54.00
Mop buckets	59.50	–	180.00
Mop bucket/wringer combo	98.50	–	306.00
Traffic control posts	233.00	–	495.00
chain or rope, per linear foot	13.15	–	21.45
signs, post mounted	270.00	–	427.00
Vacuum cleaners, upright, 12" wide	464.00	–	810.00
16" wide	605.00	–	1150.00
wet/dry	345.00	–	1640.00
heavy duty, indoor/outdoor	435.00	–	1030.00
outdoor, gas powered, manually propelled	1290.00	–	4600.00
self propelled	1860.00	–	5700.00
Water brooms	44.50	–	110.00
"Wet Floor" signs	28.00	–	64.50
Window washers, telescopic	396.00	–	660.00
Wringers	85.50	–	145.00

NOTE: For window washing platform lifts, see Section UIP 7.

CASH REGISTERS

	LOW COST	AVERAGE	GOOD
Electronic, small	420.00	1300.00	3175.00
medium	915.00	1530.00	1900.00
large	1900.00	2330.00	2600.00
Terminals, programmable,			
small	7200.00	8600.00	10400.00
large	9600.00	12500.00	15900.00
add for data collection			
(cassette, etc.)	1650.00	2800.00	4675.00
add for stand alone LCD screen	3225.00	5800.00	9950.00
add for scanner	1860.00	2675.00	3950.00
Terminal processor system	23400.00	111000.00	268000.00
Check reader	525.00	890.00	1420.00
Hand scanner	194.00	695.00	2500.00
Magnetic card reader	233.00	270.00	337.00

CHECKROOM EQUIPMENT

	COST RANGE		
Coat and hat racks, stationary,			
wall mounted, 3' to 5' long	225.00	–	427.00
portable, single face	345.00	–	665.00
double face	590.00	–	1160.00
electric, no hat shelf, 144 coat capacity	4375.00	–	7800.00
180	4975.00	–	8750.00
240	5200.00	–	9050.00
300	5300.00	–	9500.00
360	5650.00	–	9700.00
420	6100.00	–	10500.00
480	6550.00	–	11600.00
540	7150.00	–	12500.00
600	7650.00	–	13300.00
electric, one hat shelf, 144 coat capacity	5950.00	–	10300.00
180	6200.00	–	10800.00
240	6650.00	–	11700.00
300	7150.00	–	12500.00
360	7650.00	–	13300.00
420	8450.00	–	14900.00
480	9350.00	–	16400.00
540	10000.00	–	17500.00
600	10800.00	–	19000.00
electric, two hat shelves, 144 coat capacity	6350.00	–	11000.00
180	6850.00	–	12200.00
240	7650.00	–	13200.00
300	8450.00	–	14900.00
360	9200.00	–	16100.00
420	10400.00	–	18300.00
480	11300.00	–	19900.00
540	12300.00	–	21400.00
600	13300.00	–	23700.00
add for additional hand control	194.00	–	615.00
Coat and hat reels, 120 positions, bolt down	935.00	–	1390.00
self supporting	1090.00	–	1610.00
Umbrella racks, 24 umbrella capacity	306.00	–	486.00

NOTE: For surveillance equipment, see Section UIP 3.

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EQUIPMENT COSTS
SECTION UIP 12

BUSINESS OFFICES

Total furniture cost per office or workstation, excluding desk top equipment.

	STANDARD OFFICE COST RANGE	OPEN MODULAR COST RANGE
Clerical	1580.00 – 3000.00	1940.00 – 4050.00
Technical	3600.00 – 7200.00	5100.00 – 9950.00
Supervisor	8250.00 – 16900.00	13300.00 – 24600.00
Management	19000.00 – 40200.00	34600.00 – 60500.00

	LOW COST	AVERAGE	GOOD
Adding machines, electric	233.00	500.00	1090.00
Bookcases, steel,			
per square foot of face	19.75	34.75	52.50
simulated wood	13.15	21.45	42.75
wood	29.75	51.00	93.00
Calculators, desktop, programmable	187.00	337.00	630.00
non programmable	26.25	91.50	299.00
Chairs, arm	233.00	464.00	795.00
arm, swivel	172.00	560.00	1040.00
arm, executive swivel	389.00	995.00	2500.00
secretarial	217.00	405.00	805.00
settee	600.00	1080.00	1960.00
side	172.00	359.00	775.00
add for floor mats	123.00	225.00	337.00
Check writer, manual	315.00	479.00	695.00
electric	695.00	1170.00	1960.00
Computer workstations, steel	299.00	620.00	1140.00
oak or walnut	525.00	980.00	1840.00
roll top	690.00	1240.00	2180.00
laminated particle board	374.00	720.00	1310.00
standing height	337.00	760.00	1060.00
Computer security enclosures, desktop	345.00	555.00	875.00
Copiers	1060.00	3725.00	11300.00
Copyboards, built in			
electronic duplicator	4025.00	6750.00	11300.00
Credenza	540.00	1130.00	2400.00
Desk, double pedestal	570.00	1310.00	3000.00
double pedestal, executive	870.00	2460.00	5300.00
single pedestal	457.00	760.00	1270.00
secretarial/open space work surface	660.00	1130.00	1840.00
returns	486.00	890.00	1660.00
hutch/ bookcase over desk	158.00	194.00	255.00
lamp	54.00	145.00	345.00
add for miscellaneous accessories	240.00	420.00	720.00
Dictating machine	630.00	1280.00	2360.00
Facsimile machine	1270.00	2600.00	5300.00
Filing cabinets, base cost, 28” deep	217.00	330.00	540.00
add per drawer, letter size	48.00	84.00	144.00
add per drawer, legal size	84.00	122.00	180.00
add per lateral drawer	255.00	359.00	500.00
add for insulated construction,			
per drawer	262.00	359.00	500.00
Letter folding machine, electric	217.00	405.00	660.00
Letter opener, electric	91.50	187.00	359.00

BUSINESS OFFICES (CONTINUED)

	LOW COST	AVERAGE	GOOD
Machine stand,			
general purpose	233.00	420.00	790.00
Markerboard, melamine surface,			
hardwood framed	63.00	187.00	427.00
Modular office panels, steel frame, per square foot,			
fabric covered, 1¾”			
thick, straight	8.09	11.25	13.40
curved	13.85	18.25	20.70
2” thick, straight	12.80	17.55	21.45
full plexiglass, straight	23.95	33.50	46.50
curved	25.75	36.25	43.50
half plexiglass, straight	16.80	24.25	28.50
vinyl, straight	12.65	17.25	21.00
add for wood frame panels	2.98	4.21	6.11
Panel hung cabinets	217.00	405.00	660.00
bookshelves	145.00	255.00	471.00
worksurface	165.00	330.00	500.00
corner	262.00	525.00	775.00
peninsula	225.00	464.00	735.00
lateral file storage	233.00	367.00	630.00
Panel top counters, per linear foot	18.25	52.50	158.00
Paper trimmers	61.00	150.00	262.00
Paper shredders	337.00	605.00	1060.00
cross cut	680.00	1210.00	2140.00
Personal computer systems,			
including keyboard	1100.00	3475.00	8900.00
laptops	825.00	2140.00	5100.00
monitors	345.00	1210.00	3275.00
glare screens	42.75	81.00	165.00
extension arms	93.00	165.00	255.00
modems	58.00	217.00	675.00
keyboard drawers	51.00	113.00	158.00
wrist rests	26.25	48.00	84.00
Printers, inkjet	172.00	790.00	1440.00
laser	545.00	2575.00	9950.00
Rolodex, electronic	194.00	330.00	510.00
Safe, record storage	1610.00	3875.00	9200.00
money, 1,000			
to 1,800 pounds	2370.00	5200.00	11000.00
built in, wall or floor, small	635.00	1190.00	2140.00
Scales, postal	28.00	52.50	88.50
electronic, rate calculator	284.00	500.00	860.00
Table, conference	830.00	2460.00	6550.00
office	337.00	680.00	1310.00
Time recorders	145.00	323.00	780.00
Timecard racks	81.00	194.00	420.00
Typewriter	150.00	374.00	900.00
Wall storage units	690.00	1830.00	4150.00

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EQUIPMENT COSTS
SECTION UIP 12

LABORATORY

	COST RANGE	
Balances, electronic	367.00 –	1840.00
precision	2330.00 –	4050.00
moisture determination	4225.00 –	7650.00
Carts, chemical	695.00 –	1020.00
glassware	545.00 –	1180.00
pan	795.00 –	1390.00
Centrifuge, benchtop	665.00 –	1460.00
Desiccator cabinets	299.00 –	1520.00
Freezers	2925.00 –	16400.00
Fume hood units	5400.00 –	12200.00
benchtop	479.00 –	1270.00
Furnaces, benchtop	1770.00 –	3900.00
Incubators	605.00 –	2280.00
hybridization, with programmable controller	5400.00 –	10900.00
Lightplates	427.00 –	1390.00
Ovens, convection	545.00 –	2080.00
forced air	935.00 –	3325.00
Racks, drying	202.00 –	545.00
tube	84.00 –	270.00
Refrigerators, enamel exterior	1840.00 –	13300.00
chromatography	10100.00 –	18300.00
add for stainless steel interior and exterior	427.00 –	4350.00
add for glass doors	605.00 –	5300.00
Scales, bench, digital	780.00 –	1940.00
Stirrers	187.00 –	545.00
multiposition	1770.00 –	2550.00
hot plate	2300.00 –	3025.00
Ultrasonic cleaners	435.00 –	1340.00
with heater	1020.00 –	2340.00
add for timer	64.50 –	172.00
Workbenches, per linear foot	240.00 –	905.00

MORTUARIES

Autopsy table, stainless steel	15200.00 –	22800.00
Body lifter	2400.00 –	3050.00
Embalming machine	1770.00 –	4150.00
gravity unit	905.00 –	1350.00
Embalming table, porcelain with hydraulic lift	4525.00 –	6650.00
stainless steel	2800.00 –	8900.00
Layout table	1340.00 –	1840.00
truck	545.00 –	805.00
Injector aspirator unit	1340.00 –	2625.00
Refrigerator, per body tray	5250.00 –	18900.00
Casket, burial stand	1390.00 –	1690.00
carriage	735.00 –	1220.00
lowering device	3100.00 –	6300.00
sales display stand (single)	306.00 –	486.00
sales display stand (double)	367.00 –	665.00
transfer cases	605.00 –	1340.00
Mausoleum lift	23600.00 –	28900.00
Instrument cabinets	1340.00 –	4225.00

MEDICAL OFFICES

	LOW COST	AVERAGE	GOOD
Basal metabolism apparatus with stand	3750.00	6200.00	10400.00
Beds, complete, nursing home	1770.00	2525.00	3525.00
hospital, manual	2360.00	2925.00	3725.00
electric	3250.00	5100.00	7650.00
overbed table	590.00	920.00	1420.00
privacy curtain, complete, per linear foot	26.25	46.50	64.50
Cabinets, instrument and supply, per linear foot	427.00	960.00	1450.00
Nurses' station, per linear foot	575.00	695.00	810.00
Centrifuge	605.00	1370.00	2360.00
Couch, recovery	665.00	1240.00	1940.00
Diathermy or ultrasonic machine	2600.00	6350.00	10500.00
Electro surgery unit	3250.00	5950.00	10400.00
Electro cardiograph	4375.00	7000.00	11000.00
Light, examination	202.00	920.00	2250.00
operating	1220.00	3600.00	7500.00
infrared	345.00	790.00	1640.00
ultraviolet	845.00	1720.00	2700.00
Scale, with measuring rod	525.00	825.00	1160.00
Sterilizer, instrument, steam	2450.00	4050.00	6750.00
Stools, treatment or diagnosis	262.00	720.00	1220.00
Tables, operating room	15900.00	29200.00	43900.00
Tables, treatment or examination, open frame	665.00	1200.00	2140.00
with base cabinet	1450.00	3075.00	4900.00
with hydraulic base	6200.00	12500.00	19300.00
Waste receptacle	172.00	292.00	464.00
X–ray and fluoroscope unit, per 100 ma	43500.00	57250.00	75500.00
X–ray viewer, wall type, per panel	337.00	540.00	745.00

DENTAL OFFICES

Bleaching instrument	805.00	1780.00	2700.00
Chair, electric or hydraulic	5900.00	8700.00	12800.00
Compressor	4125.00	7000.00	11900.00
Cuspidor	1990.00	2750.00	3800.00
Dental unit	4825.00	9300.00	17700.00
Instrument and supply cabinets, per linear foot	374.00	845.00	1450.00
add for sink	374.00	830.00	1520.00
Instrument set	3725.00	5650.00	8200.00
Light, overhead	1160.00	1560.00	2080.00
fully adjustable	2090.00	3750.00	6750.00
Sterilizer, autoclave	2900.00	4525.00	6950.00
Stool, dentist's	590.00	1090.00	1990.00
assistant's	427.00	790.00	1400.00
Vacuum system	4800.00	7200.00	10300.00
X–ray, machine	6300.00	10800.00	18600.00
developing tank	540.00	810.00	1220.00
film processor	510.00	790.00	1160.00
automatic	2060.00	5150.00	12800.00
Viewer, wall type, per panel	337.00	435.00	545.00

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EQUIPMENT COSTS
SECTION UIP 12

MOTELS

	LOW COST	AVERAGE	GOOD
Total furniture cost, per room, excluding linen	3475.00	6350.00	11500.00
Beds, standard double or twin, complete	457.00	710.00	1130.00
Beds, oversize, complete	560.00	920.00	1520.00
Chair, upholstered arm or lounge	299.00	500.00	855.00
Chair, desk	165.00	217.00	292.00
Chest, 4 drawer	345.00	575.00	935.00
Coffee maker, wall mounted	94.50	165.00	225.00
Double dresser	540.00	805.00	1200.00
Desk	367.00	545.00	805.00
Desk spanner	330.00	389.00	471.00
Drapes, per room	337.00	560.00	905.00
Hair dryer, wall mounted	284.00	374.00	525.00
Lamps	74.50	165.00	284.00
Luggage holder	78.00	122.00	187.00
Mirror, wall	103.00	187.00	330.00
Pictures	91.50	180.00	330.00
Tables, game	172.00	306.00	570.00
nightstand	108.00	233.00	471.00
TV sets	575.00	1020.00	1740.00
platforms, wall mounted	240.00	420.00	705.00
Wardrobe closet	585.00	805.00	1070.00
Housekeeping, linen, blankets, etc., per room	457.00	615.00	860.00
maid's cart and equipment, ea.	635.00	920.00	1320.00
vacuum cleaner	471.00	745.00	1200.00
baggage carts	233.00	620.00	1680.00
room service carts	457.00	585.00	760.00
roll away beds	284.00	345.00	471.00
portable cribs	172.00	194.00	217.00
portable bars	960.00	1490.00	2340.00

LOBBY AND RECEPTION AREA FURNITURE AND EQUIPMENT

Lobby counter, per linear foot	270.00	457.00	745.00
add for registration back counter, per linear foot	202.00	457.00	985.00
Lounge chairs, upholstered (wood, deduct 65%)	585.00	1300.00	2850.00
Sofa	905.00	2220.00	5500.00
Tables, cocktail or lamp	367.00	905.00	2210.00
check writing desk	1810.00	3325.00	6100.00
patio with umbrella	299.00	1360.00	3500.00
Chaise lounge	262.00	486.00	855.00
Coat rack	84.00	233.00	585.00
Ice cube machine	2925.00	4925.00	8400.00
Key cutting machine	1070.00	1240.00	1420.00
Magazine racks	64.50	127.00	262.00
Waste receptacles	74.50	187.00	457.00
Telephone systems:			
key system base cost (individual access, intercom)	4225.00	7250.00	12500.00
add per phone station	945.00	1400.00	2060.00
PBX system base cost (central operator console, code dialing)	58000.00	136000.00	326000.00
add per phone station	337.00	540.00	820.00

RESIDENTIAL

	COST RANGE
Beds, twin, including frame, box spring and mattress	805.00 – 1380.00
double, queen	1160.00 – 2450.00
king	1460.00 – 3525.00
bunk	1040.00 – 2750.00
wall mounted	805.00 – 1770.00
rollaway	306.00 – 890.00
add for headboard	367.00 – 1270.00
Crib	605.00 – 3200.00
Bookcase	1590.00 – 6300.00
Cabinet bar, portable	306.00 – 4450.00
Chair, dining, wood seat and back	217.00 – 620.00
upholstered seat and back	299.00 – 1130.00
kitchen	149.00 – 299.00
highback	427.00 – 1380.00
patio	85.50 – 225.00
Chest of drawers	1780.00 – 6300.00
Console	745.00 – 4600.00
Desk	427.00 – 1520.00
Desk chair	187.00 – 630.00
Dinette	367.00 – 1450.00
Display cabinet	1840.00 – 8750.00
Fan, ceiling	284.00 – 1130.00
floor	141.00 – 620.00
Floor lamp	585.00 – 4450.00
Home stereo system	1450.00 – 9500.00
extra speakers, per pair	345.00 – 995.00
minisystem	323.00 – 1160.00
DVD player (Basic to blu-ray or HD)	84.00 – 1350.00
Loveseat, upholstered, fabric	1560.00 – 5700.00
leather	2150.00 – 9600.00
sleeper	1930.00 – 6300.00
Nightstand	367.00 – 1380.00
Picture, framed	172.00 – 1640.00
Recliner	2030.00 – 6900.00
Sconce, single lamp	665.00 – 2925.00
three lamps	1100.00 – 4275.00
Sofa, upholstered, fabric	1990.00 – 9500.00
leather	3250.00 – 12200.00
queen sleeper	2675.00 – 8100.00
Table, coffee, wood	435.00 – 1380.00
bronze, glass top	555.00 – 1900.00
dining, wood	2975.00 – 16400.00
stainless steel, glass top	2490.00 – 11300.00
patio, aluminum	457.00 – 1380.00
with umbrella	660.00 – 2020.00
round	1840.00 – 5650.00
TV, big screen	2575.00 – 14500.00
TV stand	420.00 – 1640.00
DVD/ VCR combo player	149.00 – 510.00

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EQUIPMENT COSTS
SECTION UIP 12

AUTOMOTIVE REPAIR AND SERVICE

Costs are for passenger car equipment normally found in small and average size repair shops. For hoists, compressors, etc., see UIP 14, Service Station equipment costs.

	LOW COST	AVERAGE	GOOD
Air conditioning repair service package	1660.00	4375.00	11500.00
Aligner, frame, complete	24000.00	44300.00	57750.00
laser measuring	12700.00	14300.00	16100.00
Aligner, wheel, complete	12500.00	21200.00	36400.00
(add 100% for 4 wheel units)			
Battery charger, slow	284.00	540.00	995.00
fast	980.00	1440.00	2140.00
Brake bleeder	590.00	635.00	720.00
drum or disk lathe	8300.00	13800.00	23000.00
Diagnostic emissions analyzer	25000.00	35400.00	50000.00
Electrical test equipment, base cost	1440.00	3250.00	7250.00
add per test unit	374.00	735.00	1440.00
Fume exhausters	2050.00	3450.00	5900.00
Grinder	323.00	710.00	1530.00
Hand tool set	1410.00	4800.00	9050.00
Jack, hydraulic, auto	1020.00	1830.00	3300.00
Jack, heavy duty, truck	2210.00	3550.00	5500.00
Lube equipment:			
hi boy, oil container, per cabinet	585.00	1020.00	1740.00
overhead reel type, per reel	1270.00	1830.00	2600.00
pump with lube dispenser and valves	5200.00	6750.00	8750.00
(supply lines to overhead reel, $7.87 per linear foot)			
miscellaneous grease guns	123.00	248.00	471.00
bucket pumps	217.00	479.00	1030.00
Oil filter crusher	1870.00	3950.00	8100.00
Overhead air, per reel	665.00	905.00	1220.00
Scope analyzer	12200.00	22500.00	41500.00
Steam cleaner	2975.00	6650.00	10300.00
Tire changer, manual	660.00	1090.00	1830.00
electric	2360.00	5750.00	14400.00
truck	11700.00	20200.00	34900.00
Wheel balancer, bubble	590.00	935.00	1440.00
on vehicle	4575.00	6950.00	10800.00
electric	5850.00	9600.00	15600.00
truck	10800.00	13700.00	17400.00
Work benches, per linear foot	84.00	165.00	292.00

NOTE: For carwash equipment, see Section UIP 11.

AUTOMOTIVE TIRE RETREADING

(Heavy truck equipment is not included.)

	LOW COST	AVERAGE	GOOD
Aligner, tread	1150.00	1560.00	2020.00
Bagger	5850.00	9500.00	15200.00
Boiler, gas fired	6800.00	11000.00	18100.00
Buffer, pedestal type, 5 to 7 H.P	3475.00	4825.00	6800.00
20 to 30 H.P.	65500.00	74750.00	85500.00
Buffer semi automatic	43900.00	61250.00	85500.00
Curing rim	374.00	510.00	665.00
Hoist, one end (compressed airjack)	1620.00	2575.00	4025.00
chain, manual	374.00	630.00	1090.00
chain, electric	1740.00	2800.00	4450.00
Matrix, full circle, standard	8050.00	9150.00	10300.00
electric	8900.00	10100.00	11300.00
non-heated	7750.00	8400.00	9300.00
Mold, segmented, steam	15500.00	20000.00	26100.00
electric	15800.00	20500.00	26700.00
Pot boiler	1900.00	2800.00	4275.00
Tire spreaders, auto	1750.00	2800.00	4450.00
Tools, tire wrench set	825.00	1560.00	2925.00
Trimmer and spinner	1830.00	2800.00	4425.00
White sidewall table	374.00	605.00	980.00

AUTOMOTIVE SPRAY BOOTHS

(Costs include exhaust fans, warning lights and installation.)

Automobile, (26' L x 14' W x 9' H)	COST RANGE
Solid back baffled exhaust	20600.00 – 33000.00
low level exhaust	26100.00 – 35900.00
filtered exhaust	27600.00 – 37100.00
Drive through, low level exhaust	29800.00 – 41400.00
Bench model	3750.00 – 7950.00
Floor model	5700.00 – 10800.00
Automobile and Truck combination, (28’ L x 14’ W x 12’ H)	
Solid back baffled exhaust	33000.00 – 47700.00
low level exhaust	39900.00 – 56500.00
filtered exhaust	42600.00 – 60500.00
Drive through, low level exhaust	44200.00 – 63500.00
Drive through, filtered exhaust	47700.00 – 68000.00
Truck, (34’ L x 16’ W x 16’ H)	
Solid back baffled exhaust	42900.00 – 65500.00
low level exhaust	50000.00 – 77250.00
filtered exhaust	52750.00 – 82750.00
Drive through, low level exhaust	56500.00 – 88500.00
Drive through, filtered exhaust	60500.00 – 94250.00

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EQUIPMENT COSTS
SECTION UIP 12

BARBER AND BEAUTY SHOPS

	LOW COST	AVERAGE	GOOD
Total equipment cost, per chair	5750.00	8700.00	13200.00
Barber pole	585.00	810.00	1030.00
Midsize	1140.00	1310.00	1530.00
Two light	1560.00	1780.00	2020.00
Chair, barber	1260.00	2300.00	4225.00
Add for electric assist	915.00	1660.00	3025.00
Chair, beauty	995.00	1920.00	3725.00
With shampooing tray, adjustable	1150.00	2360.00	4225.00
Chair, children's	620.00	1190.00	2210.00
Clippers, electric	98.50	202.00	330.00
Display case, per linear foot	108.00	217.00	337.00
Dryer	420.00	635.00	1030.00
Dryer and chair combination	705.00	1130.00	1840.00
Facial chair	1050.00	3475.00	11000.00
Facial steamer	620.00	1390.00	3125.00
Hand tool set	240.00	427.00	615.00
Lather machine	284.00	389.00	540.00
Lavatory, shampoo	620.00	1040.00	1770.00
Mirror, per square foot	14.90	21.45	34.50
Pedicure bench	420.00	1040.00	2650.00
Stool, manicure and miscellaneous	270.00	389.00	575.00
Supply cabinets, per linear foot	202.00	248.00	323.00
Table, manicure	248.00	825.00	2700.00
Vibrator	187.00	240.00	337.00

PHOTO LABS

	COST RANGE	
Darkroom door, revolving	2925.00 –	9200.00
Digital printer	615.00 –	13300.00
Enlarger	3225.00 –	30300.00
Enlarging station	2140.00 –	5450.00
Add for viewer	605.00 –	1450.00
Film dryer	4225.00 –	13300.00
Film/print drying cabinet, single door	3225.00 –	7900.00
Double door	4725.00 –	11000.00
Add for pass through cabinets	1450.00 –	3550.00
Film processor	26100.00 –	109000.00
Film scanner	1090.00 –	13300.00
Print washer, stainless steel, countertop	1930.00 –	4275.00
Free standing	3300.00 –	7950.00
Safe lights	735.00 –	2060.00
Silver recovery units, manual	915.00 –	1590.00
Automatic	1770.00 –	6750.00
Sink, steel	4275.00 –	15900.00
Fiberglass	2925.00 –	8400.00
Water control system, including filter and thermostatic mixer	4375.00 –	11600.00
Add for chiller	3050.00 –	6100.00

Complete one-hour color photo labs cost $122,000 to $205,000.

LAUNDRY AND CLEANING

The following costs are typical ranges for small commercial outlets. For large institutional or commercial plants the costs may be several times as much.

	COST RANGE	
Boiler, 10 to 20 H.P.	14500.00 –	22800.00
Dry cleaning machine, petroleum	46400.00 –	99000.00
Extractor	8700.00 –	22800.00
Ironer, 48" to 72"	5900.00 –	16900.00
Laundry machine, 44" diameter	41100.00 –	71000.00
Marking machine	1040.00 –	2270.00
Package–tying machine	5250.00 –	9050.00
Presser, hand operated	7350.00 –	15200.00
Air operated	8700.00 –	18400.00
Puffer	1720.00 –	4200.00
Scale	905.00 –	1900.00
Sewing machines	1960.00 –	3375.00
Spotting board	1590.00 –	5950.00
Tumbler, dry cleaning, steam and solvent	10200.00 –	21400.00
Washers, laundry, manual	7350.00 –	18400.00
Automatic	14700.00 –	75250.00
Conveyor system, 200 to 1,000 garment capacity	5650.00 –	14500.00
Two level system, add	2360.00 –	2750.00
Dial control, add	2450.00 –	2850.00

LAUNDROMATS

Coin operated dry cleaner, 8# capacity	21100.00 –	27600.00
25# capacity	35100.00 –	57500.00
Coin operated washer, 7# capacity	1040.00 –	1840.00
20# capacity	3975.00 –	4925.00
Gas dryer	3475.00 –	7600.00
Cart	172.00 –	413.00
Change maker	1100.00 –	5400.00
Soap vending machine	1060.00 –	2430.00
Steamer	555.00 –	1940.00
Water heater, boiler, pump, storage	11100.00 –	21400.00
Water softener, 12 to 24 units	6950.00 –	12200.00
25 to 60 units	9800.00 –	24900.00

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EQUIPMENT COSTS
SECTION UIP 12

BILLIARD ROOMS

	COST RANGE	
Balls, billiard, 3 ball set, per set	91.50 –	225.00
Pocket billiard, 16 ball set	98.50 –	345.00
Snooker, 22 ball set	248.00 –	555.00
Billiard tables, pocket or carom	5100.00 –	30900.00
Chairs, 24” high	194.00 –	435.00
Cigarette urns	93.00 –	165.00
Cue rack	91.50 –	605.00
Cues, per dozen	396.00 –	935.00
Score beads, per set	38.00 –	74.50
Snack table	165.00 –	367.00
Time recorders	605.00 –	1750.00
Computer type	2600.00 –	6350.00
Vacuum cleaner, small, for tables	202.00 –	435.00

AMUSEMENT ARCADES/CASINOS

Upright video game	5900.00 –	13300.00
Pinball machine	4025.00 –	10900.00
Cockpit game	20500.00 –	42600.00
Basketball throw	3300.00 –	7900.00
Compact disc juke box	8400.00 –	17200.00
Dart machine	3050.00 –	8300.00
Air hockey	2450.00 –	7600.00
Stick hockey	1040.00 –	7200.00
Stick soccer	1180.00 –	1920.00
Foosball	945.00 –	3875.00
Cranes	4600.00 –	11600.00
Shuffleboard table	3250.00 –	13300.00
Skee ball	4800.00 –	13600.00
Vending machines, food or drink	3625.00 –	9650.00
Blackjack table	2925.00 –	5250.00
Crap table	10200.00 –	20500.00
Roulette table	7800.00 –	10500.00
Roulette wheel	14700.00 –	31500.00
Money wheel	5100.00 –	14700.00
Slot machine	1940.00 –	5900.00
Video poker	1660.00 –	5650.00
Coin counter	1260.00 –	6000.00

NOTE: For miniature golf, batting cages, etc., see Section UIP 14.

HEALTH CLUBS

Balance beams, fixed	545.00 –	1140.00
Adjustable	1340.00 –	2270.00
Ceiling rope	525.00 –	735.00
Chinning bars, wall mounted	367.00 –	660.00
Climbers	3550.00 –	9600.00
Dumbbell sets	306.00 –	1590.00
Ellipticals	6650.00 –	11000.00
Gym divider curtains, per square foot	11.60 –	21.45
Massage belt vibrators	605.00 –	1150.00
Massage tables	735.00 –	1080.00
Mats, per square foot	11.60 –	21.45
Multi–stationed gyms	6700.00 –	21800.00
Add for leg press	1220.00 –	2925.00
Add for chin up station	367.00 –	855.00
Add for weight stack enclosures	915.00 –	2600.00
Add for low row station	660.00 –	1080.00
Parallel bars	4125.00 –	7600.00
Pulley weights, wall mounted	980.00 –	1840.00

HEALTH CLUBS (CONTINUED)

	COST RANGE	
Punching bags, 70 lbs	98.50 –	217.00
Recumbent cycles	240.00 –	5950.00
With upper body option	1220.00 –	9600.00
Rowers	1560.00 –	8400.00
Sidehorses	3400.00 –	3725.00
Ski machines	555.00 –	6000.00
Stairclimbers	980.00 –	7600.00
Stationary bikes	367.00 –	4275.00
With incline	2060.00 –	7900.00
Steppers	172.00 –	1090.00
Treadmills, level, 6 mph	725.00 –	1840.00
8 mph	1340.00 –	3050.00
10 mph	1940.00 –	6100.00
With controlled incline and programmed workout	3550.00 –	13900.00
Weight benches	306.00 –	1590.00
Add for weight set	187.00 –	486.00

BOWLING CENTERS

Bowling equipment for a new alley is generally sold at a price per lane, including all equipment necessary for operation.

Costs for a base unit of 32 lanes with 130 sets of pins, 160 balls, 320 pairs of shoes, scoring tables and chairs, bowlers’ and spectator seats and all ancillary equipment necessary for operation, including subway-ball returns, excluding:

Automatic pinsetters, per lane	35500.00 –	48200.00
Deduct for aboveground ball return	1440.00 –	2750.00
Add for automatic pinsetters	25600.00 –	45800.00
Add for computerized scorekeeper	10500.00 –	24200.00
Add for telescores (projectors)	2060.00 –	4450.00
Lobby and miscellaneous furnishings such as carpets, drapes, restaurant and bar, per lane	13800.00 –	22200.00
Cost of individual items:		
Balls	75.00 –	119.00
Ball driller, complete	8400.00 –	18900.00
Ball cleaner and polisher	6900.00 –	14600.00
Ball racks, per ball	44.50 –	61.00
Cashier’s island	5500.00 –	14600.00
Chairs	147.00 –	479.00
Floor polisher, rotary	1660.00 –	3050.00
Foul detector, 2 lane unit	805.00 –	1290.00
Lane conditioner	10200.00 –	15600.00
Lockers, bowlers’, per compartment	108.00 –	306.00
Lunch and snack tables	217.00 –	555.00
Pins, plastic coated, per set of ten	202.00 –	306.00
Wood	165.00 –	240.00
Scoring tables	605.00 –	1150.00
Seats, spectator or bowler, per space	187.00 –	545.00
Shoes, rental, per pair	61.00 –	105.00
Shoe racks, for rentals, per 12 pair	48.00 –	103.00
Shoe sterilizer, ultraviolet	525.00 –	920.00
Vacuum cleaner, industrial types	1450.00 –	2150.00
Vacuum cleaner, standard types	427.00 –	1100.00

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EQUIPMENT COSTS
SECTION UIP 12

RESTAURANTS AND SODA FOUNTAINS

	LOW COST	AVERAGE	GOOD
Booth with table, upholstered, straight seats with table, each	1190.00	1740.00	2500.00
½ circle with table, each	1690.00	3075.00	5250.00
Bench seating with table, plastic, each	855.00	1190.00	1620.00
wood, each	760.00	1380.00	2450.00
Chairs	98.50	233.00	500.00
Dining tables	233.00	495.00	1070.00
Custom designed (logo inlays, graphics, etc.)	750.00	980.00	1330.00
Dish dollies	560.00	905.00	1440.00
Counter, formica top, per linear foot	123.00	233.00	435.00
bar counter, per linear foot	202.00	374.00	735.00
Bar sink and bottle setup, per linear foot	374.00	575.00	875.00
Back bar, per linear foot	165.00	240.00	420.00
Cafeteria counter, stainless steel, per linear foot	1520.00	2020.00	2600.00
Food preparation, per linear foot	367.00	545.00	810.00
Sink and drain, stainless steel, per linear foot	306.00	479.00	750.00
Soda fountain, complete unit, per linear foot	2270.00	2800.00	3225.00
Stool, counter	225.00	323.00	486.00
Back bar cooler	2140.00	3300.00	4900.00
Barbecue machine, self basting	2975.00	4600.00	8550.00
Beer dispenser	2450.00	4375.00	8100.00
Beverage cooler, hinged top, per linear foot	1190.00	1380.00	1740.00
Bottle cooler	2090.00	3400.00	5650.00
Brewing system	153000.00	414000.00	564000.00
Broiler	6300.00	13900.00	30200.00
Can opener, electric	194.00	367.00	820.00
Cocktail workstation	1130.00	1740.00	2700.00
Condiment stand	1740.00	2210.00	2900.00
Dishwasher, rack type	15000.00	26800.00	45400.00
Undercounter	4600.00	6300.00	9500.00
Dispenser, coffee	1560.00	3450.00	7250.00
Hot chocolate	905.00	1310.00	2020.00
Multiple, carbonated	4600.00	5750.00	7150.00
Disposal	1940.00	4575.00	10500.00
Food display cart	1070.00	1520.00	2180.00
Food warmer	1380.00	2800.00	5050.00
Fryer	1660.00	3600.00	7500.00
Fryer filter system	2725.00	8050.00	20500.00
Glass washer, scrub	457.00	560.00	935.00
Glass and plate chiller	1620.00	2180.00	3250.00
Griddle	2525.00	4225.00	7250.00
Heat lamp, each	590.00	760.00	935.00
Hot food wells	855.00	945.00	1070.00
Hot shelf, electric	194.00	306.00	620.00
Ice cream cabinet	2400.00	3500.00	5450.00
Ice machine, 110# to 450# per day	3800.00	7350.00	13200.00
Keg beer dispenser	1740.00	2900.00	4525.00
Mixer, food	2900.00	6650.00	15000.00
Bar	457.00	820.00	1380.00
Microwave oven	1900.00	3025.00	5000.00
Oven	2900.00	7600.00	19900.00
Conveyor	42600.00	46600.00	51250.00
Wood burning (for gas add 50%)	18400.00	33100.00	50250.00

RESTAURANTS AND SODA FOUNTAINS (Cont'd)

	LOW COST	AVERAGE	GOOD
Range	2925.00	4725.00	7650.00
Range hood, complete, per linear foot	695.00	1130.00	1900.00
Refrigerator/freezer, Residential type	1520.00	2600.00	4600.00
Commercial, two to three doors, 17 to 72 cubic feet	6300.00	10200.00	16500.00
Roll warmers, per drawer	1070.00	1260.00	1400.00
Salad bar	6050.00	9500.00	14900.00
Scales, platform	1580.00	2140.00	2900.00
Hanging, dial	165.00	240.00	471.00
Bar inventory control	306.00	457.00	630.00
Shake/soft ice cream machine	14500.00	16500.00	18900.00
Sandwich table, refrigerated	2900.00	4375.00	6300.00
Shelving, metal, per linear foot	64.50	94.50	165.00
Slicers	2360.00	4300.00	7600.00
Smoker, wood and gas	2270.00	3225.00	4825.00
Steam table, complete, per lin. Ft.	560.00	810.00	1220.00
Toaster, commercial, 4 to 8 slice	590.00	875.00	1190.00
conveyor	1380.00	2140.00	3175.00
Waffle baker, two unit	471.00	720.00	1100.00
Water station, complete	5500.00	10100.00	19300.00

SNACK BARS

Booth, laminated, two persons	367.00	620.00	820.00
four persons	695.00	760.00	1190.00
Beverage dispenser, non carbonated	172.00	471.00	820.00
twin	1260.00	2575.00	5250.00
Condiment stand	1130.00	1520.00	2140.00
Pump dispenser	172.00	435.00	1190.00
Cotton candy machine and merchandiser*	1160.00	1870.00	3025.00
Cup dispenser	147.00	367.00	875.00
Hot food merchandiser	1260.00	2420.00	4150.00
Hot plate	1130.00	2060.00	3825.00
Hot sauce server	315.00	457.00	695.00
Hot dog merchandiser	620.00	1230.00	2420.00
Ice cream dipping cabinet, complete, incl. dispensers	3675.00	4225.00	5900.00
Menu system, indoor, including boxes and panels	457.00	810.00	1810.00
Illuminated	820.00	1620.00	3200.00
Add for photo display	98.50	233.00	345.00
Outdoor, including menu and intercom, excluding signs	2090.00	3625.00	6900.00
Microwave oven	495.00	795.00	1550.00
Nacho cheese warmer	479.00	830.00	1490.00
Pastry donut display merchandiser	367.00	560.00	820.00
Pizza oven/merchandiser	1520.00	2140.00	3675.00
preparation table, refrigerated	7750.00	9650.00	11700.00
Popcorn popper and merchandiser, small, up to 18 oz.*	1150.00	2020.00	3500.00
Large, theater machine	6150.00	8900.00	12800.00
Flavoring dispenser	225.00	367.00	810.00
Pretzel display merchandiser	479.00	1100.00	2600.00
Snow cone machine*	1560.00	2300.00	3500.00
Soft serve machine, ice cream/ yogurt (Slush, deduct 20%)	6500.00	8050.00	11000.00
Two flavors twist	11300.00	15800.00	19400.00
Straw dispenser	74.50	127.00	240.00
Trash receptacle, laminated	367.00	695.00	1130.00
Waffle cone baker	620.00	935.00	1380.00
*Add for base, cart or wagon	1130.00	1450.00	1990.00

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EQUIPMENT COSTS
SECTION UIP 12

RETAIL STORES

(Costs per linear foot except as indicated.)

	COST RANGE
Cash register stands, each	449.00 – 1220.00
Checkout, cash and wrap counter	158.00 – 330.00
Cubicle display cases, per square foot of face	74.50 – 127.00
Gondola, double, island counter, 5' high	225.00 – 345.00
Wall unit, 5' high	172.00 – 248.00
Mannequins, each, full	420.00 – 2080.00
Platform	233.00 – 635.00
Mirror alcoves, freestanding, complete, each	1220.00 – 2180.00
Mirrors, framed, complete, per square foot	24.90 – 44.50
Rack, dress or suit, wall fixture	123.00 – 217.00
Single floor unit, each	330.00 – 555.00
Double floor unit, each	471.00 – 915.00
Round unit, each	323.00 – 630.00
Rotation display bins, each unit,	
17" diameter, 4 trays	233.00 – 345.00
6 trays	323.00 – 555.00
8 trays	420.00 – 620.00
28" diameter, 4 trays	750.00 – 1210.00
6 trays	1060.00 – 1680.00
8 trays	1390.00 – 2400.00
34" diameter, 4 trays	855.00 – 1330.00
6 trays	1210.00 – 1920.00
8 trays	1530.00 – 2600.00
Add for plastic inserts, each	9.87 – 21.45
Add 20% for heavy duty bins.	
Shelving, wall display, 6' to 7' high	
With back, no drawers	194.00 – 330.00
With back, some drawers	240.00 – 540.00
With back, some drawers, glass doors	345.00 – 635.00
Without back, storage, single shelf	18.25 – 46.50
Showcase, 20" to 24" deep, glass top or full glass	248.00 – 540.00
Table, modular display, sliding doors	158.00 – 337.00
Island display, storage base	270.00 – 575.00

BAKERIES

Baker's dough scale	585.00 – 1640.00
Baker's worktable, stainless steel top,	
per linear foot, 24" wide	85.50 – 187.00
36" wide	113.00 – 217.00
Add for backsplash, per linear foot	19.75 – 64.50
Countertop, cookie/roll display case	187.00 – 705.00
Display freezer	1900.00 – 5050.00
Display cases, glass, per linear foot	323.00 – 1170.00

BAKERIES (CONTINUED)

	COST RANGE
Dessert merchandiser, 34º temperature,	
upright, with glass doors	3750.00 – 5700.00
Wall mount, pie display, 40º to 50º	3000.00 – 3750.00
Dough mixer	4850.00 – 16400.00
Mixer stand	486.00 – 1900.00
Dough roller	2120.00 – 6950.00
Oven	5100.00 – 22000.00
Shelf carts	995.00 – 3125.00
Slicer	1900.00 – 7650.00

FOOD AND BEVERAGE MARKETS

(Costs per linear foot except as indicated.)

Checkout stands, each	5200.00 – 7600.00
Gondolas, single	116.00 – 270.00
Double	187.00 – 427.00
Dairy/deli cases, single deck	745.00 – 1020.00
Multideck (add 100% for rear door loading)	920.00 – 1610.00
Reach in, glass doors	1370.00 – 1900.00
Service case, enclosed	1560.00 – 1940.00
Self contained, chest cabinet	590.00 – 1030.00
Upright, glass doors	1020.00 – 1410.00
Service case, encl. (dry cases, deduct 25%)	1090.00 – 2600.00
Floral cases, upright, glass doors	855.00 – 1530.00
Frozen food/ice cream cases, single deck	980.00 – 1530.00
Multideck	2400.00 – 3200.00
Reach in, glass doors	2180.00 – 3275.00
Self contained, chest cabinet	745.00 – 1030.00
Upright, glass doors	1620.00 – 2270.00
Ice merchandiser case, each	1840.00 – 3750.00
Night lids for open freezer, each	315.00 – 427.00
Produce cases, non refrigerated, single deck	479.00 – 725.00
Multideck	635.00 – 890.00
Island (double sided)	735.00 – 1090.00
Refrigerated, single deck	
(self contained, add 20%)	735.00 – 1150.00
Multideck	920.00 – 1220.00
Island (double sided)	1220.00 – 1650.00
Scales, checkstand/counter, electronic, each	2140.00 – 3750.00
produce, mechanical	920.00 – 1030.00
Security cases, each	1750.00 – 3625.00
Shopping carts, each	141.00 – 337.00
Turnstiles, manual, each	1030.00 – 2270.00
Walk in box, cooler, per square foot	75.00 – 172.00
Freezer	91.50 – 225.00

NOTE: For detailed walk-in box and refrigeration costs, see Section UIP 7.

Complete equipment for a typical supermarket, ready to open for business, costs $43.75 to $73.00 per square foot of gross market area.

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EQUIPMENT COSTS
SECTION UIP 12

CEREMONIAL EQUIPMENT

The following costs are typical cost ranges for standard furnishings and for custom built, highly ornamented, individually designed items. All costs include installation in quantity.

	STANDARD COST RANGE	CUSTOM DESIGNED COST RANGE
Altar, wood	2675.00 – 5500.00	6200.00 – 38200.00
Marble or granite	4300.00 – 9350.00	10500.00 – 56250.00
Ark, with curtain	2060.00 – 5450.00	7650.00 – 45500.00
With door	3475.00 – 9850.00	13000.00 – 67750.00
Baptismal font	985.00 – 2975.00	3900.00 – 7600.00
Fiberglass baptistry,		
Walk in type	6150.00 – 8300.00	9700.00 – 13600.00
Clergy chairs	284.00 – 479.00	665.00 – 1860.00
Confessionals,		
With curtain, single	6350.00 – 9600.00	11400.00 – 22500.00
Double	9150.00 – 16200.00	19000.00 – 39300.00
With door, single	9050.00 – 13300.00	16100.00 – 31100.00
Double	11700.00 – 20900.00	20900.00 – 48900.00
Lecterns	1020.00 – 2220.00	2700.00 – 4975.00
Prayer desks	540.00 – 1580.00	1970.00 – 4800.00
Pews, per linear foot		
Hardwood	108.00 – 172.00	187.00 – 248.00
Upholstered	90.00 – 124.00	147.00 – 225.00
Seat type	123.00 – 172.00	194.00 – 284.00
Add for kneelers	26.25 – 36.25	56.50 – 74.50
Pulpits	1650.00 – 5050.00	5800.00 – 18600.00
Railings, per linear foot,		
Wood	63.00 – 117.00	147.00 – 277.00
Bronze	187.00 – 277.00	270.00 – 449.00
Wrought iron	63.00 – 109.00	158.00 – 233.00
Stands	116.00 – 396.00	449.00 – 1550.00
Tables	540.00 – 1070.00	1180.00 – 3200.00
Wall crosses	420.00 – 2180.00	2975.00 – 6750.00

STAGE/THEATER EQUIPMENT

Cinema projection system, per theater	47700.00 – 56750.00
Cinema sound system,	
Per theater (excludes licenses), analog	15600.00 – 22100.00
Digital	44500.00 – 57500.00
Retrofit	13300.00 – 26700.00
Curtains, per square foot	
Fireproof, silica based	82.50 – 93.00
Main, velour	4.21 – 6.19
Main, sheer	2.21 – 3.05
Drops, velour	10.35 – 11.95
Wings, velour	6.19 – 7.34
Curtain track, per linear foot	
Straight	41.50 – 61.00
Curved	69.00 – 145.00
Add for power control, each	3175.00 – 39600.00
Poster cases, each poster	1100.00 – 1990.00
Exterior type	2270.00 – 3125.00
Projection screen, per square foot	9.43 – 15.10
add 100% for power lift.	

CHIMES AND CARILLONS

Cast bell chimes do not swing as single bells. They usually come in sets of eight to fifteen, tuned to produce a diatonic scale, manually or electrically operated. A cast bell carillon normally consists of at least two octaves arranged in a chromatic scale. The bells are fixed and can be played from a keyboard equipped for either manual or electrical operation. One very large bell on the low end of the range may cost as much as several of the smaller bells, so the total number of bells is not the only cost consideration. Electronic carillons, where the tone is produced by striking rods of solid metal and amplifying it through loudspeakers, are much less expensive than the cast bell type for the same number of tones. Costs include normal installation.

	COST RANGE
Bronze bell, 9” to 18” in diameter	3250.00 – 13800.00
30"	16100.00 – 34900.00
48”	36700.00 – 105000.00
54”	117000.00 – 209000.00
Cast bell chimes, set of 12 to 15 bells	118000.00 – 295000.00
Carillon, cast bells, 2 to 4 octaves	221000.00 – 664000.00
Chimes, electronic	12200.00 – 35700.00
Carillon, electronic	15500.00 – 127000.00

ORGANS

There are many makes and designs of organs. Each pipe organ is individually designed, so wherever possible, it is best to obtain costs from the manufacturer. Small organs cost more per stop or voice than large ones.

Organs, pipe, complete with motors,	
blowers, etc., per stop	13900.00 – 23400.00
Organs, electric, each	24000.00 – 117000.00

THEATER SEATS

(Costs include installation, in full theater quantities.)

Auditorium type, plywood or fiberglass	142.00 – 225.00
Plywood back, padded seat	217.00 – 306.00
Neighborhood theater type, upholstered	262.00 – 323.00
Good quality, upholstered, thick padded	
seats and backs	284.00 – 367.00
Good quality, hi-back	330.00 – 367.00
Deluxe, major music hall or loge quality	374.00 – 575.00

NOTE: For bleachers, see Section UIP 17.

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EQUIPMENT COSTS
SECTION UIP 12

EDUCATIONAL EQUIPMENT

	COST RANGE	
Bookcases	225.00 –	605.00
Chalk/tack boards, portable	374.00 –	1640.00
Chairs	44.50 –	103.00
chair desk	88.50 –	225.00
folding	29.75 –	91.50
stacking, plain	44.50 –	109.00
upholstered	76.50 –	150.00
study chair with writing arm	88.50 –	202.00
Chair trucks	374.00 –	830.00
Carrels, individual	306.00 –	605.00
two person	457.00 –	985.00
tabletop	149.00 –	323.00
Computer projection panels	2360.00 –	4575.00
Computer work centers, single pupil	337.00 –	615.00
three pupil, half hexagon	620.00 –	1200.00
Corkboards, aluminum framed	58.00 –	396.00
oak framed	84.00 –	457.00
Desk, single pupil	74.50 –	202.00
two pupil	149.00 –	262.00
teacher's	575.00 –	985.00
Electric chalk board eraser/cleaner	471.00 –	905.00
Erasable marker boards, wall mounted	58.00 –	277.00
with stand	389.00 –	635.00
Globes	64.50 –	187.00
custom or illuminated	471.00 –	920.00
Lecterns, steel, floor model	172.00 –	323.00
with shelf	240.00 –	435.00
adjustable height, portable	109.00 –	194.00
tabletop	158.00 –	240.00
Lecture seating, pedestal	194.00 –	500.00
add for folding arm	59.50 –	84.00
Lockers, per opening, single tier	194.00 –	374.00
double	127.00 –	240.00
triple	91.50 –	202.00
box type	66.00 –	123.00
Locker benches, per linear foot	74.50 –	100.00
Overhead projectors	420.00 –	5400.00
Overhead projector tables	315.00 –	720.00
Projection screens, tabletop	133.00 –	248.00
wall mount	98.50 –	510.00
tripod	194.00 –	486.00
auditorium, manually operated	845.00 –	1550.00
electrically operated	2420.00 –	3250.00
Tables, activity	202.00 –	449.00
cut out, wheelchair accessible	240.00 –	457.00
drafting art or library display	389.00 –	1180.00
folding	194.00 –	555.00
lecture, pedestal, per linear foot	64.50 –	110.00
round	187.00 –	337.00
Table trucks	262.00 –	760.00

LIBRARIES

	COST RANGE	
Book stacks, wood, square foot of face	44.50 –	96.00
add for double face shelves	14.90 –	58.00
steel	19.75 –	39.75
add for double face shelves	9.87 –	18.25
add for ladder, per step	108.00 –	187.00
Book carts, steel	615.00 –	1100.00
polyethylene	479.00 –	830.00
Book displayers, tabletop, 15 to 18 books	133.00 –	240.00
double faced, 20 to 35 books	277.00 –	486.00
Book returns, free standing, with shelf	2060.00 –	5150.00
with cart	2675.00 –	6550.00
built in	750.00 –	1610.00
Book stand, tabletop	147.00 –	240.00
Book truck, steel or wood	315.00 –	1490.00
Card catalogs, wood, per tray	133.00 –	209.00
plastic drawer, wood front	85.50 –	180.00
steel	67.50 –	142.00
Charging counters, per linear foot	500.00 –	820.00
Compact disc displayer, wood, 400 discs	1770.00 –	3050.00
Deluxe carrels, individual	825.00 –	1660.00
two person	1350.00 –	2430.00
Dictionary stands	1260.00 –	2460.00
Display cases, aluminum frame, sq. ft. of face	74.50 –	194.00
wood frame	165.00 –	575.00
Laminator, pouch, desktop	374.00 –	1220.00
roll, heavy duty	2900.00 –	7150.00
Magazine shelving, wood, square foot of face	58.00 –	103.00
steel, square foot of face	24.90 –	46.50
Microfiche reader, single lens	525.00 –	1150.00
double lens	705.00 –	1740.00
Microfilm reader/printer	5150.00 –	8300.00
Newspaper racks	284.00 –	750.00
On line catalog center, one–sided	427.00 –	995.00
four sided	810.00 –	1430.00
Plastic binding system, manual	545.00 –	1150.00
electric	2900.00 –	4800.00
Photo ID system	1840.00 –	3725.00
Slide viewer, tabletop	147.00 –	615.00
Step stool, wood	91.50 –	233.00
plastic	56.50 –	165.00
Videotape or audiocassette display, four way	486.00 –	1230.00
multimedia rotor stands	795.00 –	2430.00

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EQUIPMENT COSTS
SECTION UIP 12

WAREHOUSING

	COST RANGE	
Bar stock storage racks, steel,		
120” deep, per sq. ft. of face	41.50 –	68.50
210” deep	63.00 –	102.00
Bin shelving, steel, 12” deep, per sq. ft. of		
face, bins not incl.	16.35 –	40.25
18” deep	20.70 –	49.00
24” deep	27.50 –	63.50
Bulk storage racks, steel, without decking, per sq. ft. of face		
24” deep	4.86 –	6.44
36” deep	5.52 –	8.15
48” deep	6.04 –	9.05
add for wire decking,		
per square foot of face	4.86 –	7.87
add for particleboard decking,		
per square foot of face	1.72 –	3.62
Cantilever racks, single sided, each upright,		
with four arms	725.00 –	2270.00
double sided, each upright, with eight arms	920.00 –	2875.00
add for each additional arm	42.25 –	117.00
Drum dollies	82.50 –	248.00
Drum grips, hydraulic	2490.00 –	8750.00
Drum lifts, manual	2900.00 –	4800.00
Drum pallets	108.00 –	225.00
Drum storage racks, per drum	63.00 –	126.00
Drum trucks	299.00 –	870.00
Pallets, presswood	9.50 –	20.25
high density polyethylene	63.50 –	113.00
plastic, double sided	130.00 –	277.00
Pallet racks, 14 gauge steel frame, 36” deep,		
per square foot of face	5.36 –	9.65
42” deep	6.35 –	10.55
48” deep	7.43 –	11.75
Pallet turners	780.00 –	1990.00
Plastic shelving, open, 18” deep,		
per square foot of face	8.44 –	12.80
24” deep	15.25 –	23.80
Scissor lifts, portable, foot pedal,		
hydraulic pump	1520.00 –	2430.00
battery operated	2210.00 –	3950.00
Steel shelving, open, 12” deep,		
per square foot of face	7.79 –	12.85
18” deep	10.50 –	16.50
24” deep	12.85 –	19.30
30” deep	14.50 –	21.15
closed, 12” deep, per square foot of face	10.25 –	17.55
18” deep	14.35 –	21.75
24” deep	17.85 –	25.00
30” deep	20.70 –	27.00
add for framed hinged door,		
per square foot of face	13.30 –	18.25
add for shelf drawers, plastic, each	4.46 –	15.10
Trucks, platform, deck	262.00 –	1840.00
hand	54.00 –	820.00
bar and tube	920.00 –	1590.00
fork lifts (2,000 lbs. to 8,000 lbs.)	24900.00 –	75750.00
shelf	262.00 –	1690.00
security, wire mesh	1290.00 –	2550.00
pallet, hydraulic	930.00 –	2430.00
tilt, manual	464.00 –	870.00

WAREHOUSING (CONTINUED)

	COST RANGE	
Wire shelving, chrome–plated, 18” deep,		
per square foot of face	14.90 –	22.15
24” deep	18.25 –	25.75
epoxy coated, 18” deep, per sq. ft. of face	20.70 –	29.00
24” deep	24.90 –	34.75
add for door, back and side, per sq. ft. of face	32.50 –	45.25

HAZARDOUS MATERIAL STORAGE

Drum storage system, including painted steel stacking shelves and pallets, 2,000 lbs. pneumatic hoist, and drum lift,		
24 to 72 drum capacity, cost per drum	1620.00 –	3450.00
78 to 156 drum capacity	1060.00 –	1560.00
Outdoor storage cabinets, incl. vent and		
sump, cost per drum	1550.00 –	2825.00
Overpack polyethylene drums, for storage or secondary		
containment, 85-gallon capacity	217.00 –	396.00
Spill control pallets, steel, 4 to 9 drums	1460.00 –	4925.00
polyethylene	635.00 –	2030.00
Stackable drum cabinets, with sump,		
open, cost per drum	479.00 –	1330.00
closed	1130.00 –	2390.00
Storage cabinets meeting OSHA 1910.106 and NFPA 30 codes		
18-gauge stl., double wall const.,		
manual close hinged doors	1590.00 –	2800.00
bi-fold self closing doors	1710.00 –	3200.00
Steel containment platform with drain valve,		
52” x 52”	960.00 –	1740.00
Waste collection system/secondary		
containment, single drum	660.00 –	1060.00
double drum	1270.00 –	1780.00

NOTE: For truck spill containment systems, see Section UIP 14.

SHIPPING DOCK

Dock, boards	795.00 –	3800.00
plates	367.00 –	1720.00
ramps, walk	735.00 –	1770.00
yard	16200.00 –	32300.00
Dock bumpers, horizontal, per linear foot	46.50 –	110.00
vertical	74.50 –	172.00
Dock levelers, mechanical	4925.00 –	12100.00
hydraulic	8700.00 –	16600.00
edge of dock	1150.00 –	3650.00
Dock lift, hydraulic, ram, single post	13200.00 –	27600.00
double post	30500.00 –	66000.00
scissors	12500.00 –	29800.00
Dock seals, 10” projection	890.00 –	1360.00
add for each 1” of additional depth	26.25 –	56.50
add for armour plates	262.00 –	635.00
Dock shelters	1560.00 –	4675.00
Truck levelers	28000.00 –	41700.00
Trailer stabilizer jack, mechanical	337.00 –	1020.00
hydraulic	1620.00 –	2090.00
Wheel chocks, per pair	74.50 –	725.00
Wheel risers, per pair	1620.00 –	5650.00

NOTE: For strip doors, see Section UIP 5; for air-curtain doors, see Section UIP 3.

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SUBDIVISION DEVELOPMENT COSTS

SECTION UIP 13

RESIDENTIAL STREET IMPROVEMENTS

Costs for residential street improvements vary greatly due to local code requirements for different materials, street types, layouts, and utilities. The following costs are averages including ordinary charges for engineering, plans and inspection. Costs include contractors' overhead and profit but not developers', which is realized when each lot or house is sold. They do not include extensive environmental impact reports, special charges (impact or entitlement) or assessments sometimes levied against the subdivider such as annexation charges, or costs of new or existing trunk sewers. They assume that the utilities are at or near the subdivision boundary with no special connection problems or costs. Single-lot unit costs, industrial access streets or individual cul-de-sac developments can run 30% to 70% higher. Do not apply these factors to large parking lot paving.

STREET IMPROVEMENTS	COST RANGE	AVG. UNIT COST	PER LIN. FT. TYPICAL 40' STREET
*Grading and surplus disposal, sq. ft.	0.32 – 0.84	0.44	26.50
*4" rock base, sq. ft.	0.97 – 1.22	1.05	42.00
add or deduct per inch of variation	0.16 – 0.27	0.19	7.46
6" cement treated base, sq. ft.	1.24 – 1.90	1.42	57.00
add or deduct per inch of variation	0.19 – 0.27	0.22	8.64
*Paving, 4" asphaltic concrete, sq. ft.	2.92 – 3.51	3.10	124.00
add or deduct per inch of variation	0.57 – 0.72	0.62	24.75
Paving, 6" concrete, sq. ft.	4.66 – 7.71	5.51	220.00
add or deduct per inch of variation	0.59 – 1.03	0.71	28.25
Concrete curb 6", no gutter, lin. ft.	12.25 – 22.40	15.00	30.00
*Concrete curb 6", 1' gutter	18.10 – 28.25	21.00	42.00
Concrete curb 6", 2' gutter	20.85 – 32.25	24.10	48.25
Concrete curb 8", add to 6" costs	1.99 – 2.21	2.06	4.12
Asphalt curb 6" no gutter, lin. Ft.	5.48 – 6.29	5.73	11.45
berm 4" (Speed bumps, add 100%)	5.39 – 6.17	5.64	11.25
Granite curb 5", lin. ft.	38.00 – 55.00	43.00	86.00
Concrete curb, rolled, lin. ft.	10.85 – 17.70	12.75	25.50
*Concrete cross gutter, at intersection, sq. ft.	9.22 – 12.60	10.25	5.12
*Concrete sidewalk, 4" thick, sq. ft.	5.47 – 8.59	6.35	51.00
add or deduct per inch of variation	0.59 – 0.93	0.69	5.50
*Concrete aprons, 6" thick, sq. ft.	6.23 – 9.78	7.25	16.65
add or deduct per inch of variation	0.62 – 0.94	0.72	1.65
Sewer main, 9' average depth, lin. ft.			
add or deduct per foot of depth	0.88 – 3.17	1.36	1.36
*8" vitrified clay	48.50 – 58.50	-----	51.50
8" asbestos cement	31.50 – 43.00	35.00	35.00
8" plastic	25.25 – 37.25	28.75	28.75
add or deduct per inch of diameter	3.99 – 10.70	5.54	5.54
Sewer laterals, 5' average depth, lin. ft.			
*4" vitrified clay	23.65 – 28.75	25.25	25.25
6" vitrified clay	33.75 – 41.25	36.00	36.00
*Sewer clean outs, 60' o.c., each	930.00 – 1590.00	1110.00	37.00
*Sewer manholes, 400' o.c., each	3500.00 – 5800.00	4125.00	10.30
Storm drainage, lin. ft.			
18" reinforced concrete	83.00 – 106.00	90.00	90.00
18" corrugated metal	62.00 – 87.50	69.50	69.50
add or deduct per inch of diameter	2.55 – 7.29	3.61	3.61
storm manholes, 400' o.c., each	3500.00 – 4850.00	3900.00	9.76

*The cost of complete double-loaded street improvements with the components indicated above totals $635.00 to $775.00 per linear foot of street in ordinary level subdivisions, or from $401.00 to $540.00 per front foot of lot, including side street allowance. Costs may run twice as much for extreme hillside conditions. A typical fully improved industrial access street will cost $790.00 to $1130.00 per linear foot of street.

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SUBDIVISION DEVELOPMENT COSTS

SECTION UIP 13

LIGHTING AND PUBLIC UTILITIES

Water, gas and electric costs vary considerably with local requirements and codes. Often all or a portion of the initial cost, maintenance and replacement is included in an assessment or increased utility rate charged to the consumer. Sometimes an additional charge is made in remote or hilly areas for special service or additional equipment. These requirements should be checked locally. The average costs below are approximate

maximum costs for a typical subdivision, all or part of which may be borne by the community or secured against the property. Some of these costs may be refundable to the developer. Specific pipe costs can be found in Section UIP 9. For general site clearing and grading, see Section UIP 1. For septic tanks, see Section UIP 3.

PUBLIC UTILITIES	COST RANGE	AVG. UNIT COST	PER LIN. FT. TYPICAL 40' STREET
*Water main, 6” ductile iron, lin. ft.	51.00 – 58.50	53.50	53.50
6” asbestos cement	55.50 – 67.00	59.00	59.00
6” steel	56.50 – 68.00	60.50	60.50
6” plastic	34.75 – 44.25	37.50	37.50
add or deduct per inch of diameter	3.64 – 11.15	5.28	5.28
*Water lateral, 1”, lin. ft.	20.70 – 26.00	22.35	22.35
*Water meters, 60’ o.c., each	333.00 – 760.00	438.00	14.60
*Fire hydrants, 300’ o.c., each	4025.00 – 6200.00	4650.00	15.45
*Gas main, 2” steel, lin. ft.	19.95 – 22.55	20.80	20.80
3” steel	25.50 – 29.00	26.75	26.75
4” plastic	25.25 – 30.75	27.00	27.00
*Gas lateral, 3/4”, lin. ft.	13.00 – 18.65	14.65	14.65
Electricity, overhead,			
on poles, lin. ft.	23.45 – 42.00	28.50	28.50
Electrical lateral, lin. ft.	6.79 – 9.31	7.54	7.54
*Electricity, underground,			
in conduit, lin. ft.	28.75 – 47.25	34.00	34.00
*Electrical lateral, lin. ft.	17.70 – 30.50	21.20	21.20
*Telephone lateral, underground	12.75 – 20.25	14.90	14.90
Trench only, lin. ft.	7.56 – 15.45	9.59	9.59
Conduit only, lin. ft.	6.79 – 13.55	8.55	8.55
*Street light, underground			
wiring, 200’ o.c., each	2775.00 – 9700.00	4250.00	21.25
Street lights, overhead			
wiring, 200’ o.c., each	1920.00 – 4850.00	2600.00	13.00

*The cost of complete double-loaded street improvements with the components indicated above totals $635.00 to $775.00 per linear foot of street in ordinary level subdivisions, or from $401.00 to $540.00 per front foot of lot, including side street allowance. Costs may run twice as much for extreme hillside conditions. A typical fully improved industrial access street will cost $790.00 to $1130.00 per linear foot of street.

Catch basins	$2,210 to $2,700 plus $333 to $650 per foot of depth.			
Concrete headwalls	$995 plus $110 to $183 per inch of pipe diameter.			
Gunite, 2” – 3”, surfacing for open drains	$7.13	to	$11.50	per sq. ft.
Soil cement, roads	$1.27	to	$1.61	per cu. ft.
embankments	$1.87	to	$3.23	per cu. ft.
slopes	$5.35	to	$9.64	per cu. ft.
Soil dikes	$0.66	to	$1.31	per sq. ft.
Slope protection, netting, mats or fabric	$0.13	to	$1.64	per sq. ft.
Street signs, with post	$183.00	to	$615.00	each
Survey monuments	$295.00	to	$775.00	each

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YARD IMPROVEMENTS
SECTION UIP 13

PAVING\ DECKING

Typical costs per square foot, except as otherwise specified. For paved areas between 500 and 1,000 square feet, deduct 10%; 1,000 to 3,000 square feet, deduct 20%. Over 3,000 square feet, use Subdivision costs. Small separate pours of 100 square feet or less may run 25% higher. Hand mixed and spread could cost 75% more.

	COST RANGE		
2” asphalt	2.87	-	4.23
add per additional inch	0.67	-	0.84
2” aggregate base	0.93	-	1.60
add per additional inch	0.22	-	0.32
4” concrete, unreinforced	6.40	-	9.56
add or deduct per inch of variation	0.58	-	0.84
add for mesh reinforcing	0.58	-	1.51
Bar reinforcing	0.67	-	3.38
Exposed aggregate	1.09	-	4.23
Brick ribbons	1.44	-	4.73
Detectable warning surface (ADA), Stamped	3.38	-	7.40
Decorative pattern finish, stamped	7.59	-	15.00
Surface formed	6.15	-	11.40
Thin set synthetic overlay	9.31	-	20.75
Color or grits	0.93	-	2.02
Epoxy with stone or shell	6.66	-	9.46
Salt finish (cool deck)	0.58	-	1.00
Deck channel drain and grate, per lin. ft.	20.75	-	80.50
Catch basins, small, up to 24”, each	375.00	-	750.00
4” sand base	1.44	-	2.02
4” gravel base	1.60	-	2.21
add or deduct per inch of variation	0.31	-	0.39
add for 1” stone dust base	0.32	-	0.52
Open grid blocks for grass on sand base	9.31	-	12.70
Asphalt block pavers on concrete base*	12.25	-	20.20
Brick on concrete base, grouted, flat*	16.15	-	26.00
on edge	22.85	-	39.00
Concrete pavers on concrete base*	13.85	-	22.85
Flagstone on concrete base, grouted*	18.10	-	32.50
Tile, quarry on concrete base*	16.15	-	23.40
*For sand bed in place of concrete, deduct	4.04	-	7.40
Snow melting, including controls, electric	17.70	-	21.30
hydronic, large areas (excluding heat source)	9.31	-	24.95
Wood, on grade (posts, beams and joists not included)			
2” x 4” flat	8.53	-	13.85
2” x 4” on edge	12.95	-	20.75
Steps on ground, per lin. ft. of tread,			
Brick on concrete	64.50	-	132.00
Concrete	49.00	-	82.00
Approach apron, concrete	7.40	-	12.25
Concrete curb, 4” 6”, per lin. foot	17.00	-	26.50
Concrete sidewalk	6.82	-	10.35
Handicap ramps, sidewalks (retrofit, add 400%)	9.94	-	15.55
Buildings, concrete (remodel, add 200%)	29.00	-	59.50
add for railing, per lin. ft.	55.50	-	78.00
Wood	38.25	-	55.50
add for railing, per lin. ft.	21.90	-	46.50

For portable ramps, see Section UIP 7.
For synthetic surfaces, pathways, see Section UIP 14. Special stone paving, see Section UIP 6.

PATIO ROOF

(Typical costs per square foot of covered area, including supports.)

TYPE	COST RANGE		
Awning, fabric	16.60	-	37.50
Aluminum or steel, baked enamel	11.20	-	20.75
Fiberglass or screen only	8.04	-	15.00
Wood, including built up, composition	13.30	-	32.00
Open lattice, metal, vinyl or wood	9.08	-	27.75
Architectural columns, open lattice or trellis	37.50	-	127.00
Picnic shelters	24.80	-	63.00
Add for insulated metal panels	5.38	-	7.40
Add for lighting fixtures, each	123.00	-	338.00

PATIO ENCLOSURES

Typical cost ranges per linear foot of wall, 84” high, including one exterior door. Use high end of range for insulated panels or knee walls. Add for roof above.

Glassine windows	116.00	-	202.00
Acrylic windows	195.00	-	240.00
Screened only, fiberglass	39.00	-	63.50
Steel or aluminum	63.00	-	82.00
Bronze	80.50	-	121.00
Add for extra door, each	150.00	-	300.00

GAZEBOS: Typical cost each for 8’ to 20’ wood units including minimal foundation but excluding floors.

Standard	4950.00	-	26000.00
Deluxe	23100.00	-	62750.00

For Solar rooms and greenhouses, see Section UIP 14. Pool enclosures, see Section UIP 14.

TEMPLES: Typical cost each, 8’ to 12’ high (to bottom of dome) cast stone units with top ring up to 12’ in diameter. Cost does not include floors or

other ornamentation	25600.00	-	69000.00
Add for fiberglass dome	7500.00	-	13300.00
Add for masonry paver floor with no steps	5200.00	-	10300.00
Add for floor with steps	5800.00	-	13300.00

PAVILIONS: Typical cost each for

cast stone units	38600.00	-	93000.00

For individual stone columns, see Section UIP 6. For finials, urns, statuary, see Page 9.

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YARD IMPROVEMENTS

SECTION UIP 13

RAISED PATIO DECKS

Typical cost ranges per square foot of deck area, including supports. Custom installations with complex shapes, built in planters and seats can run 50% to 100% more.

TYPE	25 Sq. Ft.	50 Sq. Ft.	100 Sq. Ft.	Over 300 Sq. Ft.
Decks:				
Softwood, fir, pine, etc.	39.25 - 47.25	28.75 - 34.75	21.00 - 26.00	12.80 - 16.35
Cedar, redwood or metal	52.50 - 63.50	40.25 - 49.50	30.75 - 38.25	20.20 - 25.25
Railings:				
Softwood, fir, pine, etc.	12.40 - 15.55	8.40 - 10.50	5.78 - 6.71	2.95 - 3.81
Cedar, redwood or metal	17.45 - 21.65	12.40 - 14.95	8.40 - 10.30	4.84 - 5.78
Steps:				
Softwood, fir, pine, etc.	11.60 - 14.20	5.83 - 7.33	3.14 - 3.68	0.96 - 1.31
Cedar, redwood or metal	17.10 - 21.25	8.61 - 10.65	4.34 - 5.31	1.50 - 1.80

For each foot of height above 3 feet, increase costs by 5%. For treated softwoods increase cost by 25%.

For wood polymer composite, add 30% to softwood costs. For vinyl and tropical hardwoods, add 5% to 15% to cedar/redwood costs.

BENCH AND PICNIC UNITS

	COST RANGE
Park bench, 6'	525.00 - 2360.00
Deluxe	2070.00 - 4900.00
precast stone	1220.00 - 2410.00
Ornate	7250.00 - 17200.00
Picnic table, 6' portable	412.00 - 960.00
Stationary	930.00 - 2575.00
Wheelchair accessible	1390.00 - 3075.00
Precast stone	2410.00 - 6100.00
For 8' units, add	119.00 - 292.00
Metal cooking grills	278.00 - 930.00
Campfire rings	71.50 - 368.00
Trash receptacle with cover	359.00 - 1270.00
Drinking fountains, outdoor, steel	2800.00 - 8600.00
Precast concrete	1270.00 - 2490.00
Add for freezeproof valve	990.00 - 1610.00
Foot and body shower tower	1560.00 - 7500.00
Pet fountain	1450.00 - 5400.00

For outdoor furniture, see Section UIP 12, Page 4.

PLAYGROUND EQUIPMENT

(Cost each except as indicated.)

Balance beam	240.00 - 800.00
Baseball backstop, with hood	4900.00 - 12700.00
Arch	7500.00 - 30000.00
Basketball backstop, with pole	1610.00 - 4800.00
Bicycle rack, per lin. ft.	68.00 - 157.00
Bucketball/funnelball	1180.00 - 1560.00
Carousel, merry go round(10 ft and up)	2800.00 - 9600.00
Chin up bar	420.00 - 815.00
Climber, arc	645.00 - 1050.00
Circle	1100.00 - 2270.00
Dome	2490.00 - 4800.00
Tree	740.00 - 1160.00
Climbing wall	1440.00 - 2875.00
E–Z digger	740.00 - 1290.00
Football goal posts, per pair	3825.00 - 9600.00
Combination football/soccer goal, per pair	4900.00 - 9250.00
Horizontal ladder, monkey bars	965.00 - 2340.00
Parallel bars	300.00 - 1270.00
Sand box	780.00 - 2020.00
Seesaw, per unit	960.00 - 1590.00
Platform	1920.00 - 3975.00
Sit–up station	990.00 - 1160.00
Slides, free standing, stainless steel, per unit	2240.00 - 4325.00
Plastic	3075.00 - 5600.00
Spiral	4650.00 - 9250.00
Oval tube	5750.00 - 9900.00
Soccer goals, per pair	2575.00 - 7150.00
Nets, per pair	278.00 - 645.00
Spring rider	740.00 - 1270.00
Spring platform	930.00 - 1750.00
Swing, tire	1850.00 - 5600.00
Swing, wheel chair platform	1590.00 - 4300.00
Swings, per unit	525.00 - 930.00
Swinging bench	990.00 - 2210.00
Tennis posts, per pair	635.00 - 1220.00
Nets	262.00 - 720.00
Tetherball post	300.00 - 705.00
Vault bar	455.00 - 645.00
Volleyball posts, per pair	645.00 - 1180.00
Post ground socket	292.00 - 368.00
Whirls, 6' diameter	1670.00 - 2800.00
8' diameter	2360.00 - 3825.00
Junior	875.00 - 1440.00

FITNESS PARKS

	COST RANGE
Fitness cluster, single unit	6650.00 - 23400.00
Fitness obstacle course, per station	740.00 - 1100.00

BARBECUES

(Typical costs of brick or block barbecues, complete.)

Simple, minimum steel work	740.00 - 1400.00
Average, electric spit, good grill	1330.00 - 2575.00
Good, heavy–duty spit, many extras, with sink	2280.00 - 7500.00
With stainless steel broiler	3325.00 - 8750.00
Add for stainless steel hood	368.00 - 1160.00
Custom designed	8750.00 - 53250.00
Fire pit/chiminea	1080.00 - 9650.00
Custom fireplace	11700.00 - 21200.00

MODULAR PLAY SYSTEMS

Complete system, per sq. ft. of playground	49.00 - 79.50
Play platforms,each	3225.00 - 8300.00
Add per bridge	2210.00 - 3900.00
Ladder	990.00 - 1320.00
Pole	645.00 - 930.00
Transfer station, ADA	740.00 - 1270.00
Scramble ramp, ADA	1440.00 - 2500.00
Spiral climber	705.00 - 1100.00
Activity panel	292.00 - 930.00

SKATE PARKS

Complete skate area per sq. ft., concrete	33.50 - 44.75
modular steel	17.55 - 23.85
Quarter pipes, single (4' width), each	4700.00 - 5800.00
Double	8300.00 - 11100.00
Triple	11600.00 - 15600.00
Half pipes, 4'-8' width	10800.00 - 23600.00
12'	24600.00 - 37500.00
16'	32300.00 - 54500.00
20'	39600.00 - 65000.00
Spines	4500.00 - 16100.00
Wedges/Jumps	1790.00 - 7650.00
Rails	426.00 - 1180.00
Pyramids	10400.00 - 23000.00
Pump hump, single	5100.00 - 10400.00
Double	10300.00 - 20100.00
Launch boxes	4500.00 - 16700.00

DOG PARKS

Climb ramps/Dog walk	1390.00 - 3975.00
A-frame	1280.00 - 3200.00
Tire jump	93.00 - 1080.00
Broad jump	144.00 - 620.00
See-saw	157.00 - 1500.00
Weave poles	93.00 - 1030.00
Pause table	338.00 - 960.00
Crawl tubes/tunnel	110.00 - 2410.00
Window and fence jumps	68.00 - 1660.00
Waste station	144.00 - 1230.00

SHOOTING RANGES

Complete outdoor pistol/rifle range (includes firing line, canopy, baffles, backstop, and side berms) or sporting clay course,

per shooting position	23600.00 - 27800.00
Bullet traps, pistol/rifle, wet system	16700.00 - 46400.00
Stand alone function tester	4950.00 - 10000.00
Table top, low volume	1130.00 - 2010.00
Forensic	4000.00 - 4700.00

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YARD IMPROVEMENTS
SECTION UIP 13

SURFACE PARKING LOTS

The following are based on a cost per space and average area per space including asphalt paving, concrete aprons, striping, some lighting, landscaping and drainage. Older lots, sized for large cars or lots designed for high volume traffic tend toward the higher areas per space, while newer lots sized for smaller cars, or lots designed for maximum employee parking tend toward the lower end of the range. Costs will vary depending on the extent of grading, paving, lighting and other amenities required.

Open lot costs will vary significantly by size and amenities. Add only those components that apply and which can be mixed by quality. Reduce costs by 5% for every 100 cars over 200. For example, a small retail lot for 15 cars will be at the high end of the scale, while a 500 car lot will be at the low end minus 15%. Apply proportional costs to additional paved areas beyond the typical area per space.

	LOW	AVG.	GOOD	EXCL.
Engineering plans, survey, permits, etc.	110.00	144.00	180.00	225.00
Grading rough and finished	93.00	119.00	150.00	195.00
Drainage	217.00	240.00	271.00	292.00
Paving spaces and drives	835.00	1040.00	1370.00	1730.00
Pavement marking striping and bumpers	18.60	28.75	56.00	83.50
Buildings ticket booths	28.75	28.75	47.25	56.00
Electrical lighting and wiring	240.00	262.00	285.00	300.00
Miscellaneous landscaping, fencing, signs, etc.	195.00	217.00	262.00	285.00
Cost per car space	1730.00	2070.00	2600.00	3175.00

Average area per parking space	LOW	AVG.	GOOD	EXCL.
Square feet	285	315	345	380

PARKING LOT EQUIPMENT

	COST RANGE
Automatic pay station	137000.00 - 188000.00
Automatic ticket dispenser	12800.00 - 19200.00
with ticket reading machine	22700.00 - 28400.00
Gate operator, key	1390.00 - 1720.00
Card	1980.00 - 2550.00
Coin	3900.00 - 8050.00
Walk up slot box	2360.00 - 3625.00
Fee indicator, inside	3700.00 - 4500.00
Outside	4500.00 - 5150.00
Gate arm, automatic, one way	6200.00 - 8950.00
Two way	7700.00 - 9600.00
Traffic detectors, magnetic, each	1330.00 - 1980.00
Traffic exit spikes	2975.00 - 3250.00
Warning sign, lighted, on post	1390.00 - 1720.00

For entry grills and gates, see Section UIP 5. For toll booths, guard houses, see Section UIP 11.

PARKING LOT IMPROVEMENTS

Asphalt coating, 2" min. overlay, per sq. ft.	1.95	-	2.60
Reseal	0.29	-	0.65
Parking lot striping, per car	12.80	-	18.35
Handicap stall, each	23.40	-	31.00
Parking lot striping, preformed reflective Thermoplastics, per car	86.00	-	157.00
Handicap stall, each	307.00	-	755.00
Parking bumpers, precast concrete, per lin. ft.	7.40	-	12.70
Wood	7.00	-	14.40
Plastic	9.86	-	20.20
Sign, for handicap stall, on pole, each	300.00	-	500.00
Wall mounted	106.00	-	195.00
Painted on surface	65.50	-	150.00
Speed bumps, 10” wide, plastic, per lin. foot	30.50	-	47.25
Metal guard rail, pipe or posts, per lin. foot	32.75	-	61.50
Barrier posts or poles, each	157.00	-	479.00

FLAGPOLES

Costs are for tapered aluminum flagpoles, including concrete foundation, base, external halyard, aluminum ball and installation.

	COST RANGE		COST RANGE
20'	1290.00 - 2975.00	40'	3050.00 - 6850.00
25'	1660.00 - 3775.00	50'	4650.00 - 10200.00
30'	2120.00 - 4650.00	70’	10500.00 - 22300.00

For sectional, lightweight steel poles, deduct 18%.
For fiberglass poles, add 10%.
For an ornate base or top, add $800 to $5,100 each.
For internal halyards, add $2,020 to $4,375.
For aluminum nautical yardarms, add $2,360 to $4,650 each.

BRIDGES

The following square foot costs are national averages derived from numerous bids and contracts. The costs were trended to the present date and converted to the national base by the Local Multipliers. The high and low 5% of the cases in each category were then discarded to arrive at the reasonable high and low figures listed. The medians are derived from the total number of cases in each category.

DESCRIPTION	COST RANGE
Highway, concrete	91.50 - 300.00
Steel	99.00 - 338.00
Pedestrian, concrete	103.00 - 530.00
Steel	103.00 - 590.00
Skyway, enclosed walkway	426.00 - 860.00

NOTE: Bridge costs can vary significantly depending on the type of construction, attachment and height, span and size of deck. Skyways have varying requirements for H.V.A.C., lighting, sprinklers, enclosure and roof construction.

For light, prefabricated wood or steel foot, golf cart, etc., type bridges, see Section UIP 14, Page 1.

RAILROAD SPURS

Average costs per linear foot for a 500 foot spur. Costs include rails, ties, ballast, spikes and alignment. Low end costs represent rail replacement or installing used rails. Costs of turnouts are an additive to the rail cost.

Add 2% for each 100 feet of track under 500 feet.

Deduct 2% for each 100 feet of track over 500 feet (25% maximum deduction).

Add $7,150 for bumpers. Add $25,900 for crossing signals.

Add $164 per linear foot for 9’ concrete roadbed. Add $590 for crossing timbers.

Add $1,670 per pair for wheel stops.

WEIGHT OF RAIL (lb. per yd.)	SIZE OF RAIL (base x height)	COST INSTALLED (per ft. of track)	ADD FOR SWITCH AND TURNOUT
40#	3 1/2 x 3 1/2	99.00 - 124.00	34800.00 - 43200.00
60#	4 1/4 x 4 1/4	126.00 - 157.00	42300.00 - 51250.00
80#	5 x 5	148.00 - 180.00	48400.00 - 60250.00
100#	5 3/8 x 6 5/8	164.00 - 209.00	53750.00 - 67250.00
115#	5 1/2 x 6 5/8	180.00 - 225.00	59000.00 - 72250.00
130#	6 x 6 3/4	202.00 - 240.00	63500.00 - 77750.00

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PIPING
SECTION UIP 13

QUANTITY DEDUCTIONS

For large installations of fencing or walls, such as industrial or subdivision uses, farms and highways, deduct the following:

600 linear feet	5%	4,500 linear feet	15%
2,000 linear feet	10%	6,000 linear feet	20%

CHAIN LINK FENCES

Average cost per linear foot of galvanized steel fence, including complete installation on 2" round or "H" posts set in concrete, 10' on centers. Rails, barbed wire and gates are given as additives to the base costs. Gates are priced on a per-gate basis.

	HEIGHT				
TYPE OF MATERIAL	**4'**	**6'**	**8'**	**10'**	**12'**
2" mesh, #7 wire	19.00	27.75	36.25	44.25	52.50
#9 wire	16.30	23.70	31.25	38.50	45.50
#11 wire	14.00	20.20	26.50	32.75	39.00
Add for rail	3.11	3.11	3.38	3.38	3.38
Add for 3 strand Barbed wire	4.04	4.04	4.54	4.54	4.54
Add for barbed coils	15.55	15.55	16.60	16.60	16.60
Add for privacy slats	9.46	14.40	19.40	24.80	29.75

Add 5% to 15% for aluminum or vinyl covered wire.
For fabric wind screen, add $0.77 – $1.51 per square foot.
For security micro-mesh, add $1.09 – $2.11 per square foot.

Gates, 3' wide	368.00	471.00	555.00	-----	-----
5' wide	479.00	660.00	805.00	990.00	-----
10' wide	795.00	990.00	1140.00	1290.00	1450.00
15' wide	995.00	1280.00	1500.00	1690.00	1880.00
20' wide	1230.00	1560.00	1800.00	2070.00	2280.00
25' wide	-----	-----	1930.00	2240.00	2440.00

Add 25% for sliding gates.

SECURITY GATE TURNSTILES

(Cost each, including installation.)

	COST RANGE		
One way, manual operation, galvanized	8600.00	-	9950.00
Aluminum	11400.00	-	15300.00
Stainless steel	14800.00	-	17700.00
Pedestal type arm turnstile	2210.00	-	6350.00
Handicap gate	5400.00	-	8600.00

Add 100% for tandem gates, 25% for polycoated gates, 30% for electrically operated gates.

METAL FENCES, RAILING AND GRILLS

(Cost ranges per square foot.)

Hand forged wrought iron or modular, steel or aluminum	10.10	-	30.75
Gates	15.20	-	37.50
Add for bronze anodized or powder coated	7.59	-	14.85

For window grills and ornamental driveway gates, see Section UIP 5. Custom ornamental work can run two to three times the listed costs.

KENNEL RUNS

(Cost ranges per square foot of fencing.)

Wire mesh on steel posts (wall area)	4.73	-	8.83
Mesh cover (roof area)	5.89	-	10.80
Add for gates, each	247.00	-	471.00

MASONRY WALLS

(Cost ranges per square foot, including normal foundations.)

	COST RANGE		
4" concrete block with pilasters, reinforced	12.60	-	15.65
6" concrete block, reinforced	14.50	-	17.80
8" concrete block, reinforced	15.40	-	20.75
6' block piers or pilasters, each	195.00	-	500.00
Ornamental screen block	12.80	-	20.75
Add for each wythe of face block	1.51	-	4.29
Add for stucco finish, each side	3.62	-	5.89
Common brick, 8" thick	27.75	-	35.75
8" block back–up	23.70	-	31.75
Add for each additional 4" thickness of common brick	7.33	-	10.80
6' brick piers or pilasters, each	368.00	-	1500.00
Add for each wythe of face brick	3.46	-	5.38
Concrete, precast, posts and solid panels	11.40	-	20.75
Add for wood grain or split face	3.38	-	8.04

NOTE: For light residential subdivision installations using unskilled laborers, costs may be 50% lower.

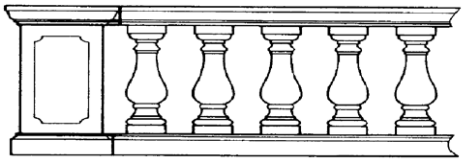
For retaining walls, gabions, see Section UIP 1. For stone walls, see Section UIP 6.

CONCRETE RAIL FENCES

(Cost ranges per linear foot of post and split rail prefabricated concrete fences, installed.)

4' high, one rail	11.30	-	16.75
5' high, two rails	16.30	-	24.95
6' high, three rails	22.55	-	34.00
6' high, four rails	27.75	-	42.25

BALUSTRADING



Precast stone balustrading found in ornamental railings around pools, patios, garden paths, stairs, etc., will cost $271 to $895 per linear foot installed, excluding urns, statuary, etc. Because of the various styles and lengths of balustrades, costs should be built up individually.

UNIT COSTS	**COST RANGE**		
Pier shaft, pedestal, pilaster, etc. (including cap and base), each	262.00	-	2800.00
Gate, fence piers, 8' each	3275.00	-	19900.00
Plinth or bottom rail, per lin. ft.	47.25	-	94.50
Baluster, each	157.00	-	525.00
Ornamental block parapet screening, per lin. ft.	127.00	-	202.00
Top rail, per lin. ft.	91.50	-	130.00
add for lighting under rail, per lin. ft.	90.00	-	129.00

For curved or stepped balustrading, add 120% to 150% to the cost.
For finials, urns, vases, statuary, etc., see Page 9.For lanterns, see Page 7.
For high density foam polymer plastic, deduct 25% to 65%.

TRASH ENCLOSURES

Masonry walled dumpster enclosures, each	3175.00	-	6300.00

CONCRETE EQUIPMENT PADS

Package air conditioning compr. pads, each	262.00	-	740.00
Transformer pads, reinforced, each	1010.00	-	3000.00

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YARD IMPROVEMENTS

SECTION UIP 13

WOOD FENCES

(Cost ranges per linear foot including complete installation.)

TYPE	COST RANGE
4’ solid prefabricated panels, lattice top	22.55 - 31.75
6’ solid prefabricated panels, lattice top	27.50 - 38.25
6’ solid board, horizontal or vertical	29.00 - 44.25
5’ solid board	26.50 - 39.25
Add for batts	2.60 - 9.38
6’ split redwood, solid palings, (grape stake)	28.75 - 45.50
5’ split redwood	27.00 - 40.25
6’ stockade, half pole	21.90 - 29.00
whole	30.75 - 37.00
6’ basketweave	34.25 - 49.50
5’ basketweave	33.25 - 46.50
12” – 24” wood picket	15.65 - 25.50
30” – 40” wood picket	20.75 - 30.75
50” – 60” wood picket	28.75 - 40.25
24” – 50” lattice, framed	11.00 - 20.75
Two rails, split, 24” – 36” high	14.40 - 20.75
Three rails, split, 36” – 48” high	15.95 - 23.20
Four rails, split, 42” – 54” high	19.40 - 29.00
Lattice, arbor, per square foot	6.24 - 11.85
Trellis (no frame)	0.93 - 1.95
Wooden gates, per square foot	15.55 - 23.70
Paint, per square foot, each coat	0.58 - 0.84

For treated wood, add 10% to the low side of the cost range.

FARM AND RANCH FENCING

(Cost ranges per linear foot, subject to quantity deductions.)

Barbed wire on steel posts, three strands	4.47 - 6.33
Four strands	4.80 - 7.17
Five strands	5.64 - 7.40
Wire mesh on steel posts	5.73 - 8.12
Snow fence	6.15 - 10.60
Add for wooden posts	0.67 - 0.84

VINYL FENCES

(Cost ranges per linear foot including installation.)

6’ spaced slat panels	33.25 - 63.50
4’ – 5’ spaced slat panels	31.75 - 57.50
5’ basketweave	46.25 - 75.00
6’ solid board panels	35.75 - 66.00
4’ – 5’ solid board panels	33.00 - 60.50
For lattice top, add	7.00 - 10.75
36” – 48” picket	22.25 - 39.75
24” – 50” lattice, framed	12.25 - 24.00
Two rails, 36” – 48” high	15.50 - 22.85
Three rails, 60” high	19.80 - 26.50
Four rails, 60” high	20.75 - 30.00
Lattice, arbor, per square foot	6.66 - 14.00
trellis (no frame)	1.60 - 2.79

	HEIGHT			
Gates, each	3'	4'	5'	6'
3’ wide	420.00	434.00	471.00	540.00
4’ wide	434.00	455.00	500.00	590.00
5’ wide	471.00	479.00	650.00	680.00

OUTDOOR LIGHTING

	COST RANGE
Floodlights, per fixture,	
High pressure sodium	1500.00 - 3075.00
Mercury vapor	1280.00 - 2490.00
Metal halide	1440.00 - 2875.00
Fluorescent or quartz–iodine	1230.00 - 1720.00
Incandescent	575.00 - 1030.00
Spotlights or luminaires, decorative, each	240.00 - 705.00
Underwater lighting	835.00 - 2010.00
Low voltage, residential	67.50 - 195.00
Poles, steel, per lin. ft. (ornamental, add 100%)	83.50 - 180.00
Concrete	51.00 - 202.00
Fiberglass	110.00 - 240.00
Add per arm bracket (aluminum, add 85%)	119.00 - 262.00
Post lantern, residential	278.00 - 800.00
Decorative pier lanterns	119.00 - 1160.00
Athletic fields (total floodlight system costs):	
Football fields, high school or park, minimum	196000.00 - 311000.00
Good lighting	311000.00 - 483000.00
Softball diamonds, minimum lighting	70000.00 - 119000.00
Good lighting	141000.00 - 221000.00

SCOREBOARDS

The costs per square foot of face are for electronic scoreboards hung or mounted flush on gymnasium walls without extra structure or foundation. See Section UIP 14 for pole costs and Section UIP 6 for extra signage.

	AREA IN SQUARE FEET		
TYPE	Under 35	35 to 75	Over 75
Outdoor	180.00 - 202.00	93.00 - 157.00	83.50 - 93.00
Indoor	202.00 - 278.00	144.00 - 202.00	110.00 - 157.00

ITEM	COST RANGE
Add for each controller	500.00 - 1260.00
Horn	800.00 - 1130.00
Shot clock, penalty timer	1880.00 - 3225.00
Separate game timer, shots on goal	3275.00 - 3750.00
Separate goal lights	500.00 - 995.00

For four faced scoreboards, add 5% to 15% to the single-face costs and apply to the area of all sides.

SOFT MODULAR PLAY SYSTEMS

Modular play systems are found in fast food chains, retail stores and pay-per-play amusement centers. Generally they consist of a padded tube frame, net walls, padded floors and a variety of playing components. The following costs are typical cost ranges for most common systems and include assembly.

Ball baths, including frame, shoe cubby, entry/exit tubes and pads.	
6 to 10–user capacity	9100.00 - 17200.00
15 to 25–user capacity	20200.00 - 38400.00
30 to 50–user capacity	32600.00 - 86500.00
add for super slide, per unit	7150.00 - 11100.00
Ball hoop, per unit	1400.00 - 2975.00
Bumper bridge, per unit	9650.00 - 17700.00
Air bounce, 10 to 20–user capacity	38100.00 - 61000.00

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YARD IMPROVEMENTS
SECTION UIP 13

RESIDENTIAL STYLE SWIMMING POOLS

Costs are averages for gunite pools constructed in frost free areas with normal soil conditions. Costs include filter, concrete coping, one light, steps, ladder and a maximum depth of 9 feet. Add or deduct 2% for each foot of deviation from the maximum. Custom free form pools may run 50% to 100% more. A negative edge rim will add 10% to 20%

to the costs. For lap pools, enter table using the perimeter. Costs do not include heaters (see table on the next page). See Pages 3, 9 and 10 for decking, waterscape, gazebos, etc. Typical decking will add 20% to 50% (double for spa only or custom pools) to the basic pool cost. Vinyl-lined are in-ground, sand-supported pools.

Size (sq. ft.)	Perimeter (feet)	Typical Cost Range	Size (sq. ft.)	Perimeter (feet)	Typical Cost Range
	60	27300.00 - 31400.00		90	41000.00 - 48300.00
	65	28500.00 - 33800.00		95	43100.00 - 50250.00
300	70	30800.00 - 35500.00	650	100	44500.00 - 52500.00
(12 x 25)	75	32700.00 - 38300.00	(18 x 36)	110	47900.00 - 56250.00
	80	34000.00 - 40000.00		120	51000.00 - 60000.00
Vinyl lined	75	25600.00 - 28800.00	Vinyl lined	110	33900.00 - 38300.00
	75	34500.00 - 40300.00		100	45400.00 - 53750.00
	80	35900.00 - 42300.00		110	48400.00 - 58000.00
450	85	37800.00 - 44200.00	800	115	50500.00 - 59500.00
(15 x 30)	90	39700.00 - 46400.00	(20 x 40)	120	51750.00 - 61500.00
	95	41000.00 - 48000.00		130	55500.00 - 65000.00
Vinyl lined	90	28400.00 - 32700.00	Vinyl lined	120	37400.00 - 41600.00
	80	37000.00 - 42500.00		110	51500.00 - 61000.00
	85	38400.00 - 44500.00		120	55250.00 - 65000.00
525	90	40200.00 - 46800.00	1,000	125	56500.00 - 66500.00
(15 x 35)	100	43600.00 - 50500.00	(20 x 50)	140	61000.00 - 72250.00
	110	46600.00 - 54000.00		150	64250.00 - 75250.00
Vinyl lined	100	31100.00 - 35100.00	Vinyl lined	140	43600.00 - 48400.00

NOTE: Larger pools will cost $68.00 to $78.00 per square foot.

Deduct for lack of filter	965.00 - 1770.00
Add for two pump spa system	725.00 - 2300.00
Add for diving stand, rock or board (add 100% for commercial diving board)	900.00 - 1770.00
Add for extra underwater lights, each	725.00 - 1560.00
Add for fiber optic lighting	1750.00 - 6950.00
Add for extra ladder	620.00 - 1590.00
Add for slide	2070.00 - 5950.00
Add for pool lip waterfall, per linear foot	12.25 - 16.45
Add for mosaic tile bottom	4700.00 - 23300.00
Add for rockscape grotto	11600.00 - 46000.00
Add for waterfalls, precast	665.00 - 4625.00
Add for swim up bar	3500.00 - 5800.00
Add for handicap lift	3225.00 - 10600.00
Add for automatic cleaner, hydrosweep	1000.00 - 4700.00
Add for automatic sanitation system, chlorinator	180.00 - 2575.00
Add for ozone generator	2360.00 - 8000.00
Add for ionizer	1160.00 - 4625.00
Add for automatic controls (spa only, deduct 50%)	2875.00 - 8000.00

In cold areas and for poured concrete pools and spas add 15% to 30% to gunite costs.

For fiberglass pools deduct 10% to 20%.

For pool or spa covers and alarms, see Solar table on page 9.

NOTE: For pool enclosures and buildings, see Section UIP 14.

SPAS AND HOT TUBS

Exterior gunite spas and tubs. For interior installations, add 5% to 15%.

Attached to pool, shared pool equipment	9800.00 - 18500.00
Detached, including equipment	13300.00 - 37100.00
Portable, self–contained fiberglass or wood units, complete	6150.00 - 12800.00
Acrylic/thermoplastic, single lounge	10200.00 - 17200.00
Double lounge	12400.00 - 23600.00
Redwood hot tubs	4000.00 - 12200.00

ABOVEGROUND POOLS

The costs are for residential, ground level, metal framed, round, plastic-lined pools including filter and pump. See Page 2 for decking.

12' diam.	2800.00 - 3325.00	21' diam.	4825.00 - 6650.00
15' diam.	3675.00 - 5400.00	24' diam.	5400.00 - 7800.00
18' diam.	4500.00 - 6300.00	27' diam.	6300.00 - 9000.00

For rectangular shaped pools, deduct 80% from the residential gunite pool table.

AQUATIC EXERCISE POOLS

The costs are averages for 8' x 18' residential, fiberglass, stationary lap or aquatic exercise pools, with a 5-HP paddlewheel propulsion system, pump, filter and electric heater. See Page 3 for decking. (For pool jets only, deduct 20%.)

DEPTH	
42", flat floor	27900.00 - 69500.00
50", side wall water return channels	35100.00 - 79250.00
60", deep water running platform	38400.00 - 99250.00
Add for natural gas heater in lieu of electric	1320.00 - 2310.00
Add for custom fit thermal blanket	300.00 - 1160.00

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YARD IMPROVEMENTS
SECTION UIP 13

SOLAR POOL AND SPA HEATING EQUIPMENT

The following are average installed costs for complete solar pool heating systems, per panel. Costs include collectors, valves, racks for flat mounting and up to 200 feet of PVC solar-loop piping. The systems utilize the existing pool pump and filter.

TYPE OF COLLECTOR	1 PANELS	10 PANELS	20 PANELS
Plastic (unglazed)	$1,180 - $1,500	$1,120 - $1,470	$1,050 - $1,450
Metal (unglazed)	1,450 - 1,850	1,360 - 1,730	1,280 - 1,610
*Metal (glazed)	2,240 - 2,750	2,140 - 2,625	2,040 - 2,490
Rubber mat	17.70 - 26.50	per sq. ft. of total panel area	

*Glazed collectors are enclosed with glass and include copper piping to and from the collectors.

	COST RANGE
Add for panel racks, to raise panel angle, per panel	434.00 - 575.00
Tedlar panel covers, per panel	83.50 - 144.00
Pool solar blanket (bubble mesh), per square foot	0.58 - 3.38
Pool safety cover, per square foot	3.97 - 10.05
Automatic pool cover, per square foot	15.20 - 35.75
Pool alarm system, complete	338.00 - 665.00
Complete with security system	1130.00 - 2875.00
Sonar system	8600.00 - 11700.00
Insulated spa covers, per square foot	13.55 - 39.00

Copper piping for solar loop; add 20% (except for glazed metal panels).

POOL HEATERS

Costs are averages of installed costs for gas or electric pool heaters. Costs do not include a base or pad. Heat pump systems cost $7,000 to $10,100 each. Small electric spa heaters (5 to 10 kw) cost $1,180 to $1,430 each.

GAS				ELECTRIC			
Output (MBH)	Cost	Output (MBH)	Cost	Output (KW)	Cost	Output (KW)	Cost
50	1990.00	150	3250.00	10	3500.00	25	4650.00
75	2280.00	250	4650.00	15	4000.00	50	5950.00
100	2725.00	500	6500.00	20	4425.00	75	7850.00

COMMERCIAL SWIMMING POOLS

The large recreational pools found in parks, schools, hotels, etc., vary widely in finish, structure and equipment, but the following costs will give a good approximation. Costs are for poured concrete and stainless steel pools and include chlorinators, filters, heaters, boards, ladders and coping. The finest quality, poured concrete, Olympic competition or municipal pools may run 100% above listed costs. Decks and bath houses should be computed separately.

POURED CONCRETE POOLS

SIZE (square feet)	TYPICAL COST RANGE (per square foot)
1,000 – 2,000	102.00 - 129.00
2,000 – 4,000	97.50 - 121.00
4,000 – 6,000	94.50 - 117.00
6,000 – 8,000	93.00 - 116.00
Over 8,000	90.00 - 111.00
Add for tiled surfaces	13.90 - 46.50

Deduct 20% for gunite pools.

STAINLESS STEEL POOLS

Stainless steel walls and recirculation system	COST RANGE
with a concrete floor	225.00 - 255.00
All stainless steel, in-ground	285.00 - 345.00
Elevated	338.00 - 448.00

SMALL COMMERCIAL POOLS AND SPAS

Motel pools, in areas where codes permit use of residential-type co struction and equipment, cost $75.00 to $110.00 per square foot including filters, boards, heater and chlorinator.

For apartments and motels, separate whirlpool baths for four to ten persons, cost $12,100 to $32,700 exclusive of decks. Portable handicapped access step platforms with railing and grab bars cost $3,825 to $7,200 plus $2,525 for deck storage locker. For water parks and slides, see Section UIP 14.

LARGE MUNICIPAL POOLS

Large municipal pools with bath houses, lighting, fences, decks and all the other features in the complete complex will cost $3,900 to $6,050 per person of rated capacity.

FOUNTAINS

Following are average costs for indoor or outdoor decorative water displays, consisting of circulating water with various spray designs and lighting. Costs include nozzles, spray rings, jets, pumps, drain, overflow system, water supply system, submerged lighting and wiring, control panel and wind control where needed. Pool is not included; see pond or conventional pool costs. Costs vary greatly due to display size, water flow rate, (commercial, 70 to 1,000 gallons in motion) and complexity of the spray design.

	LOW	AVG.	GOOD	HIGH
Residential fountains	1920.00	7500.00	18900.00	39900.00
Commercial fountains	21200.00	46400.00	99000.00	221000.00
Add for fiberglass bowls, 10” deep, up to 50 gallons	800.00	1030.00	1320.00	1670.00
Add for fiberglass pools, 8” – 16” deep, up to 500 gallons	1560.00	2020.00	2550.00	3300.00
Birdbaths, sundial pedestals, etc.	500.00	780.00	1230.00	1930.00
Precast stone, two tier, 3 1/2’ – 4 1/2’ diam.	1500.00	1750.00	2070.00	2430.00
Three tier, 4 1/2’ – 7 1/2’ diam.	1770.00	2950.00	4900.00	8050.00
Medium pool, up to 11’ diam.	8750.00	11300.00	14700.00	19200.00
Large pool, up to 26’ diam.	12100.00	16700.00	22700.00	31100.00
Add for pool fountain Centerpiece	1270.00	2750.00	5700.00	12100.00
Sculptures, brass, under 1’ or wall mount	68.00	110.00	180.00	278.00
Medium figures	338.00	555.00	875.00	1390.00
Large, 2’ – 4’ high (sundials, spheres, etc.)	1850.00	3250.00	5550.00	9250.00
precast stone, under 1’ or wall mount	368.00	471.00	585.00	740.00
Medium figures, urns, vases, Jardinieres	575.00	1100.00	2070.00	3975.00
Large finials, pinnacles	875.00	1670.00	3250.00	6300.00
Large, 2’ – 4’ high	1590.00	3075.00	5800.00	11100.00
Over 4’ high	8050.00	10200.00	12700.00	16000.00
Recirculating pump and tubing, residential	180.00	262.00	368.00	540.00

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YARD IMPROVEMENTS

SECTION UIP 13

LANDSCAPING

The square foot costs below are to be applied to the entire landscaped area and are used when actual unit pricing is not feasible. Costs for new landscaping will vary greatly due to the number of variables involved. The cost ranges include soil preparation and some post installation maintenance commensurate with the quality, excluding extremes.

LOW COST areas have a minimum number of 1 to 5 gallon shrubs, small trees and ground-cover plants proportioned for a light visual effect, manual irrigation and seeded lawn.

AVERAGE COST areas include an average number of 3 to 5 gallon shrubs, small to large trees, closer spacing of small ground-cover plants, automatic irrigation and seeded lawn.

HIGH COST areas will have a high visual content of 5 to 15 gallon shrubs, a greater proportion of large trees, closely spaced ground cover with top dressing, automatic irrigation, seeded lawn and sodded turf areas. For pathway costs, see golf course unit costs in Section UIP 14.

AREAS OCCUPANCIES	LOW	AVG.	HIGH
Small areas: single family dwellings including multiple/residential developments	6.82	9.08	12.05
Medium areas: office complexes, shopping centers, etc.	5.89	7.91	10.60
Large areas: industrial parks and educational/institutional facilities	3.11	4.29	5.56
UNIT COSTS			
Soil preparation, per sq. ft.,			
Fine grading	0.23	0.39	0.54
Mulching	0.36	0.60	1.11
Top dressing or bedding	3.56	4.25	5.12
Lawns, per sq. ft., seeding,			
With dressing	0.44	0.58	0.87
Hydroseeding only	0.15	0.23	0.49
Sodded	0.81	1.37	2.30
Ground cover, per sq. ft.,			
Gravel or wood chips	2.60	3.38	4.29
Spreading plants	0.84	1.51	2.72
Edging, per lin. ft., masonry	20.50	26.75	34.50
Cast stone	30.00	33.50	37.50
Metal	4.54	7.17	11.00
Redwood	3.55	4.80	6.66
Timber or ties	8.04	12.05	17.70
Shrubs and hedges, each, small	21.15	28.25	39.00
Medium	50.50	72.50	103.00
Large	157.00	225.00	300.00
Trees, each, small 15 gal.			
20” box	144.00	271.00	525.00
Medium, 24” 36” box	615.00	995.00	1610.00
Large, 42” box and larger	1560.00	2725.00	4425.00
Tree grates, including frame, per sq. ft.	35.00	46.25	61.50
Planter boxes, per sq. ft.,			
Concrete, precast	32.50	64.50	127.00
Cast stone, ornate troughs	404.00	575.00	740.00
Fiberglass	56.50	110.00	217.00
Wood	119.00	217.00	345.00
Planters, each, round, concrete,			
24” diameter	32.50	64.50	127.00
48” diameter	404.00	575.00	740.00
Tree size, 3’ 10’ diameter	56.50	110.00	217.00
Bollards, each, wood	119.00	217.00	345.00
Concrete	420.00	800.00	1220.00
Add for lighting	285.00	397.00	525.00

WATERSCAPING

The square foot costs apply to the entire waterscaped area and include excavation, rockscapes and all recirculation equipment. Add appropriate liner costs for lakes and streams

	COST RANGE LOW	MEDIAN	HIGH
Lakes	1.60	3.46	7.40
Collection ponds	63.50	79.50	102.00
Streams	10.75	15.00	21.30
Add for liner, asphalt	2.87	4.29	6.40
Clay	0.93	1.60	2.60
Concrete	6.15	9.08	13.20
Plastic and sand	1.86	3.38	6.66
Waterfalls, complete, per sq. ft. of face	64.00	91.50	129.00
Small, under 50 sq. ft.	164.00	247.00	359.00

LAWN SPRINKLERS

Costs per square foot, completely installed, including trenching and backfill. Intricate shrub type strip areas may be twice as high as conventional systems. For automatic controllers add $225.00 to $359.00 per station.

Conventional, residential and			
Small commercial	0.92	1.28	1.69
Commercial, medium areas	0.86	1.12	1.47
Rain Bird or Rain Jet			
systems, large areas	0.60	0.71	0.84
Sprinkler heads,			
impact or rotary	54.00	138.00	285.00
Pop–up or shrub	10.60	27.75	68.50
Control valve,			
Manual, ¾” – 1”	64.00	86.00	106.00
1¼” – 2”	150.00	195.00	240.00
for remote control, add	43.00	64.00	86.00
Gate valve, ¾” – 1”	75.00	96.00	138.00
1¼” – 2”	202.00	278.00	397.00
Valve boxes	64.00	106.00	195.00
Atmospheric vacuum			
breaker, ¾” – 1”	86.00	106.00	157.00
1¼” – 2”	180.00	247.00	345.00
Pressure backflow			
preventer, ¾” – 1”	740.00	800.00	875.00
1¼” – 2”	930.00	1100.00	1270.00
PVC pipe, Schedule			
40, ½” – ¾”, per linear foot	2.36	2.60	2.79
1” – 1½”	3.04	3.97	4.73
2”	5.56	6.15	6.66

RESIDENTIAL PONDS

SIZE (square feet)	COST RANGE	SIZE (square feet)	COST RANGE
25	7400.00 - 9450.00	150	11600.00 - 14300.00
50	8450.00 - 10500.00	200	12100.00 - 15300.00
75	9450.00 - 11600.00	250	13100.00 - 16900.00
100	10500.00 - 12700.00	300	14800.00 - 18900.00

NOTE: Large ponds will cost $59.00 – $76.50 per square foot.

COMMERCIAL PONDS

Concrete ponds used for foutain displays and for wading, including filters. Typical depth is 2’ to 4’. Use the low end of the range for commercial bumper boat ponds.

SIZE (square feet)	POURED CONCRETE PONDS	TYPICAL COST RANGE (per square ft.)
Under 500		25.75 - 35.00
500 1,000		23.55 - 31.75
1,000 2,000		22.25 - 28.75
2,000 4,000		21.45 - 27.25
4,000 6,000		21.15 - 26.50
6,000 8,000		20.20 - 26.00
Over 8,000		19.95 - 25.75

Deduct 25% for gunite pools.
Deduct 30% for fiberglass sides, free-formed.
Deduct 40% for vinyl lining.

AERATORS

Aerators pull oxygen-deficient water from depths of ponds and lakes and propel it as smalldroplets in the air to improve water quality. They help to remove algae without the use of chemicals or herbicides. Costs include floating unit, propeller pump, basic control box, 100’ power cable and installation. Add 15% for three phase equipment.

Single phase pump, 1/2 HP	2550.00	3325.00	4800.00
1 HP	3425.00	4475.00	6450.00
2 HP	4475.00	5800.00	8450.00
3 HP	5750.00	7500.00	10800.00
5 HP	7150.00	9300.00	13500.00
Add for lights	1920.00	2825.00	3900.00

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DEMOLITION
SECTION UIP 13

UNIT COSTS

(Costs per square foot except where indicated.)

These costs are of necessity very general and do not include hauling or dumping charges. Individual small jobs may involve a trade minimum that can run from $128.00 to $760.00. For detailed costs, we would recommend using our repair and claims products. For complete building demolition, see next page.

YARD	COST RANGE
Paving, asphalt	0.69 - 1.06
Concrete, unreinforced, 2" – 6"	0.90 - 3.49
Reinforced, 4" – 6"	2.05 - 4.60
Curbing, asphalt, per lin. ft.	2.90 - 3.65
Concrete	9.62 - 11.30
Fencing, chain link, per lin. ft.	2.90 - 4.36
Wood	3.15 - 4.17
Masonry	21.00 - 39.50
Piping, up to 6", per lin. ft.	5.86 - 10.60
Underground tanks, 5,000 gal. minimum and up, cost per gal.	0.21 - 0.31
Add for leak abatement and disposal	1.58 - 3.49
Dumpster costs, per week, per cubic yard, up to 10 cubic yards	51.00 - 109.00

For site clearing, see Section UIP 1.

BUILDING

Foundation, concrete, stem wall	17.95 - 33.50
Footing, per linear foot	14.05 - 21.50
Exterior wall, frame	2.00 - 6.91
Masonry	5.29 - 12.50
Floor, slab on grade, plain	3.15 - 6.72
Reinforced	5.04 - 10.40
Wood	0.96 - 2.37
Concrete, elevated (deck fill only, deduct 70%)	8.24 - 13.80
elevated, 6" – 12"	15.60 - 29.00
Floor cover, carpet	0.27 - 0.41
Wood	1.09 - 2.59
Resilient	0.47 - 0.94
Tile, ceramic/quarry	2.05 - 2.99
Partitions, studs (only)	0.68 - 1.25
Drywall	0.91 - 1.93
Plaster	1.21 - 2.21
Concrete block	3.15 - 8.63
Cabinetry, per linear foot	12.60 - 20.40
Ceiling, drywall	0.94 - 1.84
Acoustic tile	0.55 - 1.49
Plaster	1.22 - 1.96
Wood	2.37 - 4.43
Insulation	0.27 - 1.43

BUILDING (Continued)

Roofing, composition/wood shingles	0.60 - 0.93
Built up	0.91 - 2.45
Tile, concrete/clay	0.69 - 3.14
Roof sheathing, spaced boards	0.55 - 0.68
Solid	1.25 - 3.93
Fireplace chimney, masonry, per linear foot	20.85 - 63.50
Appliances, each	103.00 - 309.00
Plumbing, per each fixture	77.50 - 408.00
Piping, up to 4", per lin. ft.	2.55 - 6.31
HVAC units, each fixture, up to 150,000 BTU	166.00 - 630.00
Ducts, per lin. ft.	0.96 - 8.15
Electrical, light fixture, each	34.00 - 60.00
Receptacle/switch, each	6.72 - 16.00
Conduit/wire, per lin. ft.	5.29 - 6.98
Wire, romex, per lin. ft.	1.03 - 1.72

AMERICANS WITH DISABILITIES ACT (ADA) BARRIER REMOVAL (These costs are per occurrence where applicable.)

Enlarge doorways, replace	725.00 - 4100.00
Add for automatic doorway entries, new	6900.00 - 11900.00
Offset hinges or lever handles, replace	220.00 - 439.00
Doorknob extensions, new	61.00 - 79.00
Directional visual signage, braille (raised markings), excl. posts	31.25 - 87.50
Visual signal panels, excluding alarm (smoke, fire, etc.)	370.00 - 565.00
Audible/visual alarms, new	670.00 - 820.00
Elevator panels, replace	540.00 - 2150.00
Elevator cab, add audible or visual signal, each	915.00 - 1080.00
Interior panel control, relocate	14400.00 - 19300.00
Replace	20500.00 - 30900.00
Emergency communication system	5400.00 - 7900.00
Define traffic pattern step edging, thresholds, mats, new	61.00 - 408.00
Power relocation	225.00 - 340.00
Restrooms, accessible toilet stalls, replace	825.00 - 2700.00
Grab bars, new	103.00 - 1220.00
Raised seats, mirrors, other accessories, New or relocate	234.00 - 408.00
Handicapped fixtures, drinking fountains, Replace	1080.00 - 3075.00
Relocate sink	825.00 - 1380.00
Insulate exposed hot water pipes (trade minimum, add $110)	31.25 - 58.00
Reconfiguration of furniture, display racks, counter heights, etc.	152.00 - 825.00

Emergency egress equipment, handicap lifts, see Section UIP 7.

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DEMOLITION
SECTION UIP 13

GENERAL INFORMATION

Costs presented are average costs of removal per square foot of total building floor area, except as noted, including loading and hauling, but not dump fees. It is also assumed that the materials have no salvage value. For individual unit costs, see previous page.

Costs for removal vary greatly depending on the size and complexity of the job and extent of contamination regarding hazardous materials. The following cost ranges are in some cases based on one or only a few removal projects and should be considered as very rough guides. Due to the number of variables involved, we would suggest that, wherever possible, survey, bid or contract costs be obtained.

BUILDING DEMOLITION

(Cost range per square foot.)

	COST RANGE		COST RANGE
Class A:	7.75 - 11.35	Class D:	5.19 - 8.01
Class B:	10.10 - 13.90	Class S:	4.69 - 7.75
Class C:	6.14 - 9.13	Gutting Only:	10.75 - 33.50

Small residences (approximately 1,000 – 1,500 sq. ft. per floor):

1 Story:	5250.00 - 12200.00	2 Story:	6900.00 - 16800.00
3 Story:	9300.00 - 20300.00		

HAZARDOUS MATERIAL REMOVAL

RADON REMOVAL: Residential basement ventilation retrofit costs $1,500 to $2,250 for a passive exhaust pipe ventilation system, plus $505.00 to $845.00 for active fan and alarm.

ASBESTOS REMOVAL:	LOW	AVG.	GOOD	HIGH
Full abatement	35.50	48.25	65.00	89.00
Spot removal	13.90	18.40	24.45	33.25
Encapsulation	2.55	3.30	4.36	5.63
LEAD REMOVAL:				
Full abatement	14.95	19.10	24.45	31.75
Spot removal	12.75	14.95	17.95	21.30
Encapsulation	4.43	5.63	7.23	9.37
GRAFFITI REMOVAL:	1.69	2.25	2.83	3.40

SITE DECONTAMINATION: Biological soil remediation costs have averaged $159.00 per cu. yd. for land treatment (tilled soil conditioning farming), $295.00 per cu. yd. for bioventing vapor extraction (air stripping and soil treatment) to $461.00 per cu. yd. for full bioreactor treatment (active slurry-aeration mixing), with costs having varied plus or minus 50%.

GROUND WATER CLEANUP: Pump and treat remediation costs have averaged $0.18 to $0.27 per gallon treated per year while permeable reactive barrier treatment systems averaged $0.75 to $1.62 per gallon treated per year with costs having varied plus or minus 50%.

STORMWATER MANAGEMENT

	COST RANGE
Polyethylene chamber system, complete, per gallon	1.81 - 3.65
UNIT COSTS	
Chamber only (70 – 425 gallon), each	93.00 - 408.00
Pipe drainage system, 10", per linear foot	21.30 - 26.00
12"	24.75 - 29.75
24"	48.25 - 56.00
36"	84.50 - 93.00
48"	136.00 - 144.00
Bioswale detention system, cost per sq. ft.	6.72 - 11.15
Compost filter berm system, cost per linear foot	3.15 - 7.59

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RECREATIONAL FACILITIES

SECTION UIP 14

GOLF COURSES

Primary variables in golf course costs are type of terrain and vegetation, size and layout, amount and quality of irrigation systems and overall quality. Excluded from these studies are extensive grading, such as that required for canyon and hillside courses, special drainage problems, all structures (clubhouse, maintenance, satellite cart, starters, range buildings, etc.), including bridges, and all manmade lakes and other waterscaping. Some of these items are listed as additives, while the balance may be found in other sections of this manual. No off-site costs, feasibility and environmental studies, etc., are included.

Included in the cost per hole are normal clearing of land, including incidental grading, complete irrigation and drainage systems, planting of trees in open land, greens, tees, fairways, service roads and cart paths, builders' profit and overhead, financing during construction and architects' fees for all items except structures.

Subdivision courses which are designed to allow as many home sites as possible to border the course will require longer pipe runs and expenses greater than normal for the playing length. Many costs may be written off against the adjoining development, such as utilities which are brought in for the development and used for the course. In many cases, allocation of costs between development and golf course must be made arbitrarily.

The costs listed for a basic commercially developed course are typical ranges in four quality classifications. Actual contracts and estimates have ranged from $82,500 per hole for a sprinklered course with minimal improvements to $519,000 per hole for a course scooped out of the mountains, $695,000 per hole for a course built from marsh land and $1,333,000 per hole for a course carved out of rock with completely imported top soil. All excluded costs must be examined; in some cases total project cost, including all structures, equipment, developers' overhead and profit, startup and maintenance has reached a budget of $881,000 to $2,754,000 per hole and up, excluding land costs.

Costs of complete irrigation systems constitute approximately 15% to 30% of the cost per hole of the courses listed. Architectural design, supervision and engineering costs are included at 8% to 12%. A complex project with special engineering and a name architect can run twice the averages listed. Older courses may be at the lower end of the cost ranges where design layout and improvements had not been affected by restrictive land use and/or environmental controls.

Courses are grouped into price classifications, with a limited description of what the price includes. Courses may fit into one class by sheer length (5,500 to 7,000 yards) and another by gross area covered (100 to 170 acres) or by overall quality, in which case, interpolations may be made. Generally, simpler courses will require little clearing or grading, encompass minimal acreage, have easy playing holes and/or minimal irrigation, while good courses will include extensive sitework and/or irrigation over large acreage, with well designed holes for maximum playability.

COST RANGE PER HOLE

Class I. Minimal quality, simply developed, budget course on open natural or flat terrain, few bunkers, small tees and greens
. . . . . $106,000 - $145,000

Class II. Simply designed course on relatively flat terrain, natural rough, few bunkers, small built up tees and greens, some small trees
. . . . . $152,000 - $217,000

Class III. Typical private type club on undulating terrain, bunkers at most greens, average elevated tees and greens, some large trees moved in or clearing of some wooded areas, driving range
. . . . . $220,000 - $323,000

Class IV. Better championship-type course on good undulating terrain, fairway and greens bunkered and contoured, large tees and greens, large trees transplanted, driving range, may have name architect
. . . . . $330,000 - $529,000

Good championship course with same high cost features
. . . . . $474,000 - $732,000

Excellent championship course with extensive features
. . . . . $927,000 - $1,454,000

COMPONENT COSTS PER HOLE

The cost of the basic course as listed includes only those items designated with an asterisk(*). Courses in Classes I and II fit into the low cost range, while those in Classes III and IV fall in the medium cost range. Many courses will have components that fall into several quality levels.

	LOW COST	MEDIUM	HIGH COST
*Mobilization, proj. manage.	$6,050 - $6,700	$7,500 - $8,750	$15,000 - $34,300
*Erosion control.	650 - 1,160	1,990 - 3,250	5,300 - 9,850
*Clearing, grubbing.	710 - 945	1,490 - 2,110	2,850 - 8,050
*Rough shaping.	4,900 - 8,850	16,500 - 27,700	34,300 - 55,500
*Survey and engineering. .	1,210 - 1,840	2,550 - 3,825	4,975 - 11,000
*Architectural.	8,550 - 12,900	18,900 - 30,900	72,750 - 171,000
*Top soil, haul and spread.	1,210 - 3,100	7,350 - 17,800	22,100 - 37,400
Water features, earthwork.	4,900 - 13,700	18,900 - 31,700	36,100 - 66,500
*Drainage.	3,900 - 4,300	4,900 - 6,050	14,900 - 30,900
Storm drains.	6,700 - 19,600	29,200 - 61,500	85,500 - 136,000
*Greens.	11,000 - 19,600	34,300 - 62,000	67,250 - 82,500
*Tees.	1,890 - 3,600	7,350 - 13,700	17,700 - 44,200
*Bunkers.	3,000 - 6,050	11,000 - 19,600	24,500 - 39,800
*Finishing,shapingandgrading	3,600 - 5,400	7,800 - 13,200	16,500 - 33,000
*Cart path grading.	650 - 1,210	2,450 - 4,900	5,500 - 7,750
*Irrigation system, automatic	36,100 - 52,500	73,750 - 110,000	147,000 - 274,000
manual.	17,100 - 22,000	27,700 - 36,700	— —
add for pumping plant. .	3,900 - 9,950	14,700 - 24,900	29,200 - 36,100
Well.	2,450 - 3,100	3,900 - 4,900	5,500 - 7,750
*Grow–in seeding preparation	8,450 - 9,800	11,600 - 13,700	18,400 - 47,400
*Grassing.	18,400 - 23,000	27,400 - 34,800	41,000 - 99,500
*Finished cart paths.	11,800 - 17,700	25,500 - 36,100	48,800 - 70,500
Bridges, railings, tunnels, etc.	2,450 - 8,550	12,200 - 19,300	21,100 - 34,300
Water features.	8,550 - 23,300	43,000 - 66,250	85,500 - 218,000
*Landscaping.	1,210 - 1,890	3,100 - 5,500	16,500 - 55,500
*Fencing and walls.	710 - 1,100	1,990 - 4,425	14,400 - 47,400
*Total of indicated features	124,000 - 183,000	267,000 - 418,000	591,000 - 1,161,000
Miscellaneous:			
Clubhouse.	12,200 - 36,100	85,500 - 122,000	158,000 - 193,000
Championship entertainment facility	— —	— —	287,000 - 849,000
Maintenance, other buildings.	6,100 - 14,700	28,400 - 46,600	66,500 - 122,000
Maintenance equipment. .	22,200 - 31,700	46,600 - 55,500	61,500 - 65,750
Oper., grow–in/start–up budget.	12,900 - 20,500	31,700 - 49,600	55,500 - 111,000

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RECREATIONAL FACILITIES

SECTION UIP 14

SHORT COURSES

Pitch and putt course. Nine holes on 10 to 15 acres, 1,000 yards long, including irrigation, excluding structures and lighting . . . . . $56,250 - $79,500

Par 3 course. Nine holes on 15 to 20 acres, 1,400 yards long, including irrigation, excluding structures and lighting . . . . . $72,000 - $99,750

Executive course. Eighteen holes on 50 to 60 acres, 4,800 yards long, par 60, including irrigation, excluding structures and lighting . . . . . $109,000 - $143,000

Lighted driving ranges. Separate paved stations, fenced, irrigated, good quality, not including separate putting or pitching greens, buildings or equipment. Cost per station. . . . . $10,300 - $15,000

add for covered structure, per station.	2,925 -	4,900
add for heated tee, per station.	9,800 -	11,700
add for elevated tee, per station.	5,850 -	11,700
driving range equipment.	73,750 -	128,000

MINIATURE GOLF COURSES

Minimal quality, eighteen holes, 2,000 to 4,000 square feet on 1/4 acre, simply developed, or prepackaged budget course on flat terrain including lighting, excluding booths, and parking lot.

Cost per hole (indoor, deduct 10%). . . . . $2,550 - $7,750

Average course, eighteen holes, 4,000 to 10,000 square feet, on 1/4 to 1/2 acre, excluding booths, snack bars, and parking lot, including course plumbing and lighting, professionally designed and installed.

Cost per hole. . . . . $9,850 - $30,900

Good custom course, eighteen holes, 10,000 to 20,000 square feet, 1/2 to 1 acre or more, extensive themes with major elevation, rock and waterscape layout, excluding buildings and parking lot.

Cost per hole. . . . . $32,100 - $73,750

MISCELLANEOUS UNIT COSTS

The following costs may be used to adjust items included in the cost per hole pricing, to add for those items which have not been included (marked by an asterisk), or as a build-up method to develop a complete course cost.

	LOW	AVG.	GOOD
Site clearing, cost per acre.	$2,140	$4,575	$9,500
Fairways, cost per acre.	4,050	5,250	6,500
Rough, cost per acre.	960	2,180	5,100
Earth moving, cost per cu. yard.	3.48	5.26	7.71
Imported fill, cost per cu. yard.	5.42	10.40	19.95
Tees, elevated, cost per sq. ft..	1.44	1.78	2.55
Bunkers, (sand traps) cost per sq. ft.. . . .	4.74	7.12	10.40
Greens, flat, cost per square foot.	4.49	6.53	9.89
elevated/contoured.	8.04	11.30	16.35
Artificial putting greens.	22.45	30.25	39.75
Target greens.	5.01	6.19	7.78
Pathways, asphalt, cost per sq. ft..	3.12	4.57	6.26
Concrete.	4.74	7.12	10.60
Gravel.	0.85	1.44	1.87
Soil cement.	0.78	0.93	1.44
Bridges, pedestrian, cost per sq. ft., light wood/steel.	40.50	62.50	102.00
Golf cart.	54.00	77.50	111.00
Car or light truck.	80.50	110.00	159.00
Player shelters, open, each.	2,550	4,675	8,300
Enclosed.	11,900	17,200	23,900
Lighting, short courses, per pole.	9,800	18,300	33,600

Misc. water/landscape, paving, parking lots, lighting, fencing, drainage, etc., see Section UIP 13.

BUMPER BOAT FACILITIES

The following costs are average cost ranges of complete bumper boat facilities. Individual facilities can vary greatly depending on the type of pond construction, overall size and amenities. Boat costs are not included, and can run $3,725 to $10,200 each. Pedal boats cost $845 to $2,550 each.

Minimal quality, vinyl lined ponds with filters and wood decks, minimal railing or fence enclosure and lighting, excluding booths and parking lots. . . . . $59,000 - $92,750

Average pond and filters, some freeforming, gunite or fiberglass panel sides, concrete lip and deck, paved waiting/viewing area, adequate handrails, fencing and lighting, excluding all buildings and parking lots. . . . . $118,000 - $187,000

Good concrete pond and filters, some rock or waterscape islands, good lighting, railing and fencing, excluding buildings, canopies or parking lots. . . . . $162,000 - $255,000

WATER SLIDES

The following are typical costs for most common types of elevated water slides in straight, single or double radius configurations and include open fiberglass body flume sections, galvanized steel flume supports, stairs, decking, pump, motor, all necessary piping fittings, installation and designers' fees. All pool and land improvement costs are excluded. For swimming pools, see UIP 16. Add 15% for enclosed tubular flumes. Add 25% for serpentine configurations. Add 13% to 19% for each foot of flume over 3 feet wide. Deduct 25% to 35% for slides laid on bare earth; under 60', deduct 10% to 20%.

RIDE LENGTH (FEET)	COST RANGE	RIDE LENGTH (FEET)	COST RANGE
50	$77,250 - $123,000	250	$287,000 - $452,000
60	89,250 - 143,000	300	339,000 - 524,000
90	126,000 - 199,000	350	377,000 - 595,000
100	136,000 - 217,000	400	421,000 - 662,000
150	190,000 - 299,000	500	505,000 - 793,000
200	240,000 - 377,000	600	582,000 - 915,000

Additive costs, cost each except as indicated:

Landing pond, per square foot.	$100 -	$138
Surge reservoir.	15,800 -	23,900
Filtration and chlorination system.	48,800 -	67,250

Large aquatic parks with bath houses, lighting, fences, decks and all the other features in the complete complex will cost $2,170 to $6,500 per person of rated capacity.

WATER SPRAY PARKS

Interactive automated water spray-splash parks.

Complete area per sq. ft., concrete. . . . . $16.45 - $32.75

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RECREATIONAL FACILITIES

SECTION UIP 14

SKI LIFTS

Costs for ski lifts vary greatly due to local terrain, both the steepness and roughness of the slope. The following installed costs are averages, including drive machinery, terminals, towers and carriers, and designers' fees. Extreme costs were excluded. Equipment costs may also be affected by the speed of the lift, where requirements for the number of intermediate towers and carriers will typically decrease as the speed of the lift increases, depending upon the skill of the skier, ambient temperatures and the length of the lift, as well as the terrain.

Off-site costs are not included. These would include the costs of bringing utilities to the site, access roads, environmental impact studies, etc. On public lands governmental regulations may require the air-lifting of materials for environmental reasons, resulting in higher than normal installation costs. Any clearing or grading of the ski slopes would be considered improvements to the land.

Due to the number of variables involved, we would suggest that, wherever possible, manufacturers' costs, installers' costs or contract costs be obtained.

Typical Skier Lift Speeds

Beginner	Novice	Intermediate/Expert
340 – 360 ft./min.	420 – 460 ft./min.	500 – 550 ft./min.

Gondolas and detachable quad lifts run at 800 – 1,000 ft./min. for all classifications.

INSTRUCTIONS

Lift costs are developed by adding together the units, consisting of (1) the length of the lift in feet, (2) the length of vertical drop in feet and (3) the actual lift capacity per hour, and multiplying that total by the total unit rate costs given in the tables.

Total Units = (1) Length of lift + (2) Vertical drop + (3) Actual capacity per hour

Rated capacity may be higher than practical operation permits. Actual capacity is determined by taking (a) the cable speed per hour times the number of persons per carrier, divided by the average spacing, which is (1) the length of the lift divided by half the number of (b) carriers.

$$\frac{\text{(a) Cable speed per hour x number of persons per carrier}}{\text{Spacing : (1) length of lift} \div \text{(b) 1/2 \# carriers}}$$

For Poma or T-Bar lifts, calculate the actual capacity as you would a single chair-lift.

EXAMPLE

A double chair-lift, (1) 3,000 feet in length, (2) 850-foot vertical drop (16.5-deg incline), (a) cable speed of 450 feet per minute, with (b) 100 double chairs.

$$\frac{\text{450 (Cable speed/min.) x 60 min. x 2 (persons per chair)}}{\text{3,000' length} \div \text{50 (half the number of chairs)}} =$$

$$\frac{54{,}000}{6} = \text{900 persons per hr. (actual capacity)}$$

Unit total: 3,000' length + 850' vertical drop + 900 capacity/hr. = 4,750 units

Cost of lift from table: 4,750 units x $297.50* = $1,413,125

***NOTE:** The unit cost of $297.50 was an interpolation from the table, with 4,750 units falling halfway between 3,500 and 6,000 units ($317 and $278.)

SKI LIFT COST TABLE

Enter table with total number of units:

Type of Lift	Unit Range			Cost per Unit		
Poma Lift	1,000	-	2,500	$145	-	$131
or	2,500	-	3,500	131	-	118
TBar	3,500	-	6,000	118	-	103
Double	1,500	-	3,500	355	-	317
Chair Lift	3,500	-	6,000	317	-	278
	6,000	-	10,000	278	-	234
Triple	1,500	-	3,500	408	-	362
Chair Lift	3,500	-	6,000	362	-	324
	6,000	-	10,000	324	-	278
Quad	6,000	-	9,000	492	-	461
Chair Lift	9,000	-	12,000	461	-	439
(fixed grip)	12,000	-	16,000	439	-	408
Gondolas	9,000	-	12,000	1130	-	980
	12,000	-	16,000	980	-	810

For single chair lifts reduce double chair lift unit costs by 20%.

For detachable quad chair lifts increase fixed costs by 20%. Rope tows are usually constructed with local labor and little design, and the costs are usually determined locally.

SYNTHETIC SURFACES

Synthetic surfaces are normally applied on prepared subsurfaces, usually wood, asphalt or concrete.

Subsurface costs should be determined from Unit-in-Place or Segregated Cost sections and the cost of the appropriate synthetic surface added.

Applied costs per sq. ft. of synthetic surface:	COST RANGE		
Ice skating	$63.00	-	$65.50
Baseball and football turf, shock pad included	18.25	-	23.00
Running tracks	17.85	-	22.15
Basketball courts	14.45	-	21.80
Field houses	15.70	-	22.15
Tennis courts	11.90	-	17.10
Playgrounds	9.69	-	17.70
Golf, indoor driving ranges, no pad	2.09	-	4.52
Striping, (per lineal foot)	0.41	-	0.50

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RECREATIONAL FACILITIES

SECTION UIP 14

GRANDSTANDS AND BLEACHERS

The following are typical cost ranges of grandstands and bleachers. The ranges do not represent the lowest or highest costs that might be found in each structure. Structures built by governmental bodies have tended to be at the high end of the cost range. Costs include stairs, ramps, handicap platforms and press boxes commensurate with type and quality, as well as designers' fees.

TYPE	PER SQ. FT. (Horizontal Projection)	PER SEAT
Gymnasium bleachers, steel frame, wood benches, telescoping, manual operation;		
under 600 seats	$52.00 - $63.50	$137.00 - $197.00
over 600 seats	48.25 - 58.50	122.00 - 166.00
add for power operation	13.30 - 16.05	34.00 - 48.50
Portable bleachers, steel frame, metal, fiberglass or wood benches, erected;		
up to 800 seats	26.25 - 30.00	72.50 - 103.00
over 800 seats	24.60 - 27.25	64.00 - 93.00
Permanent bleachers, wood frame and benches;		
up to 1,000 seats	33.25 - 39.50	98.50 - 131.00
1,000 to 2,000 seats	30.25 - 38.00	87.50 - 122.00
over 2,000 seats	28.75 - 34.75	82.00 - 111.00
Grandstand bleachers, open steel frame, metal, fiberglass or wood benches, school or fairground type;		
up to 1,000 seats	52.00 - 97.00	152.00 - 324.00
1,000 to 5,000 seats	50.00 - 94.50	144.00 - 309.00
over 5,000 seats	46.75 - 89.00	131.00 - 295.00
add for roofed areas	13.00 - 20.75	38.00 - 67.00
add for pressbox box area	77.50 - 121.00	— —
Concrete or steel bleachers, no interior construction, stadium type, closed deck;		
under 5,000 seats	124.00 - 152.00	370.00 - 505.00
5,000 to 10,000 seats	121.00 - 149.00	355.00 - 499.00
over 10,000 seats	116.00 - 147.00	340.00 - 483.00
add for roofed areas	15.70 - 24.40	46.00 - 82.00
Wood bleachers, with built in dressing rooms, restrooms, pressbox, lighting, stucco exterior walls;		
under 5,000 seats	113.00 - 133.00	340.00 - 446.00
over 5,000 seats	110.00 - 125.00	324.00 - 423.00
add for roofed areas	14.05 - 16.00	41.00 - 54.50
Concrete or steel bleachers with built in dressing and training rooms, restrooms, snack bars,pressbox, lighting, college or small municipal stadium type;		
under 5,000 seats	197.00 - 242.00	580.00 - 825.00
5,000 to 10,000 seats	166.00 - 205.00	483.00 - 660.00
over 10,000 seats	159.00 - 197.00	454.00 - 620.00
add for roofed areas	18.70 - 29.25	54.50 - 93.00
Major ball parks, complete structural improvements, figured at baseball capacity;		
Older parks, minimum facilities	2,300 - 2,825	
Modern parks, minor league type	1,920 - 6,750	
major league, open stadium, sky boxes	4,225 - 10,500	
enclosed, roofed stadiums, artificial turf	5,050 - 12,700	
add for retractable roofs	3,025 - 9,050	

ICE SKATING RINKS

Artificially made ice rinks are manufactured in various forms, but all include pumps, compressors, refrigerant fluids and freezing (piping) systems. Costs include designers' fees.

Refrigeration piping systems may be found:
1. Laid on bare earth.
2. On bare earth with a concrete curbing.
3. On a concrete slab with a curbing.
4. In a formed concrete slab.

Basic mechanical equipment, pumps, compressors, fluids, and piping installed in a standard-size hockey rink:
(85' x 200') . . . . . $32.00 to $45.25 per sq. ft. of frozen area.

Smaller rinks tend to be more expensive:
Stage-size rinks, etc. $70.50 to $100.00 per sq. ft. of frozen area.

Additive costs, per square foot of rink:	COST RANGE
Excavation	$.60 - $1.25
Subsoil heating	2.81 - 3.74
Sand floor	1.38 - 2.27
Wood sleepers	.87 - 1.19
Gravel base, 4"	.88 - 1.53
Plastic water barrier	.63 - 1.00
Insulation, rigid, 2"	2.81 - 4.49
Concrete slab, 5" – 6"	6.79 - 8.99
Concrete curbing	.72 - 1.40
Dehumidifier	5.01 - 7.75
Side boards, wood frame	6.54 - 9.64
metal frame	5.85 - 12.85
add for spectator protection, clear shield	4.11 - 5.01
chain link	.87 - 1.83

MISCELLANEOUS UNIT COSTS

(Cost per ton.)

The following costs may be used to adjust the basic square foot costs above or as an alternate pricing method.

CAPACITY (Tonnage)	REFRIGERATION	PLUS PIPING PLASTIC PIPE	PLUS PIPING METAL TUBE
80	$3,900 - $4,575	$3,150 - $3,500	$3,650 - $4,100
100	3,175 - 4,025	2,675 - 2,900	2,975 - 3,175
120	2,900 - 3,500	2,170 - 2,480	2,500 - 2,900
140	2,500 - 2,975	2,020 - 2,170	2,210 - 2,480

Dehumidifier, cost per ton. . . . . $515 - $695

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RECREATIONAL FACILITIES

SECTION UIP 14

DRIVE-IN THEATERS

Costs for drive-in theaters vary greatly depending on the extent of grading, paving, natural drainage, size and type of screens, speaker equipment and service facilities provided. The costs are broken into the major cost items on a cost per space basis, excluding extremes. Miscellaneous costs such as cost of normal financing and contractors' profit and overhead are prorated to each item. Architects' and designers' fees are included in the engineering costs for all items except buildings, the costs of which will include all fees applicable to the structural improvements. Many theaters will be mixed in quality and it is often better to compute individual items from other sections of the manual. Costs for storm drains, projection and food equipment, etc., are not included.

	LOW	AVG.	GOOD	EXCL.
Engineering –				
plans, survey, permits, etc..	$197	$242	$302	$370
Grading –				
rough and finished ramps.	190	234	295	355
Paving – ramps and drives.	580	760	995	1280
Buildings – concession, restrooms,				
projection and ticket booths.	348	454	575	725
*Screens – frame and foundation.	220	302	415	575
Sound dispersement –				
serving two cars.	145	166	205	249
Electrical – lighting and wiring.	256	333	423	545
Miscellaneous – fencing, signs,				
landscaping, etc..	122	183	264	386
Cost per car space.	$2,060	$2,675	$3,475	$4,475

COST MODIFIERS

Costs of each drive in theater have been adjusted to a base number of spaces and gross area per space. Multipliers are given to adjust these costs for deviations from the base. To determine the gross area per space, divide the entire improved area of the drive in by the number of spaces.

NUMBER OF SPACES	200	300	400	500	600	700	800	900	1000
MULTIPLIER	1.25	1.16	1.08	1.00	.97	.94	.91	.89	.86

GROSS AREA PER SPACE	350	400	450	500	550	600	650	700	750
MULTIPLIER	.95	.89	.94	1.00	1.04	1.08	1.12	1.16	1.20

*For multiple-screen theaters add the cost of additional screens, and adjust the modifiers to the average number of spaces and gross improved area per screen. The following unit costs for screen structures are typical cost ranges per vertical square foot of screen area, from a simple open structure to an ornate enclosed frame with storage and office area.

MISCELLANEOUS UNIT COSTS TABLE

The following costs may be used to adjust items included in the Cost Per Space pricing, to add for those items which have not been included or as a build up method to develop a complete theater cost.

	LOW	AVG.	GOOD
Screen structures, per square foot of screen area			
Wood frame on poles with			
guy wires.	$25.50	$29.25	$35.00
Wood frame on poles set in			
concrete, timber bracing.	35.00	39.75	45.75
Wood frame on timbers,			
concrete footings, plain.	41.25	49.00	58.00
Ornate.	56.50	66.00	79.00
Steel frame,			
concrete footings, plain.	47.50	59.50	73.50
Ornate.	63.50	74.50	87.50
Tilt–up concrete, concrete frame,			
and steel–framed screen enclosure.	70.50	86.00	108.00
Sound systems, each			
Car speakers.	38.00	55.50	83.00
add for heaters.	190.00	212.00	234.00
Post and junction box.	55.00	83.00	124.00
Remote systems,			
transistor type, etc..	45.25	62.00	84.50
Lighting, low level, each			
Ramp, entry/exit, etc..	134.00	190.00	256.00
Directional signage.	324.00	461.00	670.00
Ticket booths, each.	10,600	12,800	15,800
Projection and miscellaneous equipment, excluding food service,			
Total cost per screen.	$56,250	$99,500	$177,000

Miscellaneous paving, lighting, fencing, drainage, playground equipment, etc., see Section UIP 16.

GO-CART TRACKS

The following are average costs of a complete go cart facility. Facilities will vary by the quality of construction and by size and amenities on the course. Costs are per linear foot of track, with a typical width of 22' to 30' for full size courses. Car costs are not included, and will run $4,300 to $13,000 each. For remote control shutoff, add $3,150 to $5,250 plus $960 per car.

Minimum quality facility, small, simple oval or kiddie course, flat with minimum bumpers, lights and fencing, excluding all buildings and parking lot. . . . . $166.00 - $220.00

Average facility, typically 600 to 900 linear feet of concrete track, safety rail bumper system, adequate banking, standard design, lighting and fencing, some landscape, excluding buildings and parking lot. . . . . $271.00 - $430.00

Good course, 800 feet and longer, well banked, berms, multi-oval, good concrete paving and steel safety rail system, course lighting, railing and fencing, landscaped layout, waiting/viewing area, excluding buildings, canopies or parking lots. . . . . $340.00 - $483.00

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RECREATIONAL FACILITIES

SECTION UIP 14

RADIO AND TV TOWERS

Radio and television towers are individually designed for the location, considering climatic conditions such as ice, wind, seismic and antenna loads. Costs are typical cost ranges excluding extremes. Due to the number of variables involved, it is suggested that whenever possible, manufacturers', installers' or contract costs be obtained. Included in the costs are concrete footings, erection, painting, guy wires, lighting, platforms and designers' fees. Antennas and transmission cables are not included. Multiple antenna installations and mono poles will tend to be at the high end of the range.

SELF SUPPORTING TOWERS

HEIGHT (FEET)	COST RANGE	HEIGHT (FEET)	COST RANGE
50	$21,400 - $34,200	225	$196,000 - $218,000
75	43,200 - 65,000	250	211,000 - 240,000
100	68,250 - 100,000	300	222,000 - 271,000
150	134,000 - 183,000	350	256,000 - 314,000
200	184,000 - 208,000	400	287,000 - 355,000

TRIANGULAR GUYED TOWERS

(Price per linear foot, up to 400 feet high)

10" Ham radio*, police and fire bands.	$136 - $197
20" Taxi and public service bands.	205 - 295
24" Radio, V.H.F., U.H.F. bands.	256 - 386
30" Cellular applications.	309 - 515
40" Microwave towers.	386 - 630
54" Master TV systems.	630 - 1,390

***NOTE:** Use high end of range for small crank-up self-supporting towers, add 100% for motorized operation.
Add 35% for every 100 feet of height over 400 feet.

SATELLITE DISHES

The following are average installed costs small commercial satellite dish systems. The costs include a 10' dish, receiver, descrambler, cable and all items necessary for a complete installation. The costs range from very basic systems, whose dishes are adjusted manually, to systems with many user features, extensive on-screen displays and automatically moving dishes.

LOW	AVERAGE	HIGH
$4,450	$7,350	$12,200

Deduct 20% for dishes 6' and under.

SHIP AND BOAT DOCKS

The following are typical cost ranges per square foot including fenders, railings, utilities, and miscellaneous items commensurate with the type and quality, exclusive of buildings.

LIGHT CONSTRUCTION (Wood girders, nailed)	
2" floating deck, light posts.	$39.50 - $62.50
2" decking, light posts.	42.00 - 56.50
3" decking, light piling.	48.25 - 73.00
MEDIUM CONSTRUCTION (Wood girders, bolted)	
2" floating deck, winch or light piling.	65.00 - 119.00
3" decking, light piling.	66.00 - 93.00
4" decking, average piling.	74.50 - 110.00
HEAVY CONSTRUCTION (Heavy wood girders)	
4" or heavier decking, heavy piling.	118.00 - 166.00
HEAVY CONSTRUCTION (Major shipping dock)	
Heavy concrete deck and piling.	183.00 - 271.00

SMALL BOAT MARINAS

The typical cost range for floating slips in modern commercial developments is $9,200 to $16,900 per slip, including ramps, anchor piers, utilities, lockers, designers' fees, etc., but excluding all land improvements, based on a typical mixture of boat types and sizes. Actual contracts have ranged from $4,750 per slip for small berths on calm riverfront with no utilities to $30,200 per slip for a high quality commercial complex with complete utilities and heavy anchorage.

UNIT COST	LOW	AVG.	GOOD
Deck and frame, per square foot of deck;			
Frame only, metal.	$14.65	$24.95	$42.00
wood.	6.72	15.25	34.00
Deck only, 2" treated wood. . . .	4.24	6.31	9.46
marine plywood.	10.20	11.90	13.85
concrete plank.	12.75	15.60	19.35
fiberglass panels.	16.35	19.65	23.70
metal decking.	13.30	16.95	21.20
Deck and frame, preassembled, portable dock, gangway or ramps;			
light duty pipe, wood deck. . . .	$27.00	$30.75	$36.50
medium duty truss, wood deck.	30.25	43.00	61.00
metal deck.	51.00	66.50	89.00
Drive on small craft modular dock.	37.25	77.50	159.00
Add for handrails, per linear foot. .	40.50	47.25	54.50
ladders, 4' to 6', each.	166.00	446.00	1,230.00
lockers, storage box, each. . .	461.00	890.00	1,690.00
benches, 4' to 6', each.	225.00	377.00	670.00
roof, fabric canopies, per sq. ft	16.35	22.90	32.25
metal or fiberglass.	9.02	12.50	17.10
wood.	14.05	19.45	27.00
utilities, electrical, each.	600.00	1,270.00	2,725.00
water or fire.	348.00	685.00	1,410.00
Flotation, drum or box, /sq. ft. of deck.	8.59	16.15	30.50
Anchorage system, piles, see Section UIP 1.			
metal poles, light, per lin. ft.. . .	6.72	11.90	20.85
telescoping.	55.50	62.50	70.50
stiff arm brace, per lin. ft..	60.00	66.00	74.50
winch and cable, each			
(excl. conc. anchor).	2,340.00	4,100.00	7,050.00
Swim rafts, complete, per sq. ft.. .	29.25	45.75	70.50
Boat storage racks, metal, per boat.	950.00	1,270.00	1,710.00
Boat lifts, manual, 1200# – 3000#.	2,300.00	2,975.00	4,050.00
personal watercraft.	1,490.00	2,060.00	2,775.00
motorized, add.	670.00	950.00	1,380.00
straddle hoists, cost per ton	6,600.00	10,000.00	14,700.00
add for jib mast.	29,800.00	33,300.00	37,700.00

Miscellaneous paving, lighting, fencing, see Section UIP 16; sheet piling, sea walls, Section UIP 1.

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RECREATIONAL FACILITIES

SECTION UIP 14

SPORTS COURTS

(Cost per playing court only, including design fees, excluding all building improvements.)

TENNIS COURTS (60' x 120' = 7,200 Sq. Ft.)	COST RANGE
Concrete court, posts, net, striping	$49,400 - $76,250
Add for lighting	15,800 - 21,700
Add for fencing	13,400 - 20,000
Add for resilient cushioned layer, per sq. ft	1.96 - 6.98
Additional paved area, per sq. ft.	6.06 - 8.84
Asphalt court (2" to 4"), per sq. ft.	5.63 - 9.36
Clay court, per sq. ft.	5.54 - 8.16
Umpire chair, each	875.00 - 1,320.00
add for roof	197.00 - 348.00
striping, cost per court	446.00 - 695.00

For detailed lighting, fencing, net costs, etc., see Section UIP 13.

PLATFORM TENNIS, prefabricated (30' x 60' = 1,800 Sq. Ft.)	
Metal or wood deck, metal supports, fencing, lights	$86,500 - $98,250
Deduct for wood under carriage and supports	6,700 - 12,800
Add for under deck heating	10,000 - 11,700
HANDBALL – RACQUETBALL COURTS	
(Prefabricated, 20' x 40' = 800 Sq. Ft.)	
Prefinished wall and ceiling panels, wood floor, lighting, complete	$65,250 - $96,750
Add for spectator viewing wall	19,300 - 103,000
Unit costs, per sq. ft. of surface, except as indicated:	
Wall panels, laminated, prefinished	$17.55 - $23.30
viewing window	77.50 - 94.50
clear glass wall	100.00 - 141.00
opaque glass wall	212.00 - 256.00
Flooring, hardwood	16.35 - 28.25
Ceiling panels	9.36 - 16.35
Lighting, per fixture, fluorescent	590.00 - 775.00
high–intensity discharge	1,000 - 1,240
BACKYARD PLAYCOURTS	
(concrete), per square foot	$31.00 - $42.00
add for interlocking plastic tile	9.78 - 14.45
SHUFFLEBOARD COURTS (concrete courts, complete, 52' x 6')	
Residential	$1,940 - $2,825
Commercial, parks and playgrounds	3,150 - 7,350
BATTING CAGES, cost per station	
School, ceiling mount canopy or ext. net, tube frame	$4,300 - $7,300
Add for motorized operation	10,400 - 12,200
Pitching machines, individual, each	3,025 - 6,400
coin op, auto feed	10,400 - 14,500
Protective screens, portable, each	540 - 725
Commercial, center automated range, concrete stations with chain link, canopy netting, lighting, complete	
Batting cages	$29,000 - $47,700
modular	51,000 - 108,000
Equipment, pitching, conveyor feed system	12,600 - 18,900
Misc. range supplies (bats, balls, etc.)	1,530 - 2,450
BACKBOARDS, per vertical square foot (exterior, excluding paving)	
Masonry	$30.25 - $39.75
Metal	30.00 - 33.50
Wood	41.00 - 55.00

NOTE: Playground equipment, see Section UIP 13. Synthetic surfaces, see Page 3.

RECREATIONAL ENCLOSURES

The following costs are average cost ranges per square foot of complete recreational shell type enclosures or buildings including the basic tennis and handball/racquetball courts as noted, but excluding swimming pools and decks. Costs include architects' fees. The cost for individual facilities can vary greatly depending upon the type of structure and its appointments, including the extent of lounge, refreshment, exercise and spectator areas, etc.

MINIMUM: Air supported (bubble) or light screened structures, of light translucent material with minimum lighting and plumbing.

	COST RANGE
Golf (60' to 80' clear height, incl. stations & turf)	$31.50 - $39.50
triangular domes	38.00 - 46.00
Tennis (28' to 40' clear height, including courts)	30.75 - 48.25
stressed membrane over metal frame	46.00 - 70.50
Swimming (enclosure only, no plumbing)	14.05 - 25.00
Swimming (screen enclosure only on light frame, residential)	18.65 - 23.80

LOW: Rigid steel frame with steel siding or wood frame with wood or steel siding, minimum sports lighting and plumbing, minimum restrooms, no dressing rooms or showers.

Tennis (20' to 28' clear height, including courts)	$64.00 - $80.50
Handball/racquetball (court enclosure only, no plumbing)	109.00 - 142.00
exterior masonry court (court enclosure only, no H.V.A.C.)	79.00 - 106.00
Swimming (translucent plastic or screened enclosure, structural frame, no plumbing)	42.75 - 73.50

AVERAGE: Rigid steel frame with insulated steel panels or masonry bearing walls, adequate sports lighting and plumbing, dressing rooms, showers.

Complete tennis (24' to 32' clear height, including courts)	$93.00 - $113.00
Handball–racquetball facilities only (including courts)	125.00 - 166.00
Swimming (good curtain walls, plastic, glass, motorized roof, no plumbing, enclosure only)	86.00 - 122.00

GOOD: Masonry bearing walls or insulated sandwich panels, good quality fixtures and lighting, lounge, snack bar, pro shop, health facilities, sauna.

Complete tennis (30' to 35' clear height, including courts)	$119.00 - $166.00
Handball/racquetball facilities only (including courts)	183.00 - 225.00
Complete swimming facilities (natatorium buildings)	136.00 - 271.00

SAUNA BATHS: Cost per square foot of floor area for prefabricated units including interior wall finish, door, heater and controls. Wall structure, exterior cover etc., are included under Interior Construction in the Segregated Cost sections or add from Unit-in-Place.

15 sq. ft.	$362 - $461	50 sq. ft.	$256 - $302
25 sq. ft.	302 - 362	75 sq. ft.	220 - 256
		100 sq. ft.	183 - 220

NOTE: For separate heater costs and steam generators, see Section UIP 3.

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GREEN ADDITIONS

SECTION UIP 15

Concrete

Concrete within the building envelope needs to come from within 500 miles and contain post consumer recycled material.

Costs Each	COM	RES
Insulated concrete, Form-in-place, Expanded polystyrene,		
Straight, 9-1/4"x 48"x 16", 4" core	48.25	47.50
Corner, 9-1/4"x 60"x 16", 4" core	58.00	57.00
Angle, 9-1/4"x 52"x 16", 4" core	56.00	55.00
Insulated concrete, Form-in-place, Expanded polystyrene,		
Straight, 11-1/4"x 48"x 16", 6" core	48.75	47.75
Corner, 11-1/4"x 56"x 16", 6" core	57.00	56.00
Angle, 11-1/4"x 44"x 16", 6" core	52.00	51.00
Tee, 11-1/4"x 44"x 16", 6" core	55.00	55.00
Insulated concrete, Form-in-place, Expanded polystyrene,		
Straight, 13-1/4"x 48"x 16", 8" core	51.00	50.00
Corner, 13-1/4"x 60"x 16", 8" core	60.00	59.00
Angle, 13-1/4"x 60"x 16", 8" core	59.00	57.00
Tee, 13-1/4"x 44"x 16", 8" core	57.00	57.00
Costs/Sq. Ft.		
Insulated concrete, Form-in-place, Reinforced, Expanded polystyrene form, 3,000 psi,		
4" thick core	18.85	18.30
3,000 psi, 6" thick core	20.95	20.45
4,000 psi, 8" thick core	23.70	23.05
Concrete countertop, Cast in place,		
Recycled materials, 1-1/2" thick	29.00	24.60
2" thick	38.50	33.00
Concrete countertop, Pre-cast, Various recycled materials,		
1-1/2" thick	241.00	250.00
Glass Chips, Various recycled materials,		
1-1/4" thick	295.00	304.00
Paper Fiber, Various recycled materials,		
1-1/8" thick	363.00	363.00
Concrete slab on grade, Cast-in-place, With welded wire fabric, Recycled content, 3,000 psi,		
4" thick	6.06	5.73
5" thick	7.56	7.22
6" thick	8.33	8.00
Concrete slab, Elevated, Cast-in-place, Recycled content, Reinforced, 3,000 psi,		
4" thick	22.95	20.10
5" thick	24.55	21.65
6" thick	25.75	22.80
Concrete slab, Elevated, Cast-in-place, Recycled content, Reinforced, 4,500 psi,		
4" thick	25.25	22.50
5" thick	26.75	23.90
6" thick	28.75	25.75
7" thick	30.00	27.00
8" thick	31.75	28.75
10" thick	34.25	31.25
12" thick	37.25	34.00

Costs/Sq. Ft.	COM	RES
Foundation wall, Cast-in-place, With damp proofing, Recycled content, 3,000 psi, Reinforced,		
6" thick	28.50	25.00
8" thick	30.50	27.00
10" thick	33.25	29.75
12" thick	34.75	31.75
Concrete wall, Cast-in-place, Recycled content, 3,000 psi, Reinforced,		
6" thick	28.25	25.00
8" thick	30.00	26.75
10" thick	32.75	29.50
12" thick	34.50	31.50
Costs/V.L.F.		
Concrete column, Cast-in-place, Recycled content, 4,500 psi, Reinforced,		
6" x 6"	68.00	58.00
8" x 8"	85.50	73.50
10" x 10"	103.00	89.50
12" x 12"	118.00	103.00
Concrete column, Cast-in-place, Recycled content, 4,500 psi, Reinforced,		
6" dia.	56.00	49.00
8" dia.	60.00	52.00
10" dia.	66.00	58.00
12" dia.	68.00	61.00
Costs/Sq. Ft.		
Wall, Adobe brick, 8" thick	37.00	33.75
12" thick	49.00	45.75
Concrete Masonry Block - 4"x8"x16"	12.15	11.05
Reinforced Lightweight	13.35	12.30
Concrete Masonry Block - 6"x8"x16"	14.10	13.05
Reinforced Lightweight	16.65	15.60
Concrete Masonry Block - 8"x8"x16"	16.45	15.15
Reinforced Lightweight	17.80	16.65
Concrete Masonry Block - 12"x8"x16"	23.65	21.80
Reinforced Lightweight	26.25	24.50
Concrete Masonry Block - 24"x8"x16"	47.50	43.75
Reinforced Lightweight	52.00	48.50
Glass block, With reinforcement,		
6"x 6"x 4" wide	90.50	89.50
8"x 8"x 4" wide	57.00	54.00
12"x 12"x 4" wide	49.00	47.75
Glass block, Reflective, With reinforcement,		
6"x 6"x 4" wide	62.00	59.00
8"x 8"x 4" wide	44.00	40.50
12"x 12"x 4" wide	49.00	47.75

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GREEN ADDITIONS

SECTION UIP 15

WOOD, PLASTIC, & COMPOSITES

WOOD: Material needs to be diverted from disposal in landfills through recycle and/or salvage and the material needs to have been salvaged, refurbished or reused within 500 miles. fifty percent of the wood-based materials and products must be certified in accordance with the Forest Stewardship councils Principles and Criteria.

PLASTIC: Plastic needs to contain post consumer recycled material and come from within 500 miles.

COMPOSITES: Composite material needs to be diverted from disposal in landfills through recycle and/or salvage and must contain post consumer recycled material within 500 miles.

Costs/Sq. Ft.	COM	RES
Wall panel, Structural insulated panel (SIP), OSB both sides,		
4’ x 8’ x 4-1/2” thick	11.50	11.70
4’ x 8’ x 6-1/2” thick	12.75	12.95
4’ x 8’ x 8-1/4” thick	14.20	14.35
4’ x 8’ x 10-1/8” thick	15.85	15.95
Roof panel, Structural insulated panel (SIP), OSB both sides,		
4’ x 8’ x 4-1/2” thick	11.60	11.75
4’ x 8’ x 6-1/2” thick	12.90	13.05
4’ x 8’ x 8-1/4” thick	14.40	14.50
4’ x 8’ x 10-1/8” thick	16.20	16.20
Costs/Lin. Ft.		
Engineered I-Joist, Recycled content, Southern pine,		
1-1/2”x 9-1/2”	5.60	5.62
1-1/2” x 11-7/8”	6.14	6.17
1-1/2” x 14”	7.00	7.05
1-3/4” x 9-1/2”	6.10	6.14
1-3/4” x 11-7/8”	6.85	6.91
1-3/4” x 14”	7.86	7.95
1-3/4” x 16”	7.51	7.58
2-5/16” x 11-7/8”	9.29	9.47
2-5/16” x 14”	10.30	10.50
2-5/16” x 16”	10.75	11.00

Costs/Sq. Ft.	COM/RES
Floor framing, SFI certified, Engineered laminated veneer lumber,	
1-3/4”x 7-1/4” joist, 16” o.c.	8.99
1-3/4”x 7-1/4” joist, 24” o.c.	6.90
1-3/4”x 9-1/2” joist, 16” o.c.	10.80
1-3/4”x 9-1/2” joist, 24” o.c.	8.25
1-3/4”x 11-7/8” joist, 16” o.c.	11.55
1-3/4”x 11-7/8” joist, 24” o.c.	9.06
3-1/2”x 9-1/2” joist, 16” o.c.	20.10
3-1/2”x 9-1/2” joist, 24” o.c.	15.15
3-1/2”x 11-7/8” joist, 16” o.c.	25.50
3-1/2”x 11-7/8” joist, 24” o.c.	19.35
5-1/4”x 9-1/2” joist, 16” o.c.	34.75
5-1/4”x 9-1/2” joist, 24” o.c.	26.25
5-1/4”x 11-7/8” joist, 16” o.c.	38.50
5-1/4”x 11-7/8” joist, 24” o.c.	29.00
Roof framing, SFI certified, Engineered laminated veneer lumber,	
1-3/4”x 7-1/4” rafter, 16” o.c.	9.10
1-3/4”x 7-1/4” rafter, 24” o.c.	6.98
1-3/4”x 9-1/2” rafter, 16” o.c.	10.90
1-3/4”x 9-1/2” rafter, 24” o.c.	8.34
1-3/4”x 11-7/8” rafter, 16” o.c.	12.05
1-3/4”x 11-7/8” rafter, 24” o.c.	9.16
3-1/2”x 9-1/2” joist, 16” o.c.	20.20
3-1/2”x 9-1/2” joist, 24” o.c.	15.25
3-1/2”x 11-7/8” joist, 16” o.c.	25.75
3-1/2”x 11-7/8” joist, 24” o.c.	19.40
5-1/4”x 9-1/2” joist, 16” o.c.	35.00
5-1/4”x 9-1/2” joist, 24” o.c.	26.25
5-1/4”x 11-7/8” joist, 16” o.c.	38.50
5-1/4”x 11-7/8” joist, 24” o.c.	29.00

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GREEN ADDITIONS

SECTION UIP 15

THERMAL & MOISTURE PROTECTION INSULATION

Thermal protection affects the thermal and acoustical comfort of building occupants. It also has significant impact on the amount of energy used to cool and heat the building. By controlling moisture at the building envelope the longevity, and therefore the sustainability, of the building will be increased. Proper moisture protection at the building envelope, is also an important step in controlling and preventing the growth of mold, which affects the indoor air quality of buildings. Materials must contain recycled content and future recyclability; have low embodied energy which reduces the initial energy investment of the product, not contain CFCs, HCFCs or other ozone depleting substances; nor contain formaldehyde; no VOCs and no off-gassing for promoting indoor air quality.

Costs/Sq. Ft.	COM	RES
Radiant barrier, 97% reflectivity, Double sided, Aluminum foil, Poly mesh reinforced,		
48” wide	0.44	0.42
16” to 24” wide	0.51	0.48
Laser perforated, 97% reflectivity, 48” wide	0.48	0.46
16” to 24” wide	0.56	0.53
Heavy weight, 98% reflectivity, Double sided,		
Aluminum, 48” wide	0.50	0.48
16” to 24” wide	0.60	0.58
Double bubble sided, 98% reflectivity,		
48” wide	0.82	0.82
16” to 24” wide	0.92	0.91
Reflective roof coating, Rubberized, Water based,		
White	0.96	0.88
Asphalt/aluminum, Silver	0.85	0.77
Polyester fabric, With primer and seal, Acrylic,		
Clear	1.67	1.58
Ice/water shield flashing at roof edge	-----	1.73

GREEN ROOF SYSTEM

Costs Each	COM/RES
Plant tray, Modular, Plastic, Polystyrene,	
8” x 8” x 2-1/2”	6.57
8” x 12” x 2-1/2”	7.07
22” x 11” x 2-1/2”	15.60
13” x 15” x 3-1/2”	17.65
Recycled content, Coconut coir fiber,	
11” x 21”	22.50

Costs/Sq. Ft.	COM/RES
Plant tray, Barrier, Root barrier, Bio degradable fabric, Trifluralin,	
24’ x 100’	7.08
Plant tray, Screen, Weed barrier, Moisture retention fabric, Micro-perforated film,	
6’ x 250’	7.03
Plant Growing Media, Hoisted up to 5 story,	
Clay, 2” thick	1.52
Clay, 4” thick	3.03
Clay, 6” thick	4.37
Clay, 8” thick	5.82
Recycled content, 2” thick	2.13
Recycled content, 4” thick	4.25
Recycled content, 6” thick	6.33
Recycled content, 8” thick	8.45
Sand and Gravel, 2” thick	1.07
Sand and Gravel, 4” thick	2.13
Sand and Gravel, 6” thick	2.98
Sand and Gravel, 8” thick	3.97
Roof paver, Walkway pads, Recycled rubber tire,	
19-1/2” x 19-1/2”	10.20
Walkway squares, Recycled rubber,	
16” x 16” x 1” thick	11.25
Costs/Lin. Ft.	
Plant tray, Edge restraint, Treated lumber, Wood,	
4” x 4”	5.10
4” x 6”	5.77

Costs/Sq. Ft.	COM	RES
Insulation, Batt, Foil-faced, Non-rigid,		
Walls, or Floor, 3-1/2” Thick, R11	1.21	1.16
Insulation, Blanket, Unfaced, Non-rigid,		
Walls or ceiling, 3-1/2”, R11	1.31	1.25
Insulation, Blown-in, Wall, Cellulose,		
3.5” thick, R-13	1.51	1.45
Insulation, Blown-in, Wall, Fiberglass,		
3.5” thick, R-9	1.63	1.58
Insulation, Blown-in, Ceiling, Cellulose,		
6” thick, R-22	1.16	1.13
Insulation, Blown-in, Ceiling, Fiberglass,		
6” thick, R-13	0.98	0.93
Insulation, Roof, Straw, Wire tied, 2-wire,		
Rice, 4.0 Lbs./Cu. Ft., 15” thick	0.82	0.80
Insulation, Wall, Straw, Wire tied, 2-wire,		
Rice, 4.0 Lbs./Cu. Ft., 15” thick	0.63	0.60

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GREEN ADDITIONS

SECTION UIP 15

DOOR & WINDOW OPENINGS

Openings can contribute Green credits by meeting the Minimum Energy Performance, Daylight & Views, Material Reuse and Regional Material requirements. Openings are considered part of the building envelope system design and can contribute to the Minimum Energy Performance. The Daylight requirement specifies a minimum glazing factor for a specific percentage of all regularly occupied areas. The Views requirement specifies direct line of sight for a specific percentage of all regularly occupied areas. All materials are to be extracted, processed and manufactured within 500 mile of the building site. Opening components must also be NFRC certified and contribute to the reduction in energy to be considered Green.

Costs Each	COM/RES
Skylight, Solar tube kit, With 4' flexible tube, Acrylic dome,	
10" diameter	680.00
14" diameter	770.00
21" diameter	1010.00
Skylight, Solar tube kit, With 4' rigid tube and fresnel diffuser, Acrylic dome,	
10" diameter	730.00
14" diameter	945.00
Skylight, Solar tube kit, With 6' rigid tube and fresnel ceiling diffuser, Acrylic dome, 21" diameter	1250.00
Costs/Lin. Ft.	
Weatherstripping, Door, Vinyl gasket	4.61
Weatherstripping, Window, Rubber, 3/4" wide, 7/16" thick	4.11

Costs/Sq. Ft.	COM	RES
Translucent wall panel, With aluminum frame, Triple wall, 48" wide x 5/8" thick polycarbonate panels, Clear, 8' high	35.00	34.25
48" wide x 5/8" thick polycarbonate panels, White, 8' high	35.50	35.00
6-wall, 48" wide x 1" thick polycarbonate panels, Clear, 8' high	39.50	39.00
6-wall, 48" wide x 1" thick polycarbonate panels, White, 8' high	40.50	40.00
Skyroof, Atrium, Curb mounted, Single pitch, With aluminum frame, 5/8" clear triple wall polycarbonate, 8'-0" wide	48.25	47.75
5/8" white triple wall polycarbonate, 8'-0" wide	49.00	48.50
1" clear 6-wall polycarbonate, 8'-0" wide	53.00	53.00
1" white 6-wall polycarbonate, 8'-0" wide	54.00	54.00
Skyroof, Atrium, Curb mounted, Double pitch, With aluminum frame, 5/8" clear triple wall polycarbonate, 16'-0" wide	50.00	49.00
5/8" white triple wall polycarbonate, 8'-0" wide	50.00	50.00
1" clear 6-wall polycarbonate, 8'-0" wide	55.00	55.00
1" white 6-wall polycarbonate, 8'-0" wide	56.00	57.00

FINISHES

The Low-Emitting Materials requirement limits the amount of VOC's that can be out gassed for specific product and the use of urea-formaldehyde resins. All materials are to be extracted, processed and manufactured within 500 mile of the building site. Opening components must also be NFRC certified and contribute to the reduction in energy to be considered Green.

Costs/Sq. Ft.	COM/RES
Sealer, Concrete floor, Semi-transparent, Acrylic, Colored, Semi-gloss finish, 2 coats, Spray	1.35
Paint, Floor, Interior, Concrete, Low VOC, 1 coat finish, Sprayed	0.48
Paint, Floor, Interior, Metal, Low VOC, Primer and finish coat, Sprayed	0.97
Paint, Wall, Interior, Low VOC latex paint, Average, Brush/Roller/Spray, 2 Finish coats, Flatwall	0.71
2 finish coats, Semi-gloss enamel	0.76
Paint, Ceiling, Interior, Low VOC latex paint, Average, Brush/Roller/Spray, 2 Finish coats, Flatwall	0.71
2 finish coats, Semi-gloss enamel	0.76
Paint, Wall, Interior, Zero VOC latex paint, Average, Brush/Roller/Spray, Primer and Finish coat, Flatwall	0.76
Primer and Finish coat, Semi-gloss enamel	0.82
Paint, Ceiling, Interior, Zero VOC latex paint, Average, Brush/Roller/Spray, Primer and Finish coat, Flatwall	0.82
Primer and Finish coat, Semi-gloss enamel	0.87
Wood finish, Low VOC, Polyurethane, Clear, 1 coat, Brushed	1.08
Wood finish, Low VOC, Lacquer, Clear, 3 coat, Brushed	2.10
Metal stud framing, Recycled Content, Non-load bearing, Galvanized, 25 Ga,	
1-5/8" wide, 16" o.c.	2.71
1-5/8" wide, 24" o.c.	2.02
2-1/2" wide, 16" o.c.	3.19
2-1/2" wide, 24" o.c.	2.43
4" wide, 16" o.c.	3.75
4" wide, 24" o.c.	2.85
6" wide, 16" o.c.	3.73
6" wide, 24" o.c.	2.84

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GREEN ADDITIONS

SECTION UIP 15

SPECIALTIES

This area is very broad and encompasses many items. Material or building components typically addressed in this section would be specialty items such as Green Power generation equipment, rain water harvesting, signage, bathroom hand dryers and site features to control heat island effects.

Costs Each	COM	RES
Pellet stove, Freestanding, With glass door and 3” direct vent, 40,000 Btu, Heavy steel, 24”x 21”x 20” H	-----	5250.00
45,000 Btu, Heavy steel, 22”x 27”x 29” H	-----	5800.00
With ceramic glass door, 50,000 Btu, Heavy steel, 26”x 31”x 33-1/4” H	-----	6850.00
Fireplace insert, Wood burning, With cast iron glass door, blower and 6” dia. flue liner, 69,600 BTU, Steel, Black finish	-----	3400.00
With steel glass bay door and blower, 70,000 BTU, Steel, Black finish	-----	6200.00
Automatic high efficient hand dryer, Surface mounted, With infra red sensor, High air flow, 120 V, Metal cover, Enamel finish, 11-3/4” x 12-3/4” x 6-3/4” D	810.00	-----
Satin stainless finish, 11-3/4” x 12-3/4” x 6-3/4” D	970.00	-----
Exit sign, Door mounted, Photoluminescent, Rigid plastic, 4” x 6” x 1/8” thick	43.00	39.25
Exit egress sign, With directional graphic, Photoluminescent, Rigid plastic, 8” x 4-1/2” x 5/8” thick	63.00	58.00
Egress stair signage, Wall mounted, Projecting, Double sided, With graphic, Aluminum, Photoluminescent, 6” x 5” x 3-1/2” projection	68.50	63.00
Exit sign, Photoluminescent, 15” x 8-1/2” x 1” thick	133.00	128.00
With directional arrows, 15” x 8-1/2” x 1” thick	152.00	147.00
8” lettering, Plastic, 17” x 10” x 1/2” thick	206.00	201.00
With directional arrows, Metal frame, 15-1/4” x 8-1/4” x 1” thick	241.00	236.00
8” lettering, Plastic, 17” x 10” x 1/2” thick	295.00	290.00
100 ft visibility, With directional arrows, 16” x 8-5/8” x 1/2” thick	329.00	324.00
Ceiling mounted, Double sided, With directional arrows, Metal frame, Photoluminescent, 15-1/4” x 8-1/4” x 1” thick	437.00	432.00

GLAZING/WINDOW SHADING

Costs Each	COM	RES
Awning, Fixed, Aluminum, Baked enamel finish, 2’ projection, 3’ wide	-----	620.00
Awning, Fixed, Aluminum, Baked enamel finish, 3’ projection, 6’ wide	-----	795.00
Costs/Sq. Ft.		
Awning, Retractable, Manual, Standard vinyl, 8’ projection, Up to 10’ wide.	-----	39.00
Custom acrylic fabric, 8’ projection, Up to 10’ wide	-----	41.25
Awning, Retractable, Motorized, Standard vinyl, 8’ projection, Up to 10’ wide.	-----	47.75
Custom acrylic fabric, 8’ projection, Up to 10’ wide	-----	50.00

SPECIAL CONSTRUCTION

Costs Each	COM/RES
Solar hydronic panel, Residential, Roof mounted, With 50 gal tank, photovoltaic pump and piping, Flat lexan/anodized aluminum frame panel, 20” x 144”	1900.00
Solar hydronic panel, Flat roof tilt mounted, With 52 gal steel tank, single heat exchanger, pump and piping, Flat glass/anodized aluminum frame panel, 88” x 96”	6350.00
Solar panel array. Photovoltaic, Grid-tied system, Flush mounted rack, Sloped roof, 2 kWH capacity	10700.00
Solar panel array. Photovoltaic, Grid-tied system, Flush mounted rack, Sloped roof, 4 kWH capacity	19600.00
Solar water heater, System including, Collectors/Tank/ Controller/Pump/Valves/Piping/Insulation/etc., 120 gallon	14900.00
Solar water heater, System including, Collectors/Tank/ Controller/Pump/Valves/Piping/Insulation/etc., 80 gallon	11700.00
Solar panel array. Photovoltaic, Grid-tied system, Flush mounted rack, Sloped roof, 6 kWH capacity	26200.00
Solar panel array. Photovoltaic, Grid-tied system, Flush mounted rack, Sloped roof, 8 kWH capacity	33300.00
Solar panel array. Photovoltaic, Grid-tied system, Low tilt (30deg) mounted rack, Flat roof, 10 kWh capacity	50250.00
Solar panel array. Photovoltaic, Grid-tied system, Low tilt (30deg) mounted rack, Flat roof, 15 kWh capacity	72000.00
Solar panel array. Photovoltaic, Grid-tied system, Low tilt (30deg) mounted rack, Flat roof, 50 kWh capacity	201000.00
Solar panel array. Photovoltaic, Grid-tied system, Low tilt (30deg) mounted rack, Flat roof, 75 kWh capacity	293000.00
Roofing, Solar photovoltaic, Tile, Slate, Grid-tied system, 2kWh capacity	25400.00

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GREEN ADDITIONS

SECTION UIP 15

SPECIAL CONSTRUCTION (CONT.)

Costs Each	COM/RES
Roofing, Solar photovoltaic, Shingles,	
Grid-tied system, 2kWh capacity	15900.00
Roofing, Solar photovoltaic, Shingles,	
Grid-tied system, 4kWh capacity	30200.00
Wind turbine generator, Off-grid,	
With 33' guyed pole, 9' rotor blade,	
24/36/48 VDC, 1.0 Kwatt	26500.00
Grid-tied, With inverter and 45' tower,	
12' rotor blade, 120/240V,	
1.8 Kwatt	36900.00
Electric backup power storage,	
With inverter and steel enclosure,	
8-200aH 6V batteries,	
120VAC, 9.6 KwH	13800.00
Electric backup power storage, Add-on	
with steel enclosure, 8-200aH 6V batteries,	
240VAC, 9.6 KwH	5800.00
Wind turbine generator, Grid-tied,	
With inverter and 60' tower, 12' rotor blade,	
120/240V, 2.4 Kwatt	79750.00
Electric backup power storage,	
With inverter and steel enclosure,	
4-200aH 6V batteries,	
120VAC, 4.8 KwH	11900.00
12-200aH 6V batteries,	
120VAC, 14.4 KwH	14400.00
12-200aH 6V batteries,	
240VAC, 14.4 KwH	8950.00

PLUMBING & FIXTURES

Plumbing has a very broad impact on Green credits primarily covering the efficient management, use and reuse of water and energy use. Typically items covered may include Water Efficient Landscaping, Innovative Waste Water Technologies, Water Use Reduction, Minimum Energy Performance, Recycled Content, On-site Renewable Energy and Fundamental Refrigerant Management.

Costs Each	COM	RES
Reverse Osmosis/Deionization water purification system,		
3 stage, 24 gpd	270.00	250.00
3 stage, 36 gpd	275.00	255.00
3 stage, 50 gpd	285.00	265.00
3 stage, 75 gpd	290.00	270.00
3 stage, 100 gpd	373.00	358.00
4 stage, 75 gpd	462.00	447.00
5 stage, 50 gpd	605.00	590.00
5 stage, 75 gpd	620.00	605.00
6 stage, 50 gpd	650.00	635.00
6 stage, 75 gpd	660.00	645.00
Greywater system, For water closet, With pump and filter,		
5-1/2 gallon tank, 12VDC	-----	790.00
Greywater tank, Residential, Up to 3 persons,		
39 gallons, 22-1/2" dia. x 46"H	-----	5350.00
Up to 6 persons, 66 gallon tank,		
22-1/2" dia. x 60"H	-----	5950.00
Up to 9 persons, 92 gallon tank,		
30-1/2" dia. x 53"H	-----	6750.00
Up to 12 persons, 119 gallon tank,		
30-1/2" dia. x 60"H	-----	7200.00

PLUMBING & FIXTURES

Costs Each	COM	RES
Atmospheric water system, Generator,		
110v 50Hz, 3.17 gallon capacity	-----	2130.00
3.6 gallon capacity	-----	2500.00
Faucet aerator, Kitchen, 2 spray setting, With pause valve,		
Chrome finish, 1.5 GPM	-----	26.25
Showerhead, Shutoff valve, Brass,		
Chrome finish	-----	30.00
Costs/Lin. Ft.	**COM**	**RES**
Insulation, Pipe, Fiberglass, P/c with ssl,		
1/2"o.d. x 1/2"wall	8.33	7.26
3/4"o.d. x 1/2"wall	9.13	8.04
1"o.d. x 1/2"wall	9.51	8.37
1-1/4"o.d. x 1/2"wall	9.97	8.78
1-1/2"o.d. x 1/2"wall	10.45	9.24
2"o.d. x 1/2"wall	11.15	9.82
2-1/2"o.d. x 1/2"wall	12.10	10.70
1/2"o.d. x 1"wall	8.80	7.69
3/4"o.d. x 1"wall	9.80	8.68
1"o.d. x 1"wall	10.35	9.22
1-1/4"o.d. x 1"wall	11.70	10.50
1-1/2"o.d. x 1"wall	12.65	11.55
2"o.d. x 1"wall	13.40	12.35
2-1/2"o.d. x 1"wall	14.35	13.20
3"o.d. x 1"wall	15.00	-----
4"o.d. x 1"wall	16.00	-----
1/2"o.d. x 1-1/2"wall	11.40	10.35
3/4"o.d. x 1-1/2"wall	12.50	11.50
1"o.d. x 1-1/2"wall	13.05	12.00
1-1/4"o.d. x 1-1/2"wall	13.75	12.65
1-1/2"o.d. x 1-1/2"wall	14.50	13.35
2"o.d. x 1-1/2"wall	15.30	14.10
2-1/2"o.d. x 1-1/2"wall	16.05	14.80
3"o.d. x 1-1/2"wall	16.85	-----
4"o.d. x 1-1/2"wall	18.30	-----
1/2"o.d. x 2"wall	13.95	12.95
3/4"o.d. x 2"wall	15.15	14.20
1"o.d. x 2"wall	15.85	14.85
1-1/4"o.d. x 2"wall	16.50	15.48
1-1/2"o.d. x 2"wall	17.60	16.47
2"o.d. x 2"wall	18.70	17.54
2-1/2"o.d. x 2"wall	19.85	18.60
3"o.d. x 2"wall	21.40	-----
4"o.d. x 2"wall	23.35	-----
Insulation, Pipe, Rubber, Closed cell foam,		
3/8"o.d. x 3/8"wall	7.98	6.73
1/2"o.d. x 3/8"wall	8.44	7.14
5/8"o.d. x 3/8"wall	8.76	7.44
3/4"o.d. x 3/8"wall	8.97	7.63
7/8"o.d. x 3/8"wall	9.29	7.91
1-1/8"o.d. x 3/8"wall	9.74	8.33
1-3/8"o.d. x 3/8"wall	10.10	8.70
1-5/8"o.d. x 3/8"wall	10.75	9.33
2-1/8"o.d. x 3/8"wall	11.80	10.35
3/8"o.d. x 1/2"wall	8.89	7.60
1/2"o.d. x 1/2"wall	9.29	8.00
5/8"o.d. x 1/2"wall	9.58	8.28
3/4"o.d. x 1/2"wall	9.74	8.43
7/8"o.d. x 1/2"wall	10.00	8.68
1-1/8"o.d. x 1/2"wall	10.50	9.16
1-3/8"o.d. x 1/2"wall	11.05	9.63
1-5/8"o.d. x 1/2"wall	11.75	10.30

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GREEN ADDITIONS

SECTION UIP 15

PLUMBING & FIXTURES

Costs/Lin. Ft.	COM	RES
Insulation, Pipe, Rubber, Closed cell foam,		
2-1/8"o.d.x1/2"wall	13.20	11.75
2-3/8"o.d.x1/2"wall	14.55	13.00
2-5/8"o.d.x1/2"wall	15.35	13.85
3-1/8"o.d.x1/2"wall	16.95	15.40
3-1/2"o.d.x1/2"wall	17.75	16.15
4-1/8"o.d.x1/2"wall	21.20	19.65
4-1/2.d.x1/2"wall	22.30	20.70
5/8"o.d.x3/4"wall	10.65	9.34
7/8"o.d.x3/4"wall	11.55	10.20
1-1/8"o.d.x3/4"wall	12.90	11.60
1-3/8"o.d.x3/4"wall	13.80	12.45
1-5/8"o.d.x3/4"wall	15.85	14.60
2-1/8"o.d.x3/4"wall	18.00	16.70
5/8"o.d.x1"wall	13.75	12.50
7/8"o.d.x1"wall	16.00	14.85
1-1/8"o.d.x1"wall	17.75	16.65
1-3/8"o.d.x1"wall	19.45	18.30
1-5/8"o.d.x1"wall	21.00	19.95
2-1/8"o.d.x1"wall	24.20	23.15

Costs Each	COM	RES
Pipe fitting cover, Insulated, Pvc, Ips, 90 degree elbow, #2,		
3/8"o.d.x1/2"wall	12.60	10.80
1/2"o.d.x1/2"wall	12.95	11.10
3"o.d.x1/2"wall	19.55	17.45
3-1/2"o.d.x1/2"wall	20.75	18.60
4"o.d.x1/2"wall	22.65	20.45
Pipe fitting cover, Insulated, Pvc, Ips, 45 degree elbow, #2,		
3/8"o.d.x1/2"wall	12.60	10.80
1/2"o.d.x1/2"wall	12.95	11.10
3"o.d.x1/2"wall	19.55	17.45
3-1/2"o.d.x1/2"wall	20.75	18.60
4"o.d.x1/2"wall	22.65	20.45
Pipe fitting cover, Insulated, Pvc, Ips, 45 or 90 degree elbow, #4,		
3/4"o.d.x1/2"wall	13.40	11.50
1"o.d.x1/2"wall	13.70	11.75
1-1/4"o.d.x1/2"wall	14.25	12.25
1-1/2"o.d.x1/2"wall	14.75	12.70
2"o.d.x1/2"wall	16.05	14.00
2-1/2"o.d.x1/2"wall	17.70	15.60
5"o.d.x1/2"wall	31.00	29.00
6"o.d.x1/2"wall	36.75	34.75
7"o.d.x1/2"wall	41.50	39.50
8"o.d.x1/2"wall	51.00	50.00
9"o.d.x1/2"wall	59.00	57.00
10"o.d.x1/2"wall	65.00	63.00
11"o.d.x1/2"wall	79.50	78.50
Pipe fitting cover, Insulated, Pvc, Ips, Tee, #2,		
3/8"o.d.x1/2"wall	18.90	16.15
1/2"o.d.x1/2"wall	19.15	16.40
3/4"o.d.x1/2"wall	20.05	17.30
1"o.d.x1/2"wall	20.45	17.70
1-1/4"o.d.x1/2"wall	21.95	19.10
1-1/2"o.d.x1/2"wall	22.35	19.50
2"o.d.x1/2"wall	24.15	21.25
2-1/2"o.d.x1/2"wall	26.50	23.55
3"o.d.x1/2"wall	28.75	25.75
3-1/2"o.d.x1/2"wall	30.75	27.75
4"o.d.x1/2"wall	34.75	32.00
5"o.d.x1/2"wall	44.25	41.50

Costs Each	COM	RES
Pipe fitting cover, Insulated, Pvc, Ips, Tee, #2,		
6"o.d.x1/2"wall	68.00	66.00
7"o.d.x1/2"wall	73.50	71.50
8"o.d.x1/2"wall	93.50	91.50
9"o.d.x1/2"wall	103.00	98.00
10"o.d.x1/2"wall	108.00	108.00
11"o.d.x1/2"wall	123.00	123.00
Pipe fitting cover, Insulated, Pvc, Ips, Mechanical groove, 90 degree elbow,		
2"o.d.	35.00	34.00
2-1/2"o.d.	36.50	35.50
3"o.d.	43.50	42.50
4"o.d.	52.00	51.00
5"o.d.	61.00	60.00
6"o.d.	74.50	74.50
8"o.d.	93.50	93.50
10"o.d.	123.00	123.00
12"o.d.	-----	196.00
Pipe fitting cover, Insulated, Pvc, Ips, Mechanical groove, 45 degree elbow, #9,		
2"o.d.	34.00	32.75
2-1/2"o.d.	34.50	33.50
3"o.d.	42.25	41.25
4"o.d.	50.00	50.00
5"o.d.	59.00	58.00
6"o.d.	71.50	70.50
8"o.d.	88.50	89.50
10"o.d.	113.00	113.00
12"o.d.	191.00	162.00
Pipe fitting cover, Insulated, Pvc, Ips, Mechanical groove, Tee,		
2"o.d.	56.00	54.00
2-1/2"o.d.	59.00	58.00
3"o.d.	67.00	65.00
4"o.d.	88.50	88.50
5"o.d.	103.00	98.00
6"o.d.	123.00	123.00
8"o.d.	142.00	142.00
10"o.d.	177.00	182.00
12"o.d.	221.00	226.00

Costs/Lin. Ft.	COM	RES
Pipe cover, Jacket, Pvc, Ips, Cc with ssl,		
1"o.d.	8.71	7.95
2"o.d.	9.69	8.86
3"o.d.	10.75	9.81
4"o.d.	12.20	11.15
1-1/2"o.d.	9.17	8.40
2-1/2"o.d.	10.10	9.22
3-1/2"o.d.	11.40	10.40
4-1/2"o.d.	12.75	11.75
4-3/4"o.d.	13.55	12.45
5-1/2"o.d.	13.90	12.75
5-3/4"o.d.	14.45	13.25
6-1/2"o.d.	15.15	14.00
6-3/4"o.d.	16.05	14.85
7-1/4"o.d.	17.25	15.90
8-1/2"o.d.	17.85	16.50
8-3/4"o.d.	18.70	17.30

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GREEN ADDITIONS

SECTION UIP 15

PLUMBING & FIXTURES

Costs/Lin. Ft.	COM	RES
Pipe cover, Jacket, Aluminum, Ips, Cc with ssl, Smooth, 16mil,		
1-1/2"o.d.	9.82	8.43
2"o.d.	10.50	9.07
2-1/2"o.d.	11.05	9.55
3"o.d.	11.60	10.05
3-1/2"o.d.	12.25	10.65
4"o.d.	12.85	11.25
4-1/2"o.d.	13.65	12.05
5-1/2"o.d.	14.80	13.05
7"o.d.	16.40	14.55
8"o.d.	17.55	15.55
9"o.d.	18.75	16.70
10"o.d.	19.95	17.75
11"o.d.	21.05	18.85
12"o.d.	22.45	20.10
13"o.d.	23.65	21.25
15"o.d.	25.50	22.85
15-1/2"o.d.	26.50	23.85
Pipe cover, Jacket, Aluminum, Ips, Cc with ssl, Stucco embossed, 16mil,		
1-1/2"o.d.	9.87	8.49
2"o.d.	10.60	9.17
2-1/2"o.d.	11.10	9.59
3"o.d.	11.75	10.20
3-1/2"o.d.	12.30	10.75
4"o.d.	13.00	11.40
4-1/2"o.d.	13.85	12.20
5-1/2"o.d.	15.00	13.25
7"o.d.	16.65	14.75
8"o.d.	17.75	15.80
9"o.d.	19.00	17.00
10"o.d.	20.25	18.05
11"o.d.	21.35	19.15
12"o.d.	22.80	20.45
13"o.d.	24.00	21.60
15"o.d.	25.75	23.20
15-1/2"o.d.	26.75	24.25
Pipe cover, Jacket, Aluminum, Ips, Cc with ssl, Smooth, 20mil,		
1-1/2"o.d.	10.25	8.91
2"o.d.	11.05	9.63
2-1/2"o.d.	11.65	10.15
3"o.d.	12.30	10.75
3-1/2"o.d.	12.90	11.40
4"o.d.	13.65	12.10
4-1/2"o.d.	14.60	13.00
5-1/2"o.d.	15.85	14.20
7"o.d.	17.65	15.80
8"o.d.	18.85	16.95
9"o.d.	20.25	18.20
10"o.d.	21.55	19.45
11"o.d.	22.80	20.60
12"o.d.	24.30	22.10
13"o.d.	25.75	23.35
15"o.d.	27.50	25.00
15-1/2"o.d.	28.75	26.25

Costs/Lin. Ft.	COM	RES
Pipe cover, Jacket, Aluminum, Ips, Cc with ssl, Stucco embossed, 20mil, 1-1/2”o.d.	10.35	8.99
2"o.d.	11.15	9.76
2-1/2"o.d.	11.75	10.30
3"o.d.	12.40	10.90
3-1/2"o.d.	13.05	11.55
4"o.d.	13.80	12.25
4-1/2"o.d.	14.80	13.20
5-1/2"o.d.	16.05	14.40
7"o.d.	17.80	16.00
8"o.d.	19.10	17.20
9"o.d.	20.45	18.50
10"o.d.	21.85	19.80
11"o.d.	23.10	20.95
12"o.d.	24.55	22.45
13"o.d.	26.00	23.75
15"o.d.	28.00	25.50
15-1/2"o.d.	29.25	26.75
Costs Each		
Pipe fitting cover, Insulated, Aluminum, Ips, 90 degree elbow,		
1-1/2"o.d.	18.00	16.25
2"o.d.	19.00	17.25
2-1/2"o.d.	20.70	18.95
3"o.d.	21.45	19.70
3-1/2"o.d.	23.45	21.70
4"o.d.	25.75	24.05
4-1/2"o.d.	29.50	27.75
5-1/2"o.d.	37.00	35.75
7"o.d.	41.25	39.75
8"o.d.	44.75	43.25
9"o.d.	55.00	54.00
10"o.d.	57.00	55.00
11"o.d.	65.00	64.00
12"o.d.	81.50	81.50
13"o.d.	84.50	84.50
13-1/2"o.d.	98.00	98.00
14"o.d.	103.00	103.00
Pipe fitting cover, Insulated, Aluminum, Ips, 45 degree elbow,		
1-1/2"o.d.	17.75	16.00
2"o.d.	18.70	16.95
2-1/2"o.d.	19.60	17.80
3"o.d.	21.15	19.35
3-1/2"o.d.	22.05	20.25
4"o.d.	23.30	21.45
4-1/2"o.d.	25.75	23.95
5-1/2"o.d.	32.00	30.25
7"o.d.	35.00	33.25
8"o.d.	40.00	38.50
9"o.d.	45.25	43.75
10"o.d.	53.00	51.00
11"o.d.	58.00	57.00
12"o.d.	65.00	64.00
13"o.d.	72.50	70.50
13-1/2"o.d.	88.50	87.50
14"o.d.	95.50	94.50

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GREEN ADDITIONS

SECTION UIP 15

PLUMBING & FIXTURES

Costs Each	COM	RES
Pipe fitting cover, Insulated, Aluminum, Ips, Tee,		
1-1/2"o.d.	47.75	46.50
2"o.d.	48.50	47.00
2-1/2"o.d.	49.00	47.50
3"o.d.	50.00	48.00
3-1/2"o.d.	50.00	48.50
4"o.d.	51.00	49.00
4-1/2"o.d.	53.00	51.00
5-1/2"o.d.	60.00	58.00
7"o.d.	62.00	61.00
8"o.d.	64.00	63.00
9"o.d.	71.50	70.50
10"o.d.	74.50	72.50
11"o.d.	76.50	74.50
12"o.d.	79.50	77.50
13"o.d.	82.50	80.50
13-1/2"o.d.	93.50	91.50
14"o.d.	95.50	93.50
Costs/Lin. Ft.		
Insulation, Pipe, Polyethylene, P/c with ssl,		
3/8"o.d.x3/8"wall	7.32	6.16
1/2"o.d.x3/8"wall	7.68	6.48
3/4"o.d.x3/8"wall	8.14	6.90
1"o.d.x3/8"wall	8.63	7.36
1-1/4"o.d.x3/8"wall	9.15	7.85
1-3/4"o.d.x3/8"wall	10.15	8.79
1/4"o.d.x1/2"wall	7.84	6.67
3/8"o.d.x1/2"wall	8.19	6.98
1/2"o.d.x1/2"wall	8.64	7.39
3/4"o.d.x1/2"wall	9.13	7.85
1"o.d.x1/2"wall	9.62	8.30
1-3/4"o.d.x1/2"wall	10.25	8.89
1-1/4"o.d.x1/2"wall	11.55	10.10
2"o.d.x1/2"wall	12.05	10.55
2-1/4"o.d.x1/2"wall	12.75	11.30
2-3/4"o.d.x1/2"wall	14.35	12.80
3/8"o.d.x3/4"wall	9.29	8.09
1/2"o.d.x3/4"wall	9.87	8.68
3/4"o.d.x3/4"wall	10.75	9.54
1"o.d.x3/4"wall	11.55	10.30
1-1/4"o.d.x3/4"wall	13.10	11.90
1-3/4"o.d.x3/4"wall	15.10	13.90
Pipe Insulation - Fiberglass Wrap,		
1/2"pipe	6.68	5.44
3/4"pipe	7.07	5.79
1"pipe	7.46	6.14
1-1/2"pipe	8.07	6.68

HVAC

This area focuses on the efficient use of energy for habitant comfort as it pertains to climate control of a building. Typically items covered may include Minimum Energy Performance, On-site Renewable Energy and Fundamental Refrigerant Management.

Costs Each	COM	RES
High efficiency gas furnace, Forced air, With single stage burner, 92% AFUE, 45 MBH	-----	2575.00
With 2 stage burner, 95% AFUE,70 MBH	-----	2750.00
With 2 stage burner, Variable speed blower, 95% AFUE, 115 MBH	-----	4575.00
Heat pump, Geothermal, Forced air, Heat and cooling,		
2.5 tons	-----	8200.00
3 tons	-----	8350.00
4 tons	9250.00	9000.00
5 tons	9800.00	9550.00
Heat pump, Geothermal, Hydronic, Heat and cooling,		
2.5 tons	-----	10100.00
3 tons	-----	10300.00
4 tons	12400.00	12100.00
5 tons	14000.00	13600.00
6 tons	16800.00	16300.00
Heat pump, Geothermal, Hybrid, Hydronic and forced air, Heating and cooling,		
3 tons	-----	14900.00
4 tons	17100.00	16800.00
5 tons	18900.00	18400.00
6 tons	21600.00	21000.00
Geothermal, Horizontal ground loop system, With tubing and heat transfer fluid,		
3 tons	-----	8450.00
4 tons	10600.00	10500.00
5 tons	13000.00	12900.00
6 tons	14900.00	14700.00
Costs/Sq. Ft.		
Radiant floor, Sub-floor, With heat transfer plates, Circulating pump, Brass manifolds,		
3/4” id BPex tubing at 8’’ o.c.	-----	9.16
With aluminum covered grooved OSB panels,		
1/2” id BPex tubing at 8’’ o.c.	-----	10.75
Embedded, Light weight concrete,		
3/4” id BPex tubing, 12’’ o.c.	-----	10.75
4” concrete, 2” polystyrene insulation,		
3/4” id BPex tubing, 12’’ o.c.	-----	13.80
Costs Each		
Heat recovery ventilator, 4 port, 56 Max CFM,		
18-1/2”x 17-1/4”x 10-1/2”	-----	1420.00
4 port, 160 Max CFM, 3 speed,		
28”x 17-1/2”x 25-1/2”	-----	2140.00
4 port, 240 Max CFM, 3 speed,		
28”x 20-1/2”x 25-1/2”	-----	3575.00
Combination exhaust fan, With 2- 18W compact fluorescent bulbs, Ceiling mounted, Low profile housing,		
80 CFM, 25 Watts, 13” x 13”	-----	275.00
80 CFM, 25 Watts	-----	304.00
110 CFM, 36 Watts	-----	329.00
Combination exhaust fan, With 26W compact fluorescent bulb, Ceiling mounted, Perimeter grill,		
110 CFM, 13” x 13”	-----	373.00
150 CFM	-----	491.00

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GREEN ADDITIONS

SECTION UIP 15

HVAC (CONT.)

Costs Each	COM	RES
Fan, Exhaust, Ceiling mounted, Wall timer, Low profile housing, With 2- 18W compact fluorescent bulbs, 80 CFM, 25 Watts, 13” x 13”	-----	339.00
Fan, Exhaust, Ceiling mounted, Humidity sensor, 130CFM, 90 Watts.	-----	265.00
Attic fan, Roof mounted, Solar powered, With thermostat switch,		
10 watt, 850 CFM	-----	820.00
1200 CFM, 20 Watts	-----	1170.00
Attic fan, Gable mounted, Solar powered, With thermostat switch,		
10 watt, 850 CFM	-----	765.00
1275 CFM, 20 Watts	-----	1100.00

ELECTRICAL LIGHTING

Costs Each	COM	RES
Floodlight, Exterior solar powered, Motion/Photo-cell activated, With photovoltaic panel and 4V battery, 2 - 3W LED conical heads, Cast aluminum housing,		
8” x 9” x 6-3/4”H	-----	157.00
6V battery, Plastic housing, 80 LED,		
7-7/8” x 7-3/4” x 8-5/8”H	-----	211.00
6V battery, Plastic housing, 30 watt,		
8”x 7-3/4”x 9”H	-----	265.00
Light fixture, Wall mounted, Automatic dusk-to-dawn, With photovoltaic module and battery, 8 LEDs, Aluminum housing, Beveled glass,		
6-7/8”x 13”H	-----	280.00
Light fixture, Ceiling mounted, Vandal resistant, Motion sensor, With photovoltaic module and battery, 11 watt compact fluorescent, Aluminum housing,		
8”x 8”x 8”H	-----	1030.00

EXTERIOR IMPROVEMENTS

Exterior improvements are items typically independent of the building itself, such as equipment associated with Public Transportation Access, Bicycle Storage, Restoration of Habitat, Stormwater Quantity & Quality Control, Non-Roof heat Island Effect and Water Efficient Landscaping.

PERMEABLE/POROUS PAVING

Costs/Sq. Ft.	COM	RES
Paving, Reclaimed Brick, With 2” sand bed,		
4”x 8”x 2-1/4”	-----	18.05
Herringbone pattern	24.40	21.85
Paving, Reclaimed, Granite, With 2” sand bed,		
Random, 4”x 6” to 6”x 10”	27.00	25.00
Herringbone pattern	31.75	29.00
Permeable pavers, interlocking, 2”gravel base and fill,		
Precast concrete	18.85	17.30
L-shaped, precast concrete	16.90	15.65
Turf block, 8 open cells,		
Precast concrete	13.40	12.30

EXTERIOR IMPROVEMENTS

WALK/ROAD/PARKING APPURTENANCES

Costs Each	COM	RES
Bollard, In ground, Round, Hollow, Galvanized,		
Steel, 5-1/2” dia. x 72”long	466.00	-----
Recycled plastic,		
6” square x 33” high	152.00	-----
Bollard, Sphere, Concrete,		
18” diameter	670.00	-----
24” diameter	1100.00	-----
30” diameter	1470.00	-----
Barrier, Delineator, Plastic,		
21” high x 60”long	295.00	-----
Recycled plastic, HDPE,		
8’ high x 8’ long	378.00	-----
Bicycle rack, Powder coat, Galvanized steel,		
7 bike	447.00	-----
9 bike	575.00	-----
Double powder coat, Galvanized steel,		
9 bike	1050.00	-----
Thermo plastic covered, Galvanized steel,		
2 bike	221.00	-----
10 bike	2650.00	-----
Recycled plastic, 10 bike	1320.00	-----
Wave design, 12 ga.steel, 1-5/8” diameter,		
5 bike	580.00	-----

DRIP IRRIGATION

Costs Each	COM	RES
Drip irrigation, Micro sprayer kit, Small lawn,		
Up to 450 sf	201.00	187.00
Medium lawn, Up to 675 sf	285.00	260.00
Large lawn, Up to 900 sf	412.00	388.00
System timer, Hose end, Electronic,		
Single station	51.00	48.50
Digital dial, Dual station	74.50	71.50
Digital programmable, Multi station	84.50	80.50

IRRIGATION WEATHER SENSOR

Costs Each	COM	RES
Irrigation control, Rain/Freeze sensor,		
Wireless	147.00	137.00
Rain/solar/freeze sensor, Automatic seasonal		
adjustment, With 40 ft wiring	241.00	147.00
Sensor, Mini weather station, Automatic		
seasonal adjustment	319.00	304.00

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GREEN ADDITIONS

SECTION UIP 15

EXTERIOR IMPROVEMENTS

RAINWATER HARVESTING SYSTEM

Costs Each	COM	RES
Rain barrel, Brass spigot, 54 gallon	-----	383.00
Rain barrel, Recycled wood, Brass hose bib, 55 gallon	-----	780.00
Stained, 55 gallon	-----	1070.00
Rainwater storage tank, Recycled plastic, Modular, 50 gallon	-----	845.00
Cistern, Above ground,		
100 gallon, HDPE, 28” wide x 45” high	309.00	
150 gallon, 36” wide x 41” high	422.00	
300 gallon, 46” wide x 50” high	491.00	
500 gallon, 46” wide x 76” high	675.00	
1000 gallon, 72” wide x 66” high	1120.00	
2000 gallon, 90” wide x 79” high	1610.00	
3000 gallon, 96” wide x 109” high	2080.00	
4000 gallon, 96” wide x 141” high	3950.00	
10,000 gallon, 141” wide x 160” high	8650.00	

PLANT PREP-MULCH

Costs/Sq. Ft.	COM	RES
Mulch, Straw, Hand spread, 1” deep	2.17	2.00
Bark, Mixed hardwood	7.00	6.49
Bark, Pine mini nuggets	8.40	7.95
Bark, Aromatic red cedar	10.10	9.76

GROUND COVER AND VINES (NO PREP)

Costs Each	COM	RES
Ground Cover, Standard pot, Euonymus fortunei	6.98	6.54
Vinca major	13.55	13.45
Kamtschaticum	12.60	12.45
Xeriscape, Drought resistant, Potted, Plant ready, Sedum, 2” pot	6.72	6.34
4” pot	8.61	8.33
1 gallon	11.25	11.05
24 plug tray	128.00	118.00

SHRUBS AND TREES

Costs Each	COM/RES
Tree, Evergreen, Potted, DeGroot’s spire, 5 gal, 1-2’ tall	128.00
DeGroot’s spire, 10 gal, 2-3’ tall	206.00
Fir, 15 gal, 4-5’ tall	260.00
Fir, 5-6’ tall	550.00
Fir, 6-7’ tall	745.00
Juniper spartan, 15 gal, 4-5’ tall	403.00
Pine, Austrian, 15 gal, 4-5’ tall	275.00
Pine, Austrian, 5-6’ tall	339.00
Pine, Austrian, 6-7’ tall	595.00
Adler, 15 gal, 2-3’ tall	309.00
Ash autumn purple, 15 gal, 2-3’ tall	270.00
Ash marshall, 15 gal, 2-3’ tall	250.00
Hawthorn, 15 gal, 2-3’ tall	290.00
Maple, 15 gal, 2-3’ tall.	290.00
Oak, 15 gal, 4-5’ tall	319.00
Willow weeping, 15 gal, 4-5’ tall	280.00

PLANTING ACCESSORIES

Costs Each	COM/RES
Planter box, Recycled plastic, White, 12” long	295.00
Cedar, 17” long	373.00
Planter box, Recycled plastic lumber, Small, 19” x 19” x 17”	300.00
Large, 24” x 24” x 19”	373.00

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GENERAL INFORMATION

SECTION UIP 16

PERCENTAGE BREAKDOWN OF BASE COST

AVERAGE QUALITY FAST FOOD RESTAURANT

The following percentages indicate the approximate portions of the total cost attributable to each component in a typical fast food restaurant of average size and quality.

Plans, engineering and supervision	2.5%
Permits	1.8%
Survey	.7%
Clearing, excavation and fill	2.3%
Concrete	.8%
Footings	1.2%
Foundations	2.0%
Slab on grade	1.9%
Masonry	4.3%
Wall insulation	.7%
Steel frame and decking	8.4%
Doors, frames and hardware	1.6%
Carpentry	5.5%
Roofing and roof insulation	7.0%
HVAC	10.6%
Electrical, including fixtures	8.6%
Plumbing	6.4%
Sprinkler system	.3%
Toilet partitions and equipment	.6%
Glass storefront	4.9%
Exterior ornamentation	2.7%
Drywall and partitions	2.3%
Tile, (FRP) and marlite	4.0%
Acoustic ceilings	1.6%
Floor coverings	4.6%
Painting	1.8%
Cleanup and miscellaneous	.5%
Tax and freight	.5%
Contractors' overhead and profit	9.9%
TOTAL	100.0%

The 9.9% listed for the general contractors' overhead and profit is the percentage of total cost. This is the equivalent of 11.6% of labor, material and subcontracts, excluding plans, engineering, permits and survey.

AVERAGE QUALITY NEIGHBORHOOD SHOPPING CENTER

The percentages listed below indicate typical costs attributable to various components in an average quality and size neighborhood shopping center.

Plans, engineering and supervision	5.5%
Clearing and excavation	3.1%
Foundations	6.5%
Steel frame and decking	11.5%
Carpentry	3.9%
Insulation and waterproofing	1.5%
Masonry	13.6%
Aluminum storefronts	4.0%
Roofing	6.1%
Doors and frames	1.7%
Interior finish: gypsum board	5.7%
Floor cover	2.5%
Painting	.5%
Acoustic ceilings	1.3%
Plumbing	5.6%
Toilet accessories and partitions	.3%
Heating and air conditioning	7.7%
Electrical, including fixtures	10.2%
Contractors' overhead and profit	8.8%
TOTAL	100.0%

The 8.8% listed for the general contractors' overhead and profit is the percentage of total cost. This is the equivalent of 10.3% of labor, material and subcontracts, excluding plans, engineering and supervision.

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GENERAL INFORMATION

SECTION UIP 16

AVERAGE QUALITY MULTIPLE RESIDENCE

The following percentages indicate approximate portions of the total cost of average quality multiple residences attributable to each component listed, as derived from an analysis of several 10 to 40 unit, 1 to 3 story, wood frame buildings. Cost of plans are normally on a flat fee basis and includes simple details with no supervision by architect.

Plans and engineering	1.7%
Plan check and permit	1.1%
Survey	.5%
Water meter and temporary facilities	.4%
Excavations, forms, concrete and backfill	4.5%
Lightweight concrete	1.2%
Rough carpentry (material)	8.9%
Rough carpentry (labor)	8.9%
Roofing	1.1%
Insulation and weatherstripping	1.5%
Siding, stucco and masonry veneer	5.7%
Plaster and drywall	5.8%
Sash and doors	3.6%
Finish carpentry (material)	1.5%
Finish carpentry (labor)	2.2%
Hardware, rough	.3%
Hardware, finish	.7%
Cabinets	3.9%
Countertops/tile	2.1%
Floor covering: carpeting	2.2%
Resilient	.9%
Plumbing and sewer connections	9.2%
Tub enclosures and toilet accessories	.3%
Electrical	8.3%
Light fixtures	.7%
Built-in appliances	2.9%
Heating	3.7%
Sheet metal	.7%
Ornamental iron	.8%
Painting	2.2%
Miscellaneous (TV antenna, etc.)	.6%
Cleanup	.4%
General contractors' overhead and profit	11.5%
TOTAL	100%

The 11.5% listed for the general contractors' overhead and profit is the percentage of the total cost. This is the equivalent of 13.5% of the labor, material and subcontract costs, excluding costs of plans, survey, plan check and permit, and engineering, with a range from 8.6% to 18.5%.

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GENERAL INFORMATION

SECTION UIP 16

HOTEL DESIGN AND CONSTRUCTION DEVELOPMENT EXAMPLE

PLANNING DESIGN ANALYSIS		GROSS AREA
SITE AREA (1 acre/50 rooms)		10 acres
HOTEL		
	Net Area	
500 Room facility	(sq. ft.)	
223 Doubles @ 310	69,130	
160 Kings @ 375	60,000	
32 Suites @ 585	18,720	
85 Various @ 555	47,150	
Circulation (net to gross) 1.4 x	195,000	273,000
Public facilities		
Lobby, 18 sq. ft./room		9,000
Function, ballroom,meeting, conference space, 1,200 seats @ 15 sq. ft.	18,000	
prefunction and support (net to gross)	x 2	36,000
Food and beverage,dining, lounge and bar space, 600 seats @ 25 sq. ft.	15,000	
Preparation and support (net to gross)	x 1.4	21,000
Retail, 500 sq. ft. (net to gross)	x 1.35	675
Recreational, 2,000 sq. ft. (net to gross)	x 1.4	2,800
Circulation		11,525
Staff support facilities		
Housekeeping @ 25 sq. ft./room	12,500	
Misc. support and storage @ 20 sq. ft./ room	10,000	
Administration @ 15 sq. ft./room	7,500	
Circulation (net to gross) 1.4 x	30,000	42,000
Mechanical facilities		
H.V.A.C., electrical, maintenance areas @ 30 sq. ft./room	15,000	
Circulation	1,500	16,500
TOTAL SQUARE FEET		412,500
Gross area per room (square feet)		825

SITE AMENITIES

Swimming pools, 15 sq. ft./room, 7,500 sq. ft.

Tennis courts, 4

Gazebo, whirlpool spa

Surface parking, 1.4 spaces/room, 700 spaces

Landscaped and waterscaped

CONSTRUCTION PERIOD, approximately 24 months

PROJECT COST ANALYSIS		AMOUNT	COST/ROOM
LAND, $850,000/Acre		$ 8,500,000	$ 17,000
BASE BUILDING COSTS			
Sitework			
Clearing, grubbing, fill, drainage		$ 737,250	$1,475
Site amenities			
Pools and decks, gazebos		1,029,000	2,058
Tennis Courts		236,250	473
Parking and paving		743,250	1,487
Landscape and waterscape		1,213,250	2,427
Exterior signs and lighting		135,000	270
HOTEL HARD CONSTRUCTION COSTS (Percent of base costs)			
Clearing and excavation	(.8)	$ 379,250	
Structural steel	(12.0)	5,811,500	
Miscellaneous metals	(2.9)	1,388,000	
Reinforcing steel	(2.3)	1,110,250	
Exterior walls	(9.7)	4,683,500	
Carpentry	(.8)	382,250	
Millwork	(4.0)	1,929,750	
Concrete	(8.8)	4,259,000	
Roofing and sheetmetal	(1.3)	627,750	
Metal decking	(.5)	241,000	
Windows, doors and hardware	(1.0)	480,500	
Plaster and drywall	(10.2)	4,908,000	
Painting	(1.3)	617,250	
Elevators	(3.9)	1,885,500	
Marble and tile	(2.7)	1,304,000	
Plumbing	(10.9)	5,266,750	
Heating and air conditioning	(7.8)	3,785,250	
Electrical	(7.8)	3,773,000	
Miscellaneous and cleanup	(.6)	284,750	
General contractors' overhead and profit	(10.7)	5,166,000	
TOTAL	(100%)	$48,283,250	$ 96,567
SOFT CONSTRUCTION COSTS			
Architectural, engineering and design fees, project management, surveys, inspections		$3,348,000	$6,696
Contingency		1,658,750	3,318
INDIRECT COSTS			
Construction interest and fees		$5,529,000	$11,058
Utility connection, impact fees		338,000	676
Real estate taxes		255,000	510
Special consultants/feasibility		405,750	812
Legal, title, insurance		169,250	339
Developers' overhead, venture accounting misc. reimbursables, etc.		586,500	1,173
CAPITAL COSTS			
Furniture, fixtures and equipment		$15,788,250	$31,577
Operating equipment		845,000	1,690
Pre-opening expenses		2,933,750	5,868
Working capital		1,920,000	3,840
TOTAL PROJECT COST		$94,654,500	$189,309

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GENERAL INFORMATION

SECTION UIP 16

OFFICE DESIGN AND CONSTRUCTION DEVELOPMENT EXAMPLE

PLANNING DESIGN ANALYSIS	GROSS AREA
SITE AREA	5 acres
SUBURBAN OFFICE BUILDING (Gross sq. ft.)	180,000
Net to gross @ 80%	
10 stories, atrium entry	
Floor system, underfloor duct with cells	
Floor loading, 80 and 100 lbs/sq. ft.	
H.V.A.C., central VAV and perimeter system	
Four elevators	
SITE AMENITIES	
Parking, surface and 3-level structure	
3 spaces/1,000 sq. ft.	
Landscaped and waterscaped	
CONSTRUCTION PERIOD	
Approximately 18 months	

PROJECT COST ANALYSIS	AMOUNT	COST/SQ. FT.
LAND, 900,000/acre	$4,500,000	$25.00
BASE BUILDING COSTS		
Sitework		
Clearing, grubbing, fill, drainage	$ 957,500	$ 5.32
Site amenities		
Parking and paving	4,640,250	25.78
Landscape and waterscape	810,250	4.50
Exterior signs and lighting	95,500	.53
OFFICE HARD CONSTRUCTION COSTS		
Building shell (Percent of base costs)		
Clearing, excavation and foundation (2.4)	$ 445,250	$ 2.47
Steel frame and decking (14.0)	2,598,750	14.44
Miscellaneous steel and iron (1.4)	260,000	1.44
Concrete, floors and roof (6.5)	1,206,750	6.70
Masonry backup and partitions (.7)	129,750	.72
Roofing and roof insulation (.9)	167,000	.93
Building wall panels (4.1)	761,000	4.23
Sash and glazing (8.6)	1,596,500	8.87
Marble, tile and terrazzo (.7)	129,750	.72

PROJECT COST ANALYSIS (CONTINUED)		AMOUNT	COST/SQ. FT.
Building shell (continued)	(Percent of base costs)		
Toilet accessories and partitions	(.6)	$ 111,500	$ .62
Elevators, conveying systems	(4.1)	761,250	4.23
Plumbing	(4.5)	835,000	4.64
Heating and air conditioning	(7.3)	1,355,000	7.53
Electrical	(5.3)	983,750	5.47
Miscellaneous and cleanup	(1.4)	260,000	1.44
General contractor's overhead and profit	(8.5)	1,577,750	8.77
Subtotal	(71.0)	$13,179,000	$73.22
Tenant improvements, work letter			
Acoustical ceilings, suspended	(1.7)	$ 315,500	$ 1.75
Resilient floor coverings	(.6)	111,500	.62
Carpeting	(1.7)	316,500	1.76
Frame and drywall partitions	(3.8)	705,250	3.92
Carpentry	(.6)	111,500	.62
Doors, frames and hardware	(.9)	167,000	.93
Paint and wall covering	(.8)	148,500	.83
Sprinklers and life safety	(1.7)	309,500	1.72
Heating and air conditioning	(4.0)	742,500	4.13
Electrical	(4.2)	779,500	4.33
Tenant upgrades			
Interior finishes	(2.7)	$ 499,500	$ 2.78
Partitions and cabinetry	(3.2)	589,250	3.27
Electrical and mechanical	(2.0)	369,500	2.05
Security and life safety	(1.1)	199,000	1.11
TOTAL	(100.0)	$18,543,500	$103.02
SOFT CONSTRUCTION COSTS			
Architectural, engineering and design fees, project management, surveys, inspections		$1,281,250	$7.12
Contingency		515,250	2.86
INDIRECT COSTS			
Construction interest and fees		$1,657,500	$9.21
Utility connection, impact fees		243,500	1.35
Real estate taxes		81,000	.45
Special consultants, feasibility, space planning		198,250	1.10
Legal, title, insurance		117,000	.65
Advertising, leasing commissions		832,250	4.62
Developers' overhead, venture accounting, etc.		479,250	2.66
CAPITAL COSTS			
Furniture, fixtures and equipment		$3,647,250	$20.26
Operating expenses		2,416,250	13.42
TOTAL PROJECT COSTS		$41,015,750	$227.87

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GENERAL INFORMATION

SECTION UIP 16

FLOOR LOADS

The following are typical live floor loads of some of the buildings used in determining cost averages. A majority of the buildings will have more than one design floor load to accommodate various uses. Thus, a hotel will be designed for heavy floor loads in halls and public areas and lighter floor loads in room areas; banks will need heavier floors in computer and equipment rooms than in office areas.

The listed loads are typical of design specifications of a number of buildings built or under construction during recent years.

OCCUPANCY	POUNDS PER SQUARE FOOT	
Apartments and dormitories	40	
Attics (no storage)	10	(20 lbs. for limited storage)
Auditoriums and churches	100	(50 lbs. for fixed seating)
Balconies (exterior), stairs and fire escapes	40	
Banks	100	
Computer floors	150	
Corridors and stairways	100	
Dwellings	40	(30 lbs. for sleeping rooms)
Equipment rooms	125	
Garage repair	100	
Garages and automobile parking	50	
Governmental buildings	80	
Gymnasiums	100	(150 lbs. for drill floors)
Hospitals	50	(40 lbs. for patient rooms)
Hotels and clubs	40	(100 lbs. in public areas)
Industrials, light	100	
Medium	150	(200 to 350 lbs. for cleanrooms)
Heavy	200	
Libraries	60	(150 lbs. in stacked areas)
Lofts	100	
Meeting rooms	100	
Offices, low cost	50	(plus 20 lbs. for partitions)
Average	50	(plus 20 lbs. for partitions)
Good	60	(plus 20 lbs. for partitions)
Excellent	80	(plus 20 lbs. for partitions)
Restaurants	100	
Roofs, unoccupied	20	(plus snow and drift)
School, classrooms	50	
Stores, retail	75	(125 lbs. in appliance floors)
Wholesale	125	
Theaters	100	(150 lbs. for stage floors)
Warehouses, light	125	
Medium	175	
Heavy	250	(and up)

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GENERAL INFORMATION

SECTION UIP 16

CONSTRUCTION TIME

The following table of average periods of construction lists points on smoothed empirical curves, which have been developed from figures for actual construction jobs. The data was adjusted for time lost due to labor shutdowns with extreme and/or staged construction cases being discarded. No adjustments were made for holidays, inspection delays, weather or other minor shutdowns. Figures are the number of contract (calendar) days from ground breaking to completion of project.

DESIGNED OCCUPANCY	TOTAL COST OF PROJECT (in thousands of dollars)												
	100	200	500	1,000	2,000	3,000	5,000	10,000	20,000	30,000	50,000	75,000	100,000
Apartments and retirement homes, low rise	100	145	200	240	285	310	340	385	-----	-----	-----	-----	-----
Apartments and retirement homes, high rise	-----	-----	-----	250	325	370	425	500	575	620	675	-----	-----
Auditoriums and clubhouses	105	135	185	235	300	345	410	520	665	765	910	1050	-----
Banks	95	120	165	210	270	310	375	475	-----	-----	-----	-----	-----
Churches	120	160	235	315	420	495	610	815	1090	-----	-----	-----	-----
Department stores and shopping malls	-----	-----	-----	280	330	365	415	490	575	635	720	-----	-----
Dormitories	120	150	205	260	330	375	445	565	710	-----	-----	-----	-----
Garages and parking structures	105	135	195	250	325	380	460	600	-----	-----	-----	-----	-----
Governmental buildings	120	155	220	280	365	425	515	665	860	995	1200	1400	1550
Hotels and motels	115	145	195	245	305	350	415	520	655	750	885	1010	1110
Hospitals	-----	-----	270	325	395	445	510	620	755	845	975	1090	1180
Industrial buildings	100	125	165	205	250	285	335	410	510	575	675	-----	-----
Libraries	105	190	300	385	470	515	580	665	750	-----	-----	-----	-----
Medical office buildings	120	180	260	320	375	410	455	515	575	-----	-----	-----	-----
Nursing homes	160	195	250	305	370	415	480	585	710	795	-----	-----	-----
Office buildings	110	135	185	225	280	315	370	460	570	645	760	860	940
Town houses	95	125	180	240	315	-----	-----	-----	-----	-----	-----	-----	-----
Restaurants	120	155	220	285	365	425	515	-----	-----	-----	-----	-----	-----
Retail stores, markets and discount stores	105	160	235	290	340	375	415	470	-----	-----	-----	-----	-----
Schools and colleges	145	180	240	295	365	415	485	600	740	840	985	-----	-----
Veterinary hospitals	130	175	255	340	455	535	665	-----	-----	-----	-----	-----	-----
Warehouses	90	110	150	185	235	265	315	390	-----	-----	-----	-----	-----

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